MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION FOR PIPERACK FOUNDATION -49W1 Raw Water Treatment Plant**

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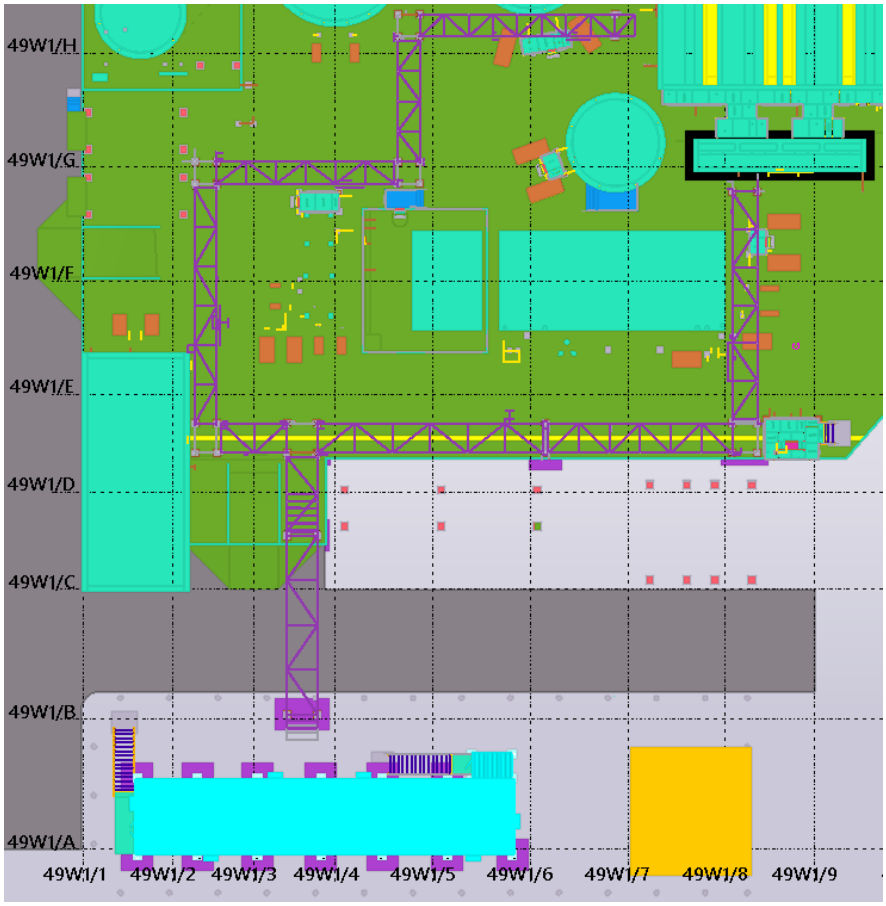
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1. INTENT

This calculation report is to design and analyze pipe rack foundations in Raw Water Treatment Plant area. The report describes the methodology and the design parameters used in the design calculation.

2. SCOPE



The design follows and verified with requirements in Saudi Building Code:

- SBC_301 (Structural – Loading and Forces)
- SBC_304 (Structural – Concrete Structures)
- SBC_303 (Structural – Soil and Foundations)

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Concrete structures are modeled as surface and member in Dlubal RFEM 5.21, both strength and allowable stress calculations are included.

3. MATERIAL PROPERTY

Concrete compressive strength at 28 days: f_{ck} = 35 N/mm^2

Reinforcing bars ASTM A615- Grade 60, yield strength f_{yk} = 420 N/mm^2

4. LOAD CALCULATIONS

4.1. Design loads

Basic load values and their calculation are described in document MD-216-49W1-EG-ST-CAL-1001. Input load values are given in Attachment 1. Basic load cases include:

1	Dead load (D)
2	Live load (L)
3	Wind load in ±X and ±Y direction (W)
4	Temperature ±25 (T)
5	Seismic load (E)

4.2. Load combinations

For strength calculation:

Load Combination	Expression
101	1,40 D
102	1,40 (D +T) + 1,70 L
103	1,20 D + 1,00 L
104	1,20 D + 0,80 W
105	1,20 D + 1,60 W + 1.00 L
106	1,20 D + 1,00 E + 1,00 L
107	0,90 D + 1,60 W
108	0,90 D + 1,00 E

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For allowable stress calculation:

Load Combination	Expression
201	D
202	D + L + T
203	D + W
204	D + 0,70 E
205	D + 0,75 W + 0,75 L
206	D+ 0,75 (0,70 E) + 0,75 L
207	0,60 D + W
208	0,60 D + 0,70 E

5. SOIL BEARING CAPACITY

Vertical modulus of subgrade reaction K_s = 33,750 kN/m^3 at depth below FGL 0.6m is given in geotechnical report MD-216-3000-EG-CV-RPT-1008-C01, BH08 and applied in RFEM as spring constant in z direction. Horizontal modulus of subgrade reaction K_s = 155,000 kN/m^2 is applied as line support in x and y directions.

Recommended allowable bearing capacity q_a =375 kN/m^2 is given in geotechnical report. Therefore, the contact stress results shall not exceed the limit.

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6. CALCULATION ANALYSIS

6.1. Load cases

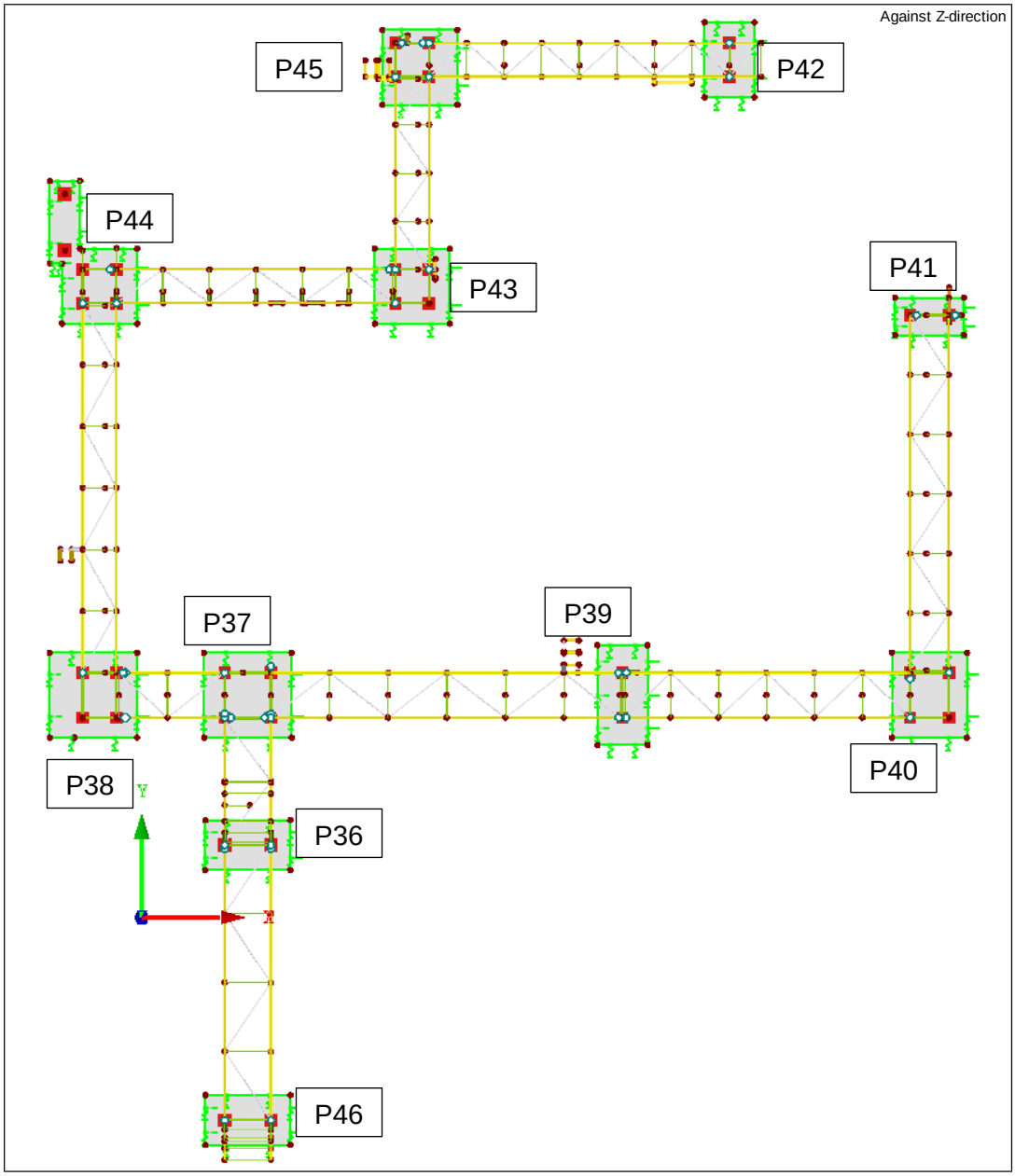


Figure 1 Calculation model in RFEM (pad marks)

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Table 1 Pad reinforcement summary

pad footing	top reinforcement	bottom reinforcement	size [mm]		
			dimension Bx	dimension By	thickness
P36	T16/150mm	T16/150mm	3400	2000	400
P37	T16/150mm	T16/150mm	3500	3400	400
P38	T16/150mm	T16/150mm	3500	3400	400
P39	T16/150mm	T16/150mm	2000	4000	400
P40	T16/150mm	T16/150mm	3000	3400	400
P41	T16/150mm	T16/150mm	2700	1500	400
P42	T16/150mm	T16/150mm	2000	3000	400
P43	T16/150mm	T16/150mm	3000	3000	400
P44	T16/150mm	T16/150mm	max. 3498	max. 5707	400
P45	T16/150mm	T16/150mm	3000	3000	400
P46	T16/150mm	T16/150mm	3400	2000	400

dimensions can be found on page 83.

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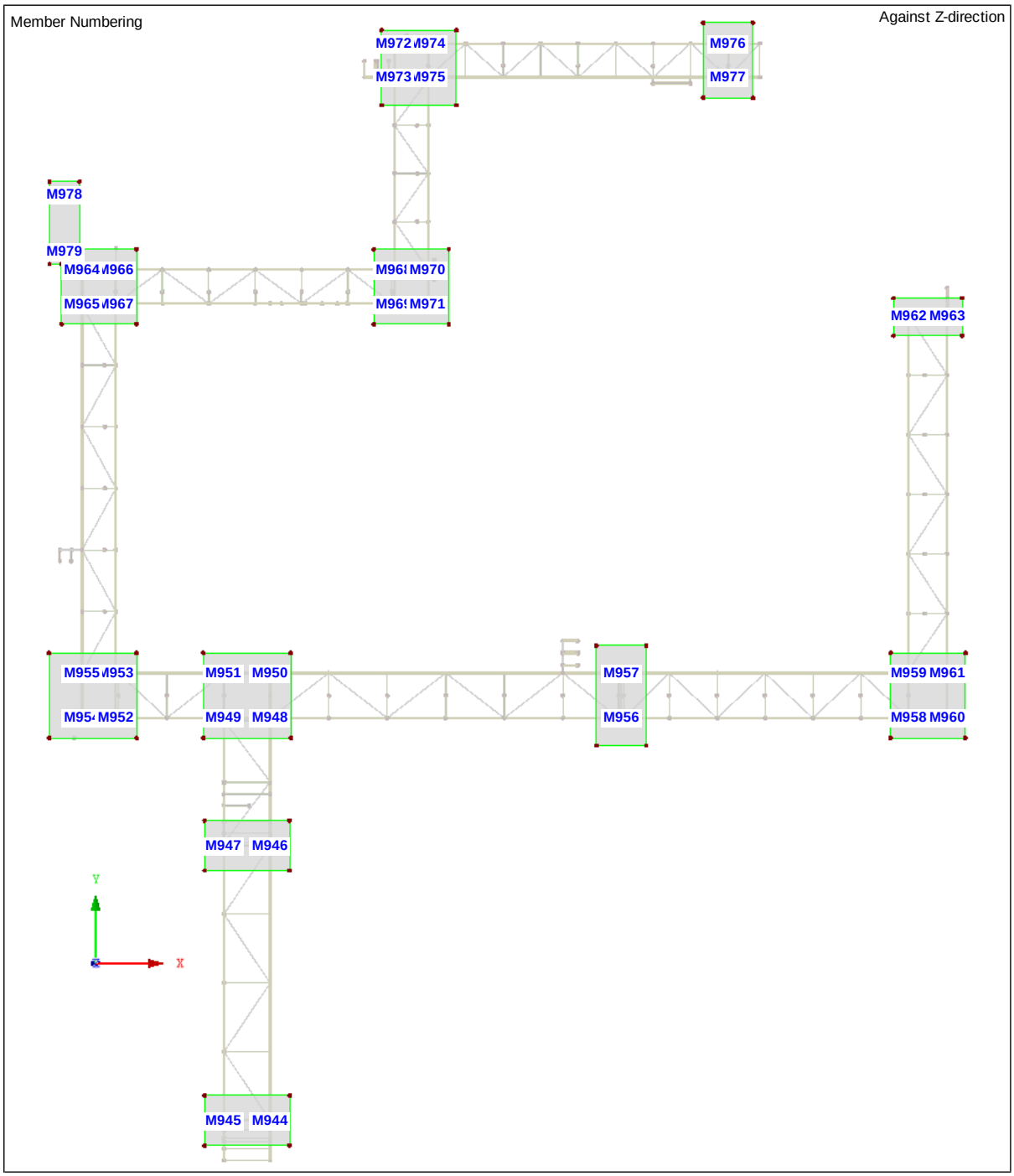


Figure 2 Calculation model in RFEM (pedestal marks)

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Table 2 Pedestal reinforcement summary

pedestal Nr.	cross-section [mm]	height [mm]	longitudinal reinf	shear reinforcement
944	550x550	1150	8T16	T10@200mm
945	550x550	1150	8T16	T10@200mm
946	550x550	1150	8T16	T10@200mm
947	550x550	1150	8T16	T10@200mm
948	500x500	1150	8T16	T10@200mm
949	500x500	1150	8T16	T10@200mm
950	500x500	1150	8T16	T10@200mm
951	500x500	1150	8T16	T10@200mm
952	500x500	1150	8T16	T10@200mm
953	500x500	1150	8T16	T10@200mm
954	500x500	1150	8T16	T10@200mm
955	500x500	1150	8T16	T10@200mm
956	500x500	1500	8T16	T10@200mm
957	500x500	1500	8T16	T10@200mm
958	500x500	1500	8T16	T10@200mm
959	500x500	1500	8T16	T10@200mm
960	500x500	1500	8T16	T10@200mm
961	500x500	1500	8T16	T10@200mm
962	500x500	1150	8T16	T10@200mm
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966	500x500	1150	8T16	T10@200mm
967	500x500	1150	8T16	T10@200mm
968	500x500	1150	8T16	T10@200mm
969	500x500	1150	8T16	T10@200mm
970	500x500	1150	8T16	T10@200mm
971	500x500	1150	8T16	T10@200mm
972	500x500	1150	8T16	T10@200mm
973	500x500	1150	8T16	T10@200mm
974	500x500	1150	8T16	T10@200mm
975	500x500	1150	8T16	T10@200mm
976	500x500	1150	8T16	T10@200mm
977	500x500	1150	8T16	T10@200mm
978	550x550	1280	8T12	T10@200mm
979	550x550	1280	8T12	T10@200mm

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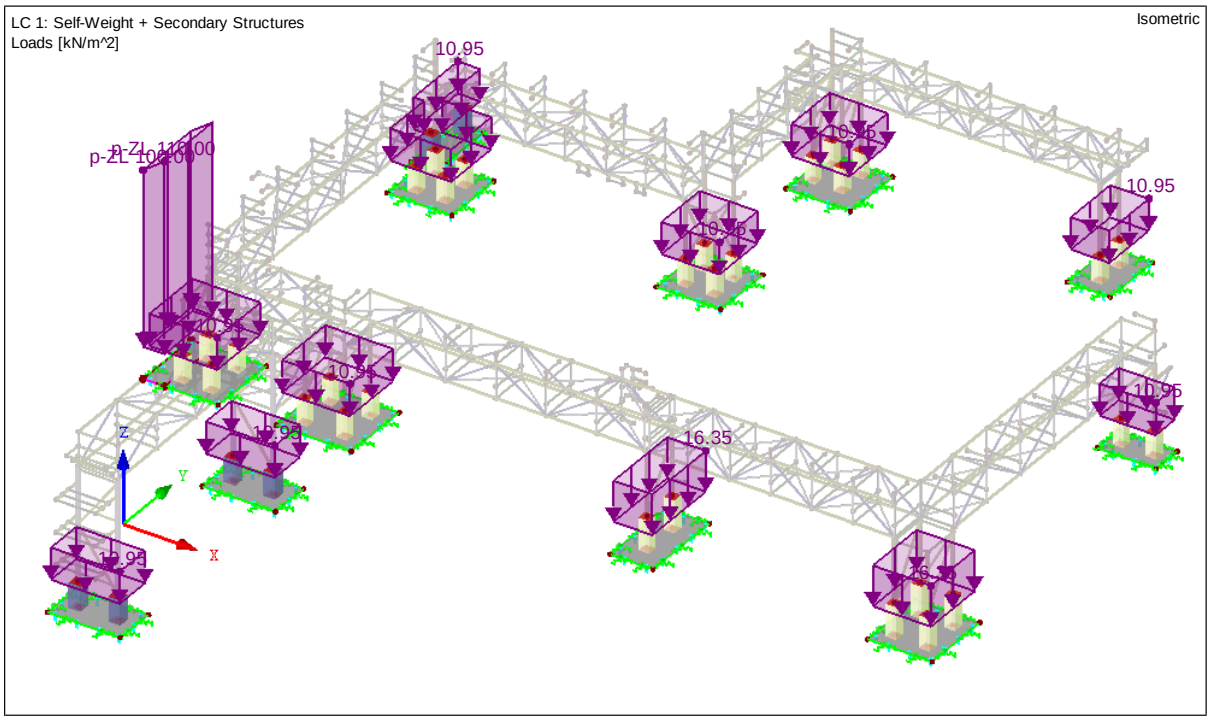


Figure 3 Dead load

Dead load includes self-weight, backfill soil and ground slab weight 18kN/m^3*0.4m + 25kN/m^3* 0.15m=10.95kN/m^2 or 18kN/m^3*0.7m + 25kN/m^3* 0.15m=16.35kN/m^2.

For foundation P38, a surface load 110kN/m2 is applied to count for the upper structure weight on footing.

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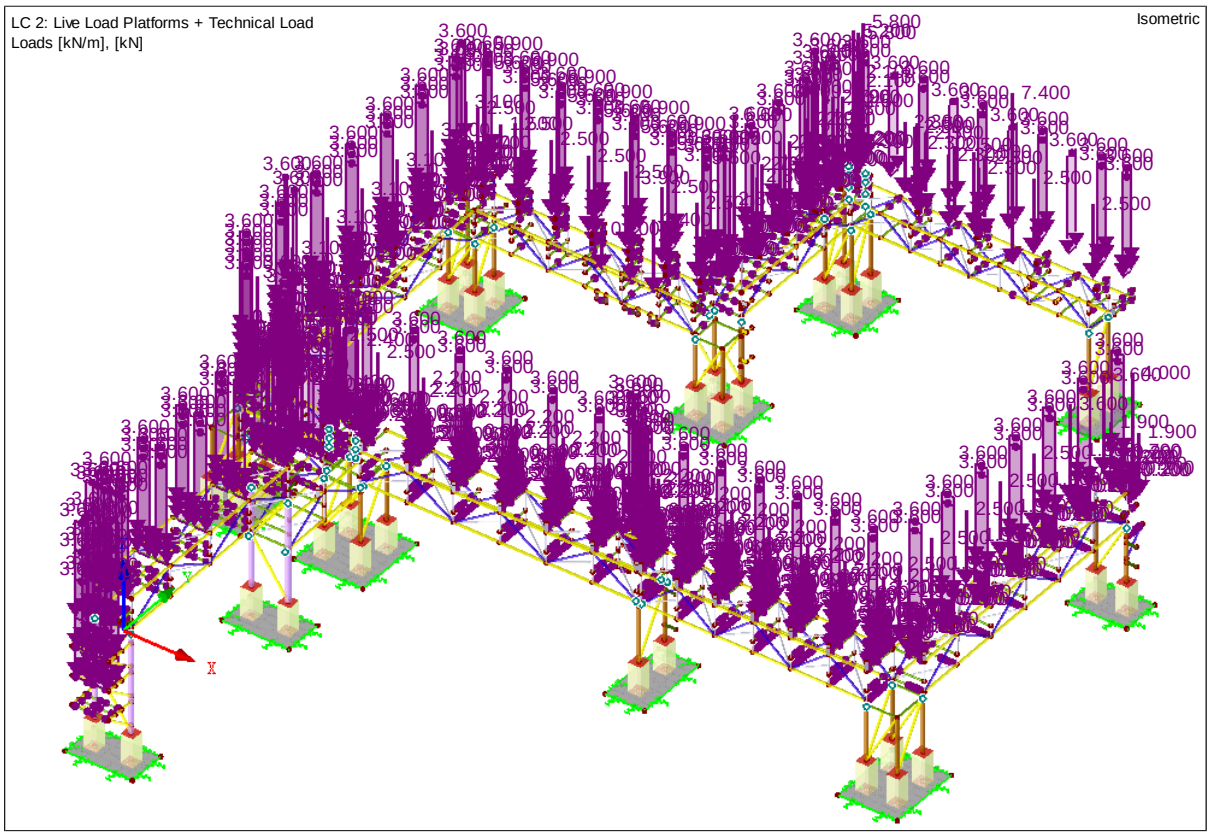


Figure 4 Live load

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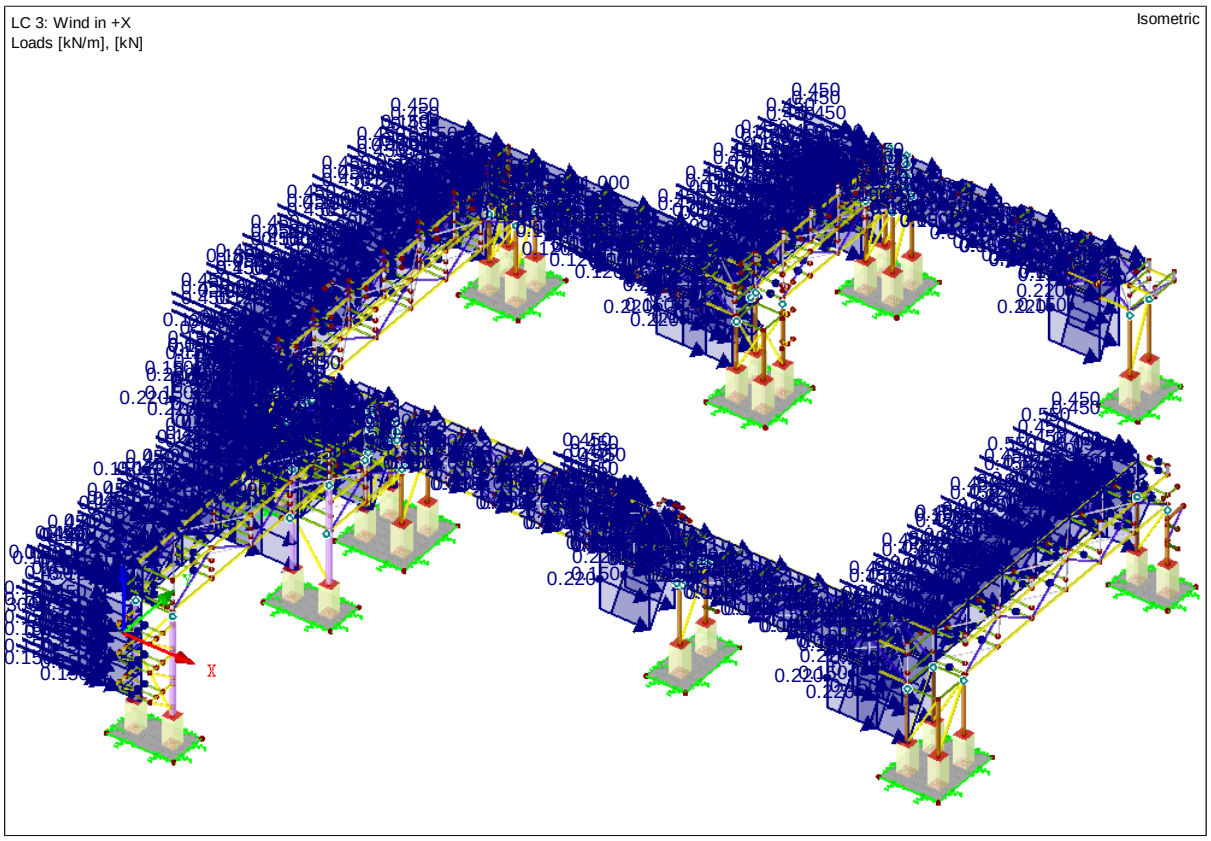


Figure 5 Wind load in X+

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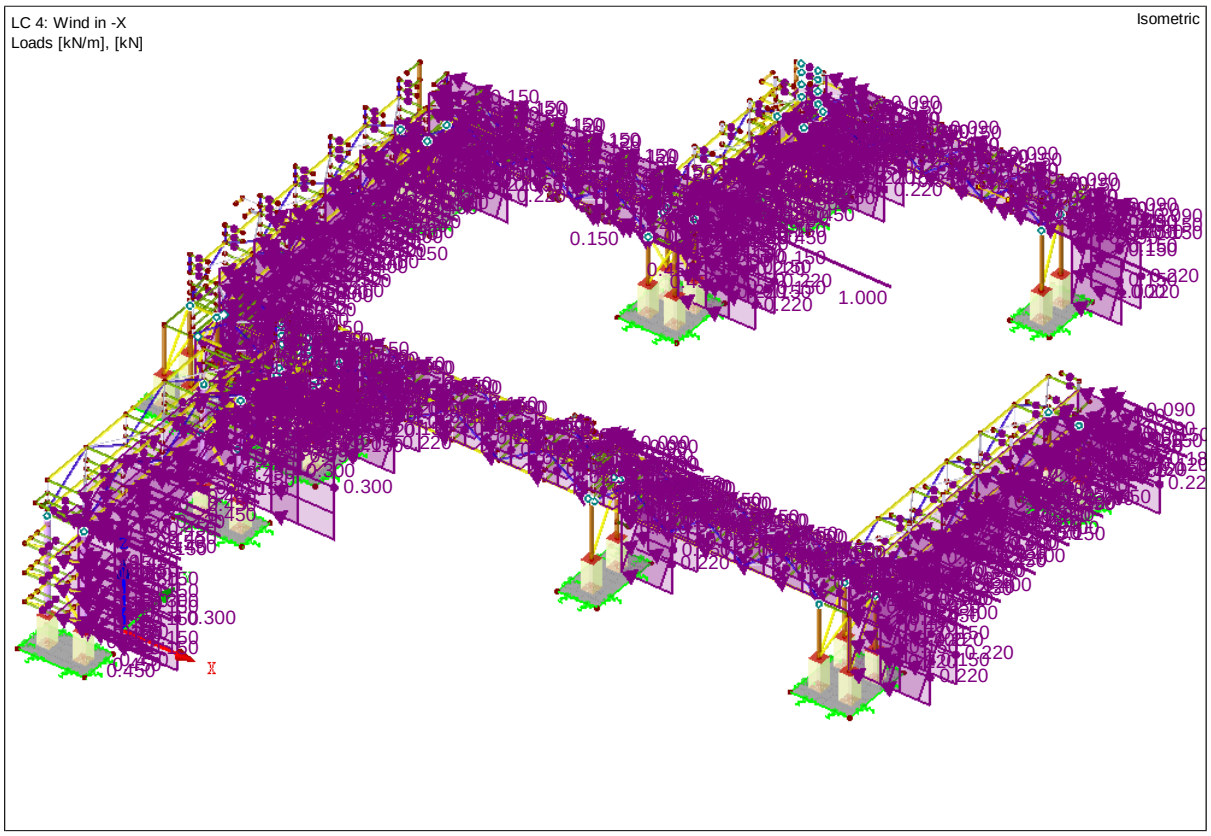


Figure 6 Wind load in X-

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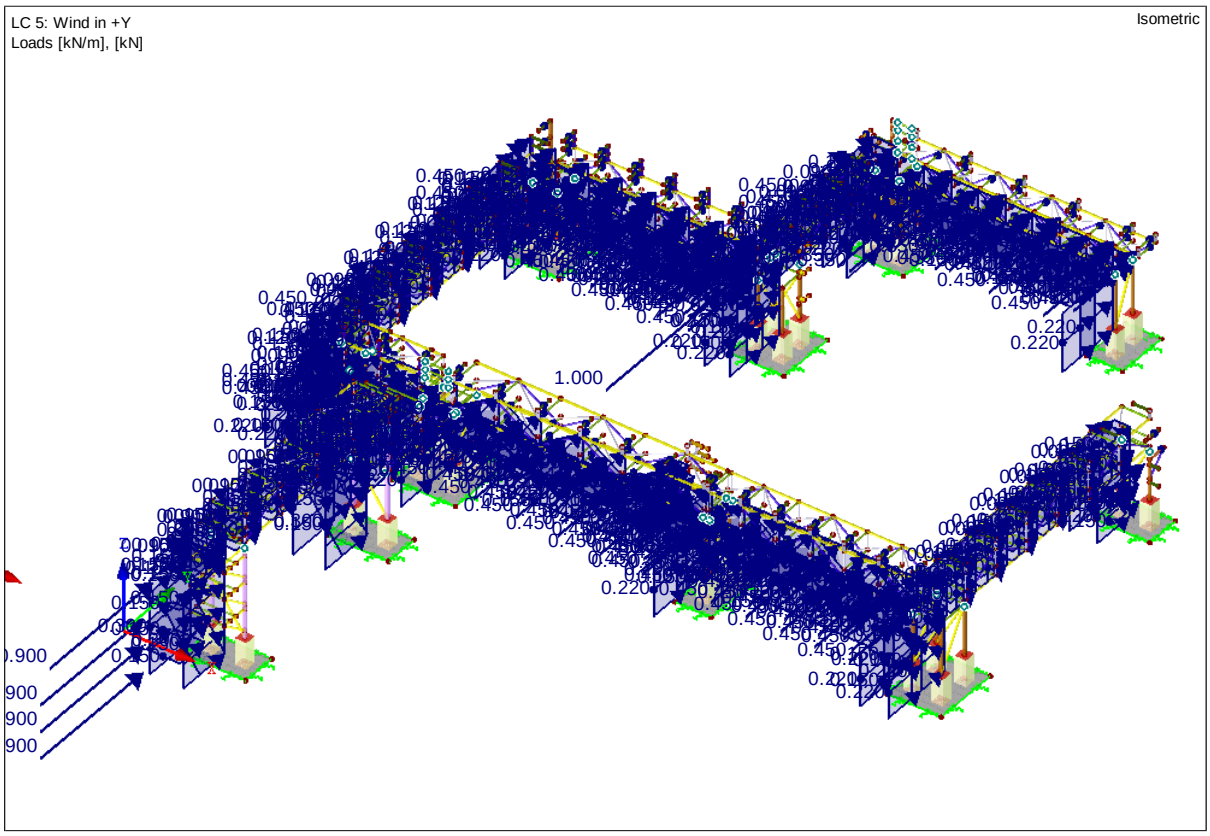


Figure 7 Wind load in Y+

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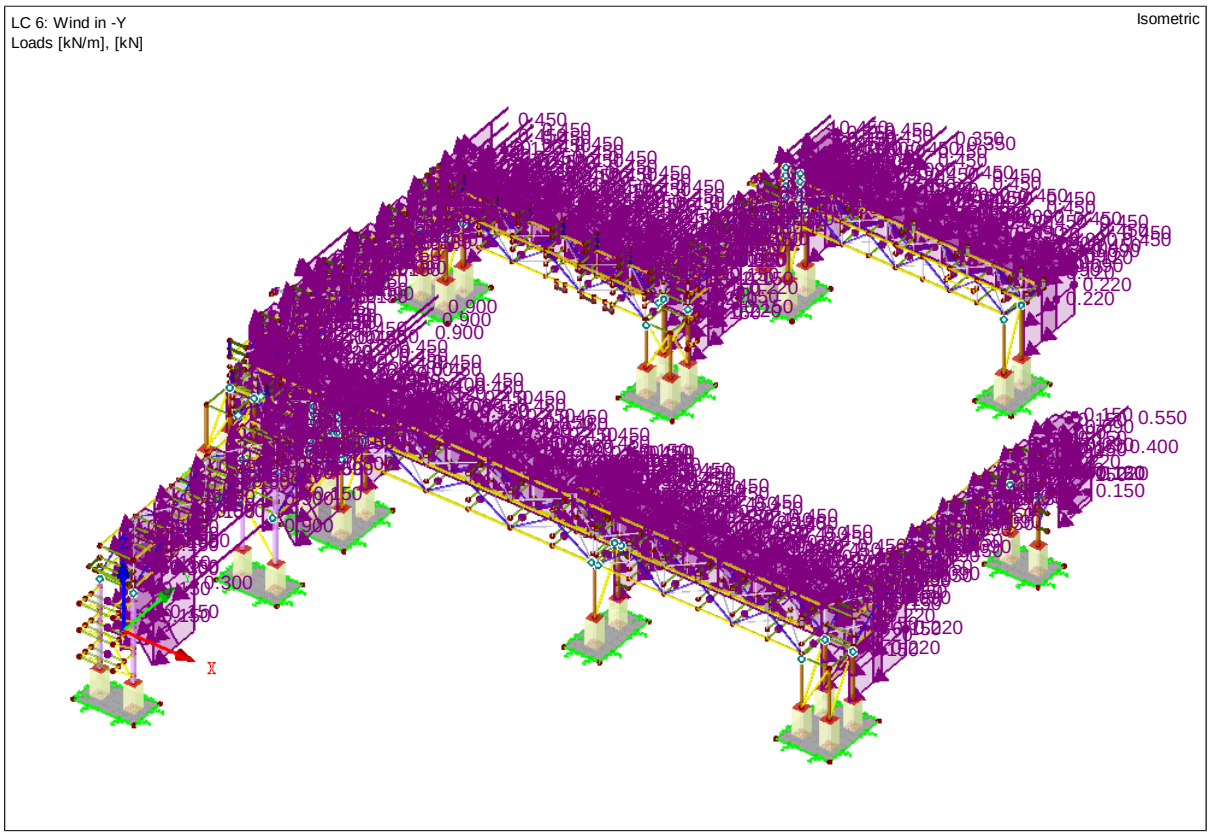


Figure 8 Wind load in Y-

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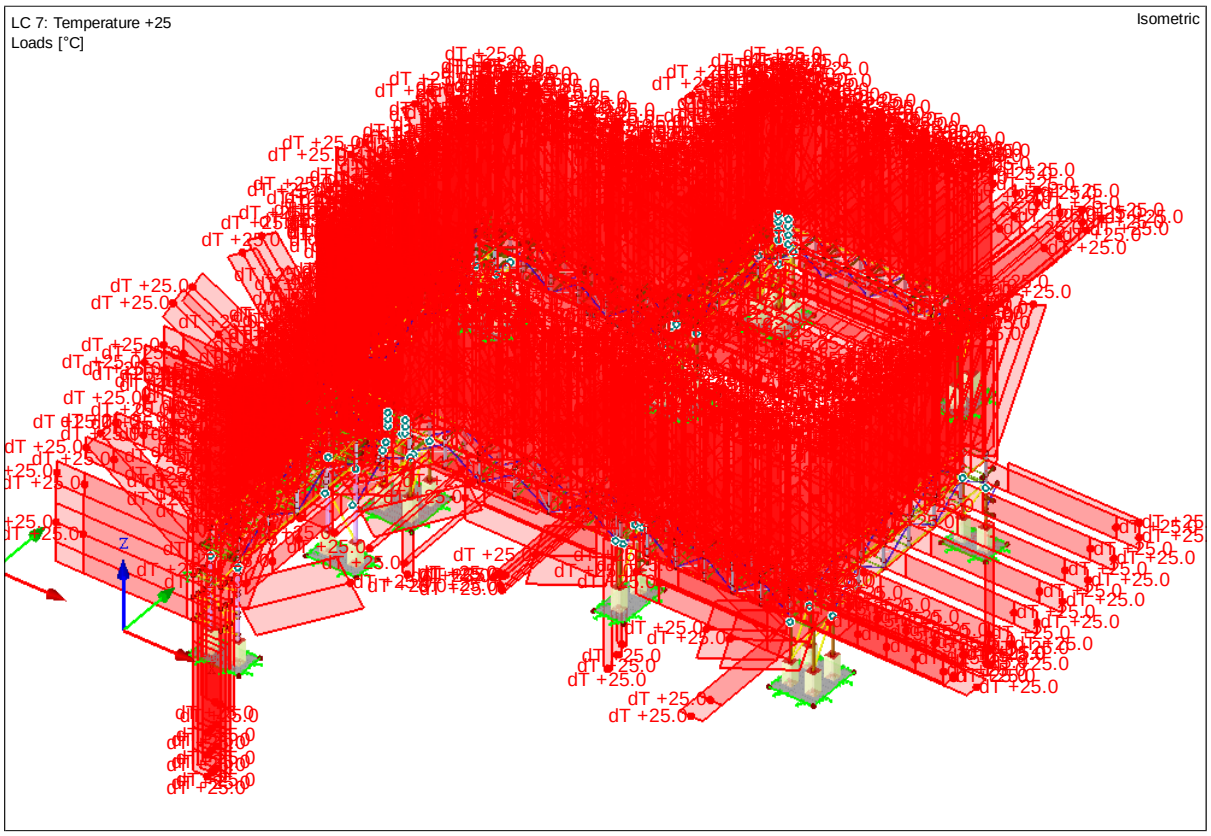


Figure 9 Temperature load +25

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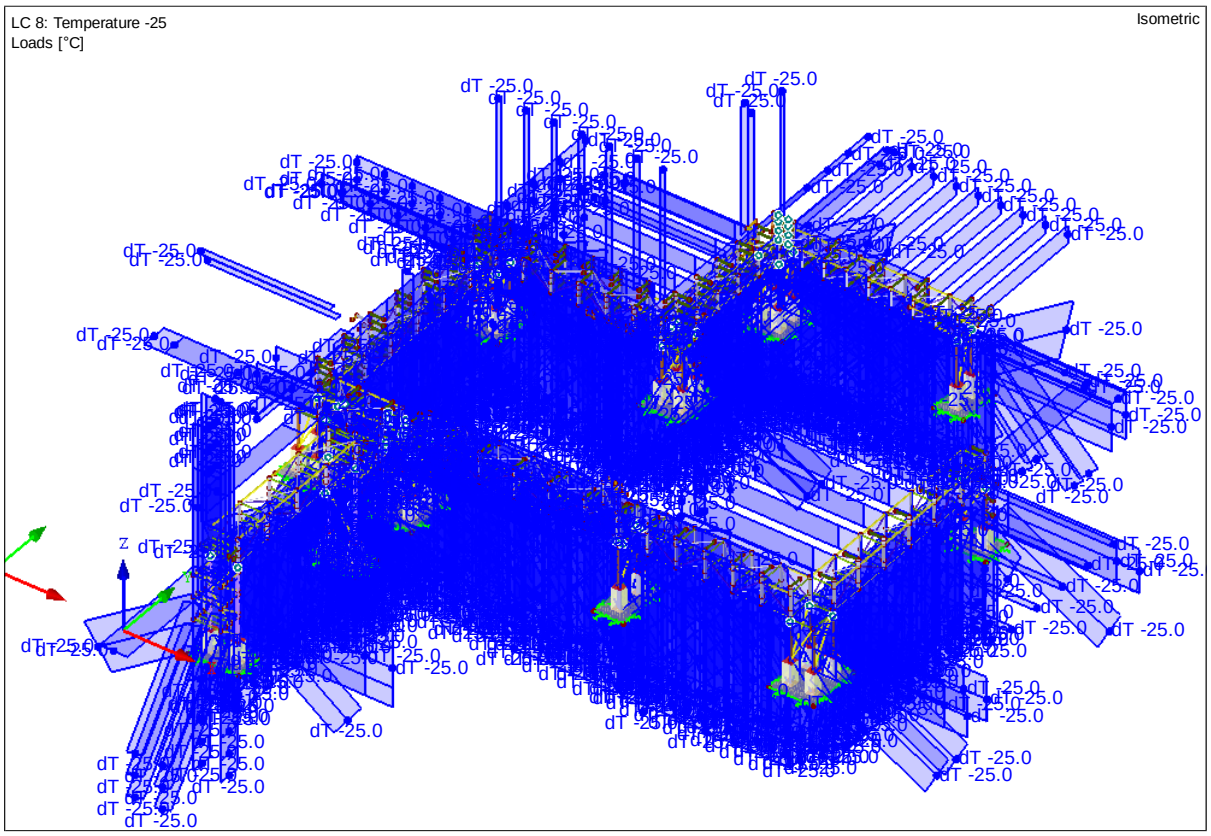


Figure 10 Temperature load -25

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6.2. Bearing pressure check

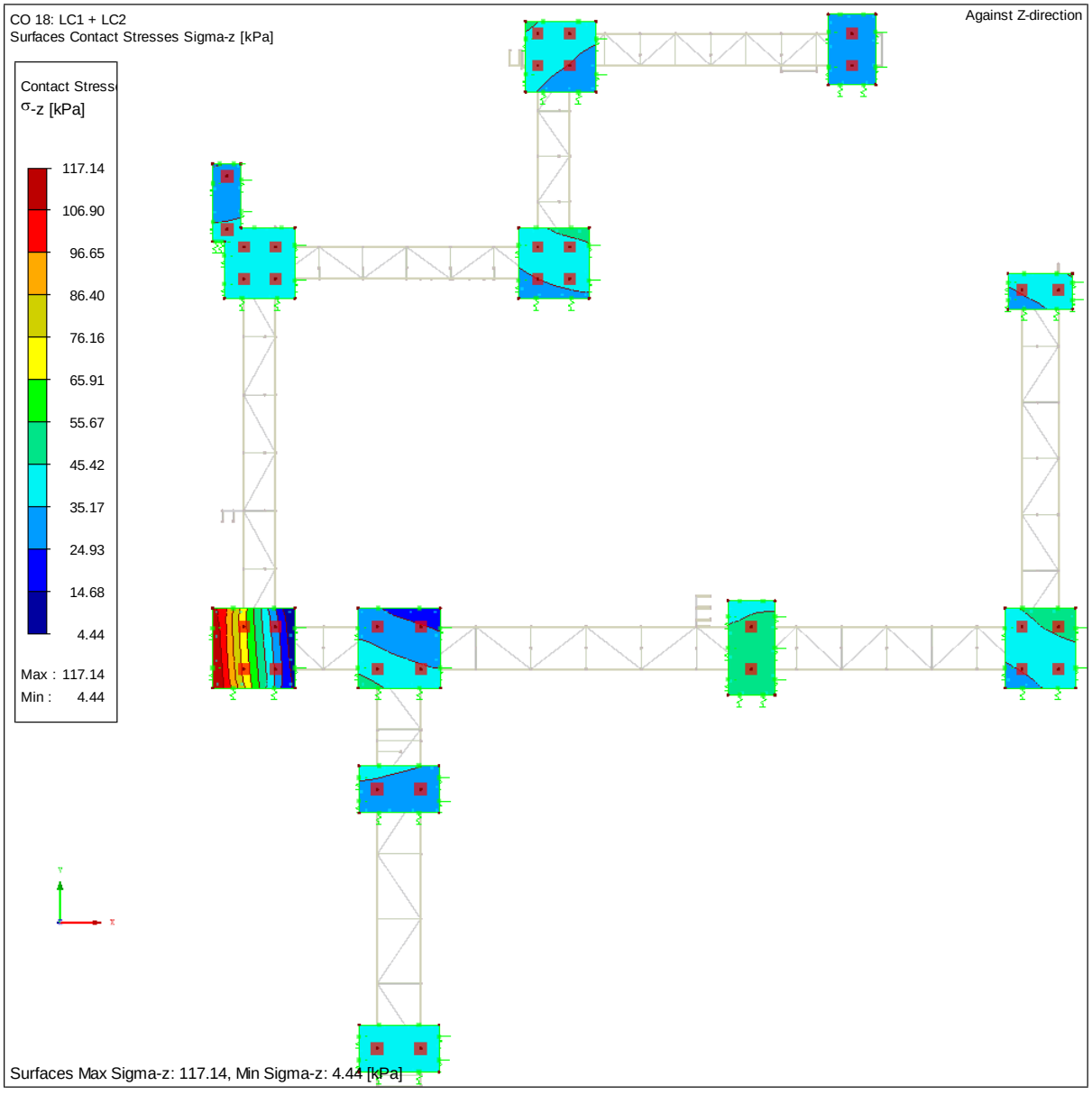


Figure 11 Soil contact pressure for load combination 18

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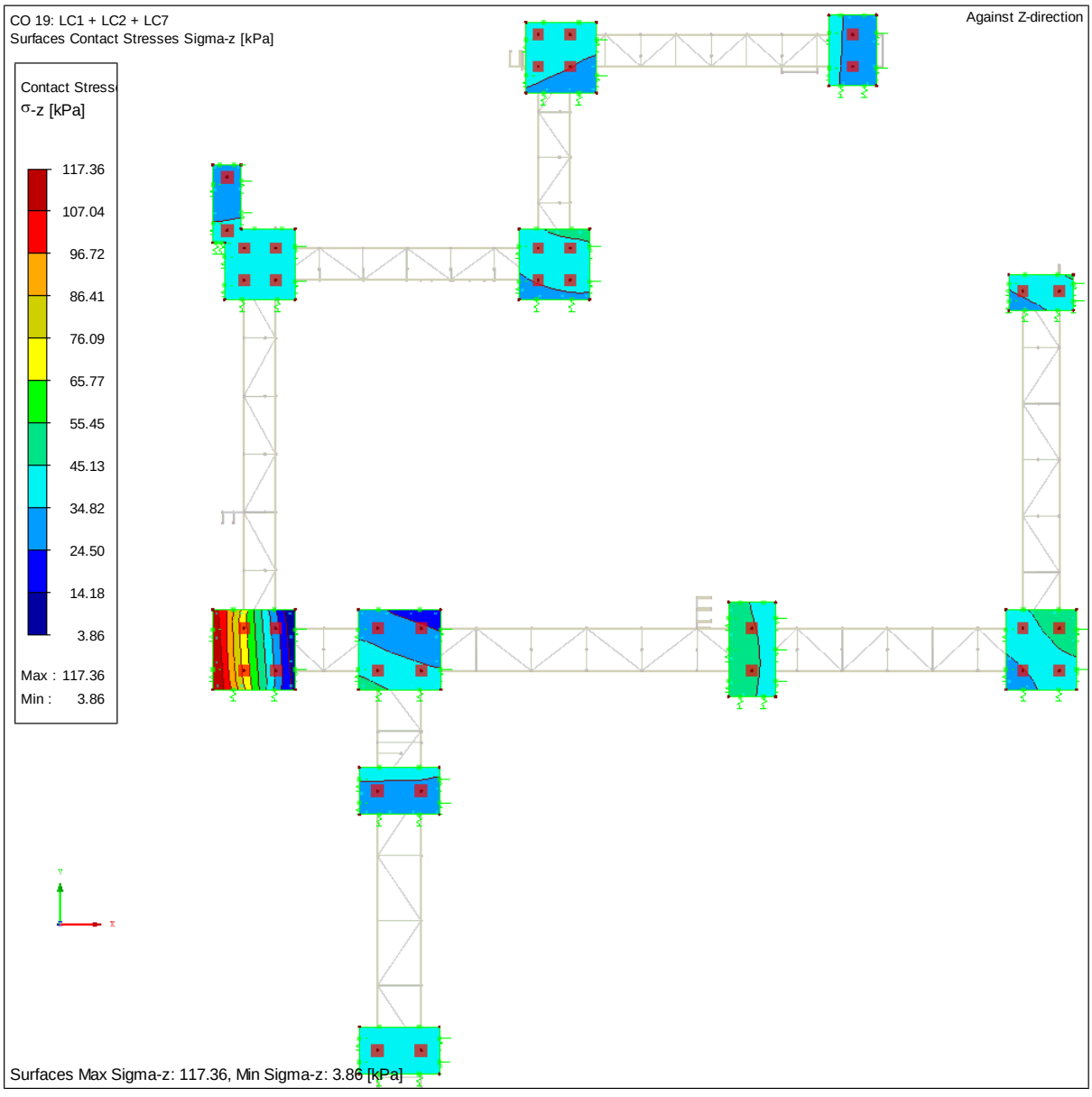


Figure 12 Soil contact pressure for load combination 19

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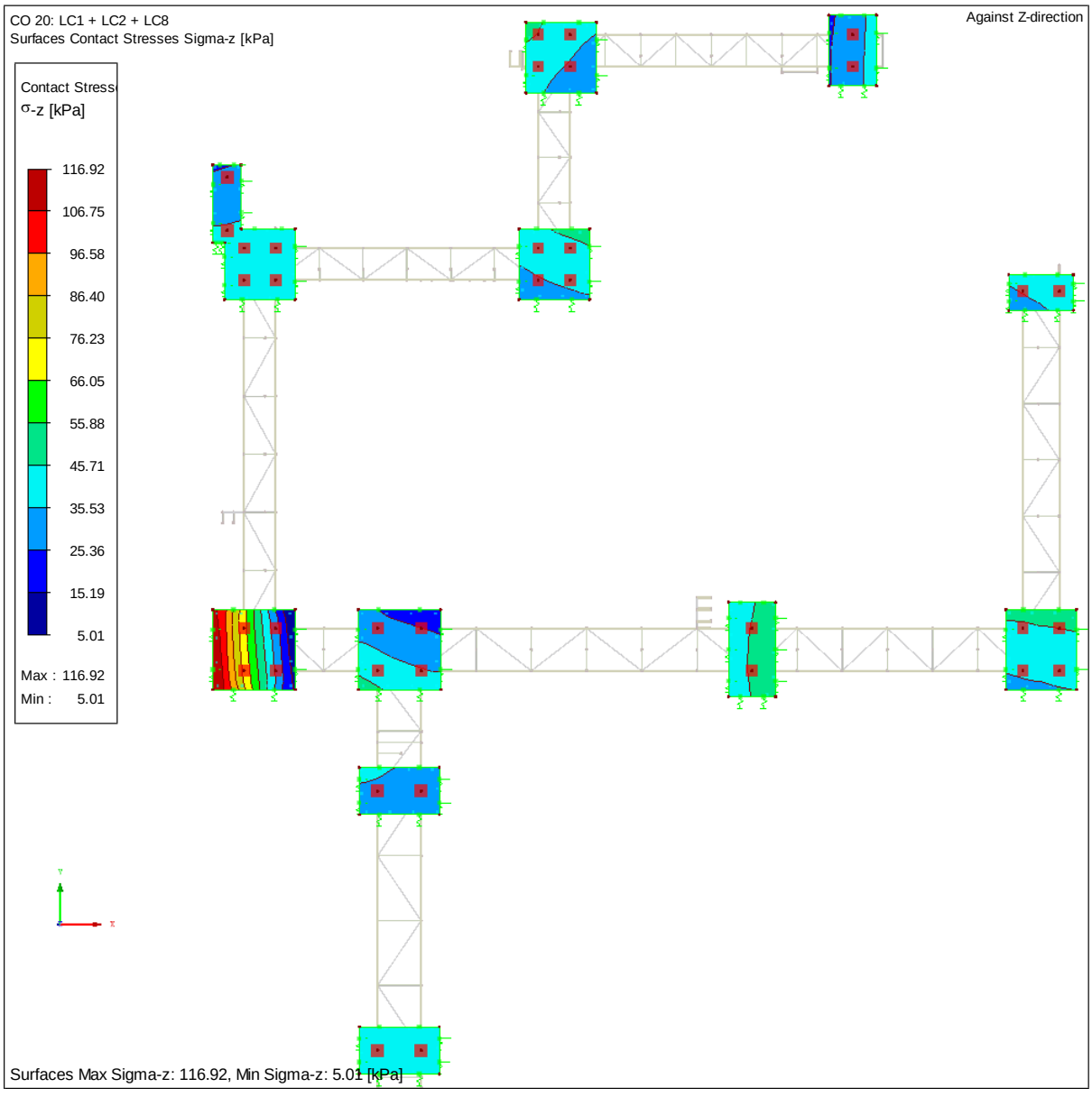


Figure 13 Soil contact pressure for load combination 20

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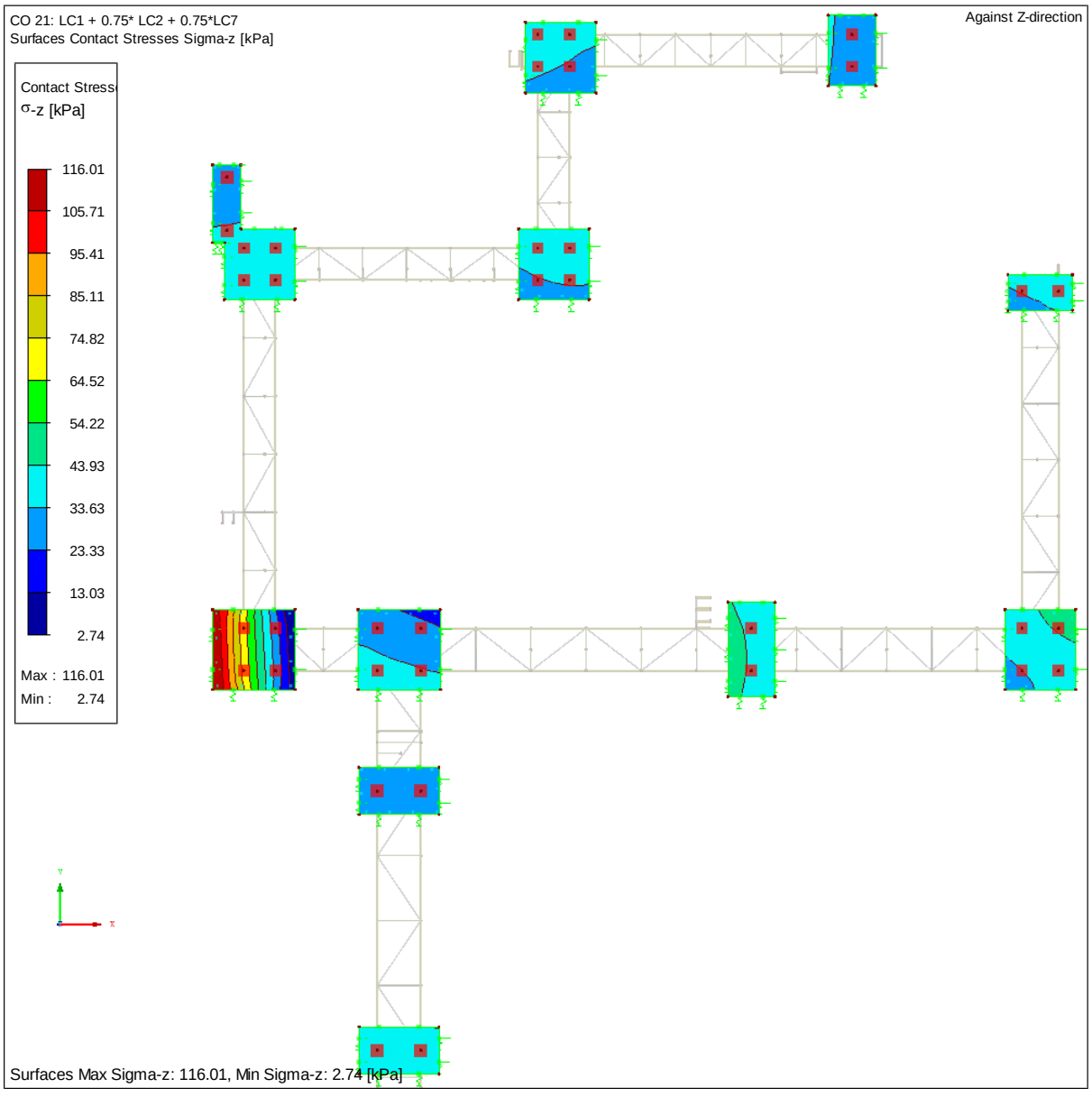


Figure 14 Soil contact pressure for load combination 21

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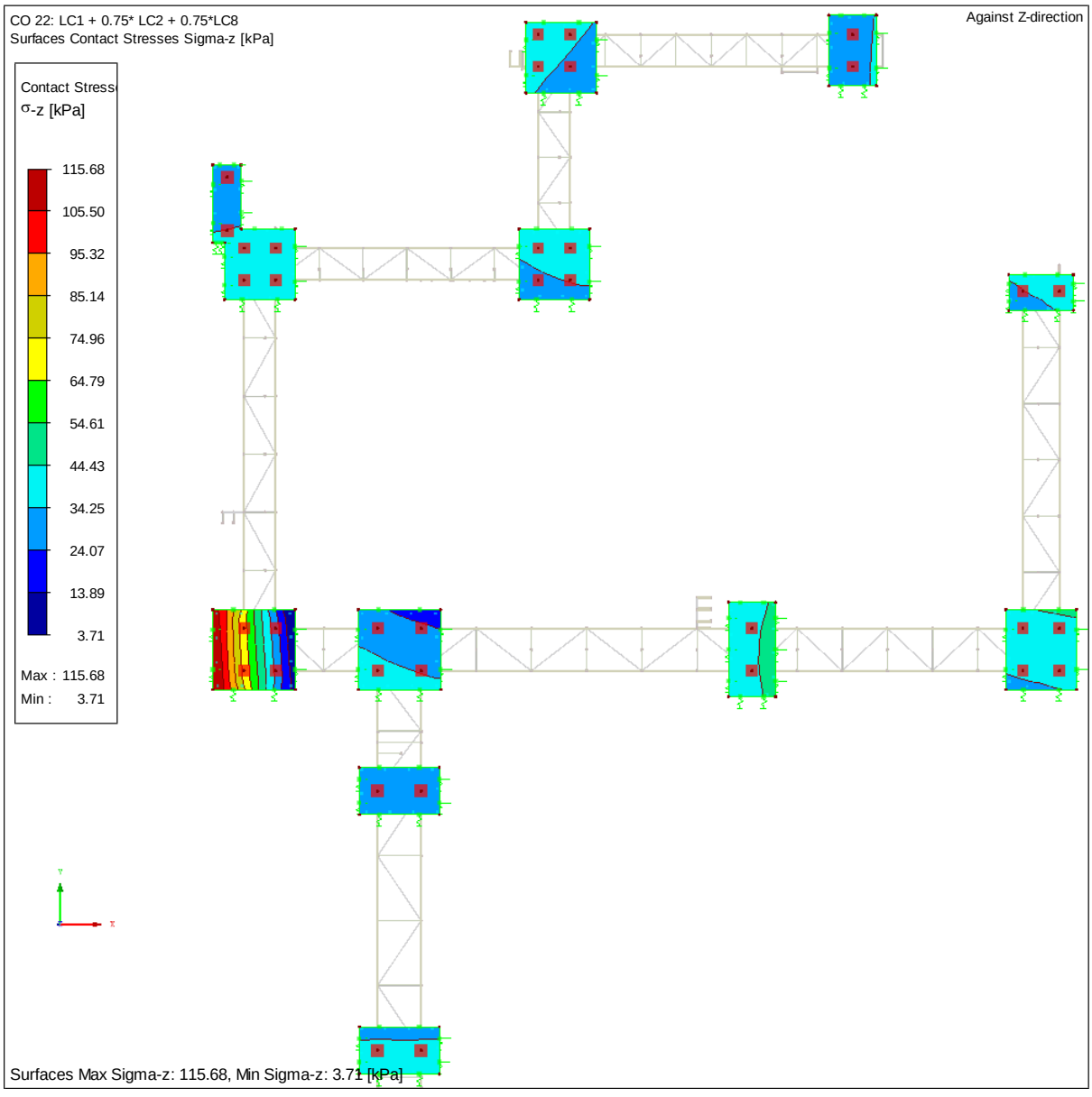


Figure 15 Soil contact pressure for load combination 22

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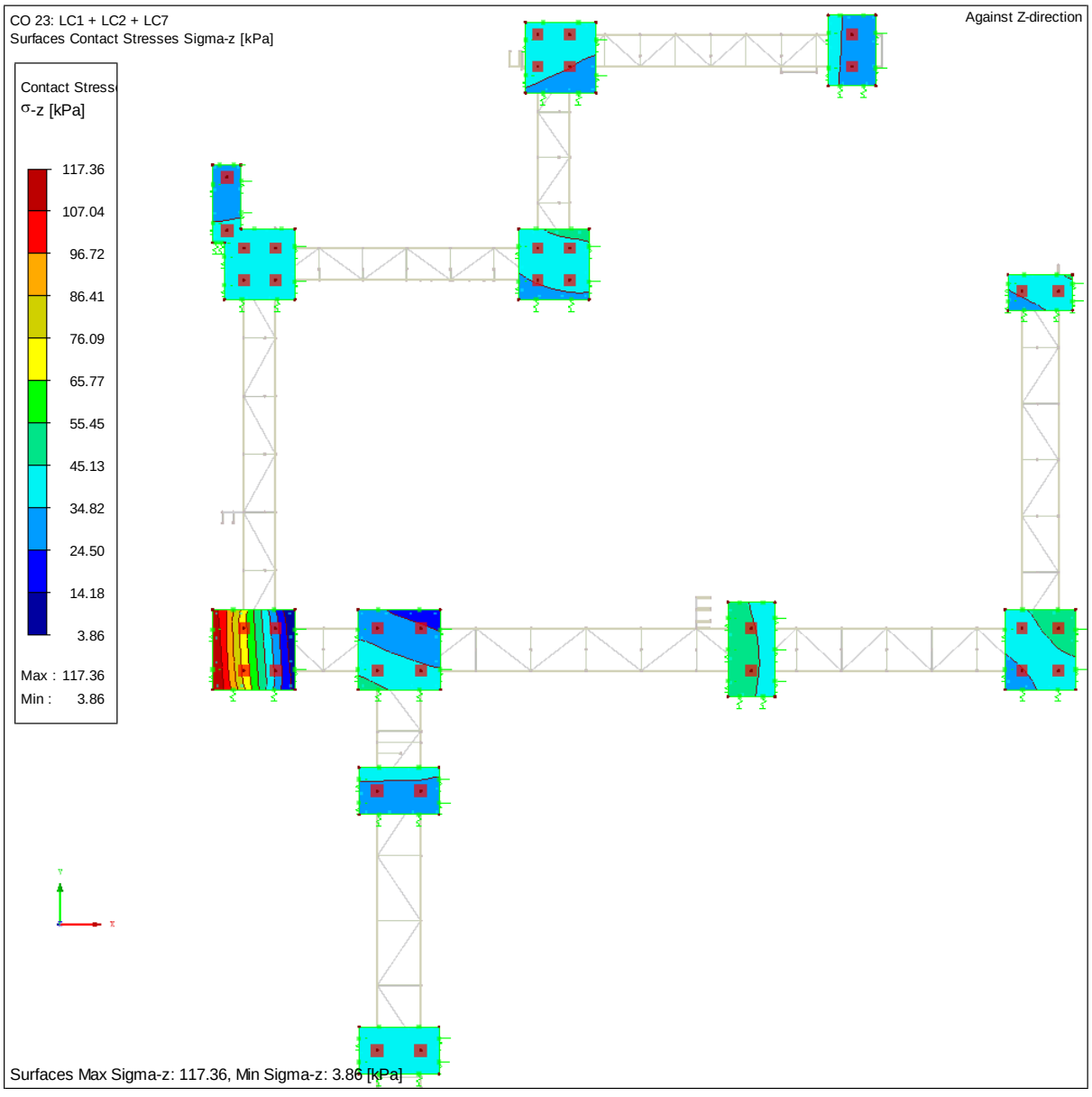


Figure 16 Soil contact pressure for load combination 23

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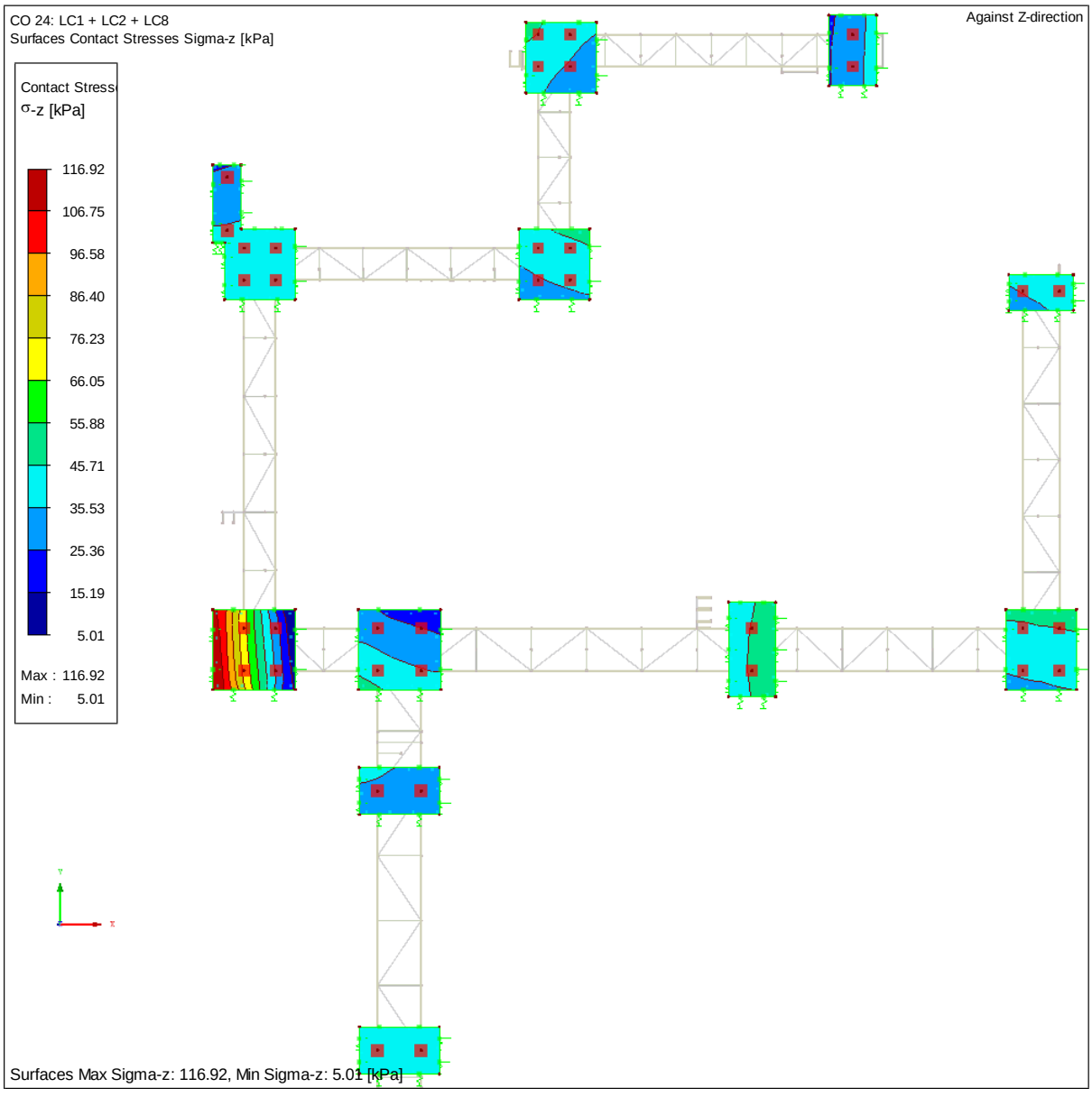


Figure 17 Soil contact pressure for load combination 24

$\sigma_s < q_a = 375$ kN/m^2, ok!

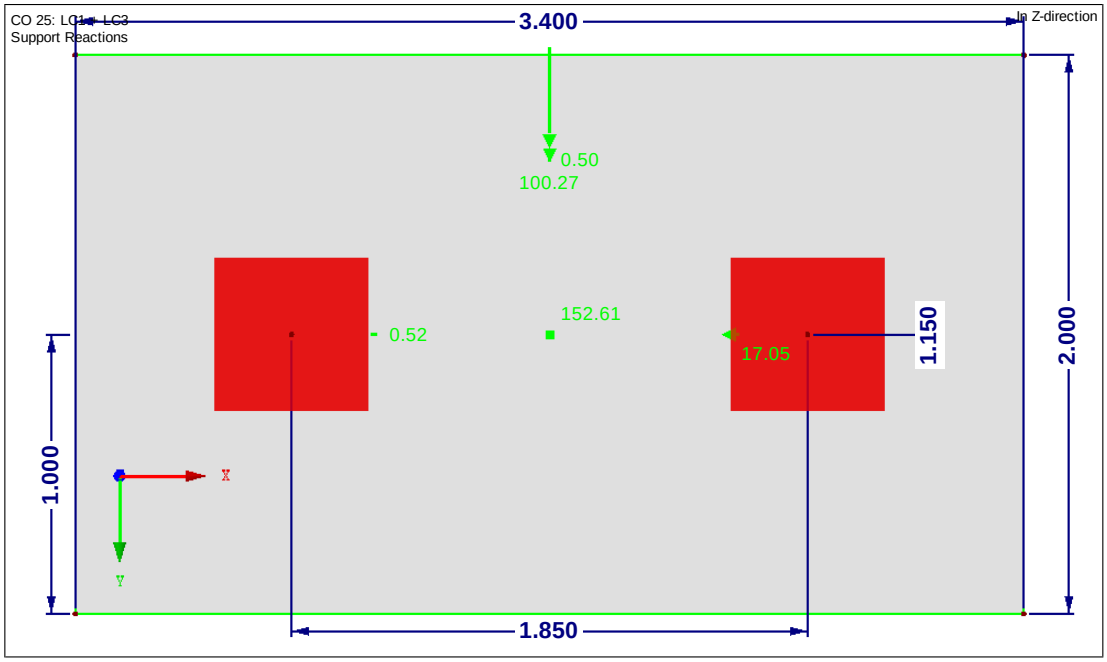
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6.3. Stability check for 2-pedestal foundation P36, P39, P41, P42, P46

combined footing P36

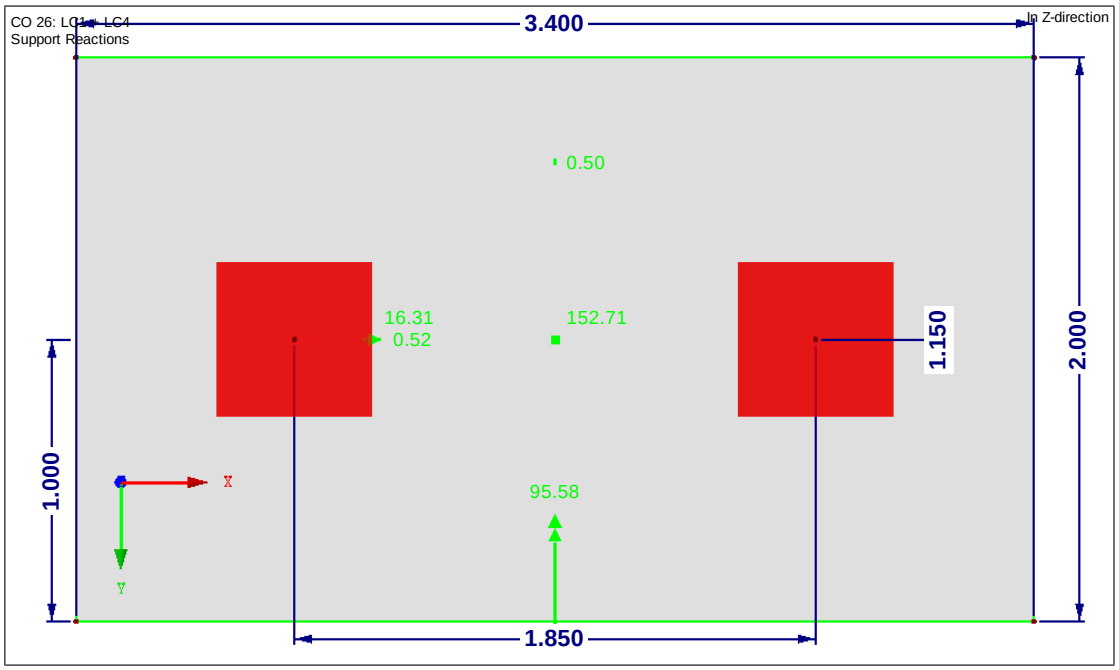


OVERTURN:	Mx=	100.27	kNm	overturn moment				
	Fz=	152.61	kN	total vertical load				
	e=	0.657	m	eccentricity				
	B/2=	1.7	m					
	F.O.S=	2.587	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	0.5	kN	lateral force in y				
	Fy=	17.05	kN	lateral force in x				
	Hd=	17.06	kN	sliding force				
	Rd=	61.044	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	3.579	>1.50	Factor of safety against sliding				

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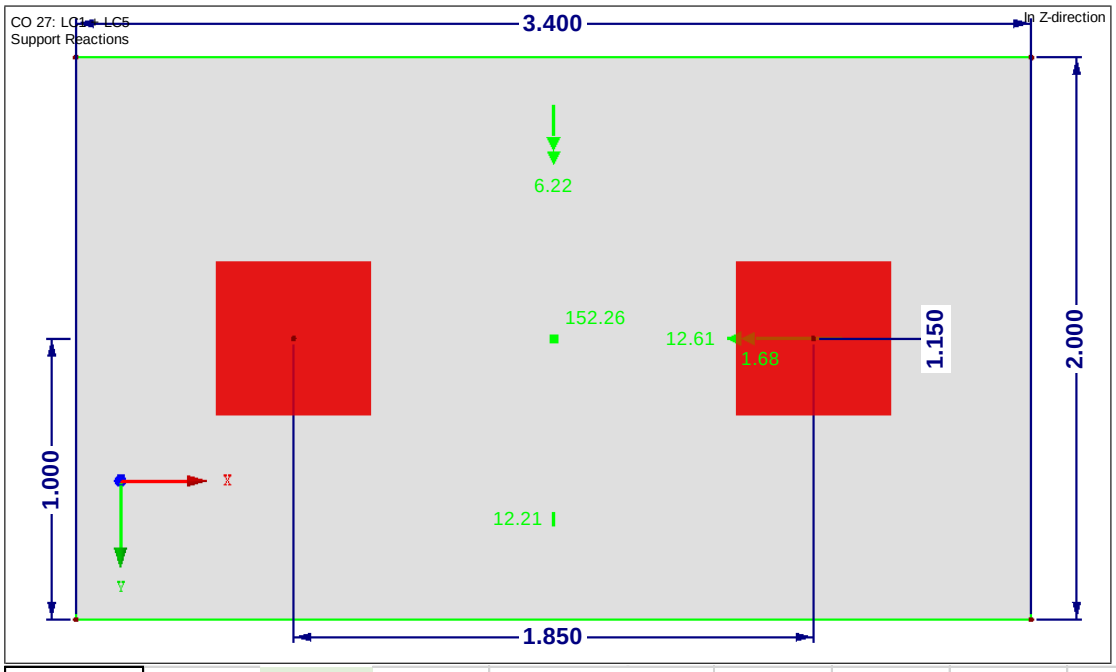


OVERTURN:	Mx=	95.58	kNm	overturn moment					
	Fz=	152.71	kN	total vertical load					
	e=	0.626	m	eccentricity					
	B/2=	1.7	m						
	F.O.S=	2.716	>1.50	Factor of safety against overturning					
SLIDING:	Fx=	0.5	kN	lateral force in y					
	Fy=	16.31	kN	lateral force in x					
	Hd=	16.32	kN	sliding force					
	Rd=	61.084	kN	resistance to sliding =coefficient of friction* total vertical load					
	F.O.S=	3.743	>1.50	Factor of safety against sliding					

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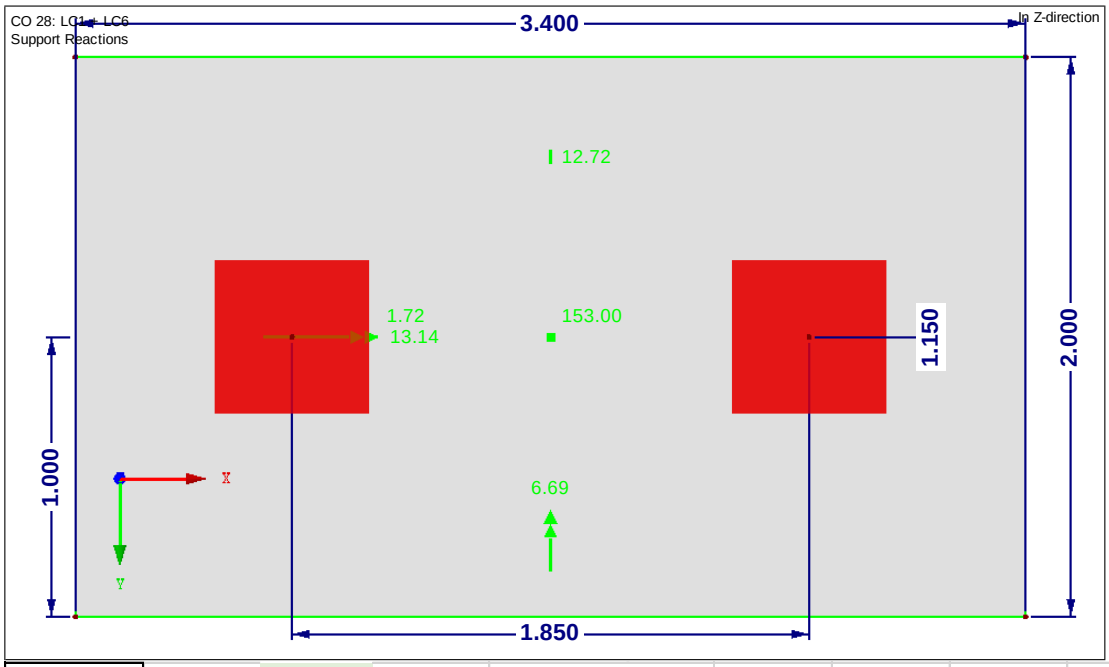


OVERTURN:	Mx=	12.61	kNm	overturn moment				
	Fz=	152.26	kN	total vertical load				
	e=	0.083	m	eccentricity				
	B/2=	1	m					
	F.O.S=	12.075	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	12.21	kN	lateral force in y				
	Fy=	1.68	kN	lateral force in x				
	Hd=	12.33	kN	sliding force				
	Rd=	60.904	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	4.941	>1.50	Factor of safety against sliding				

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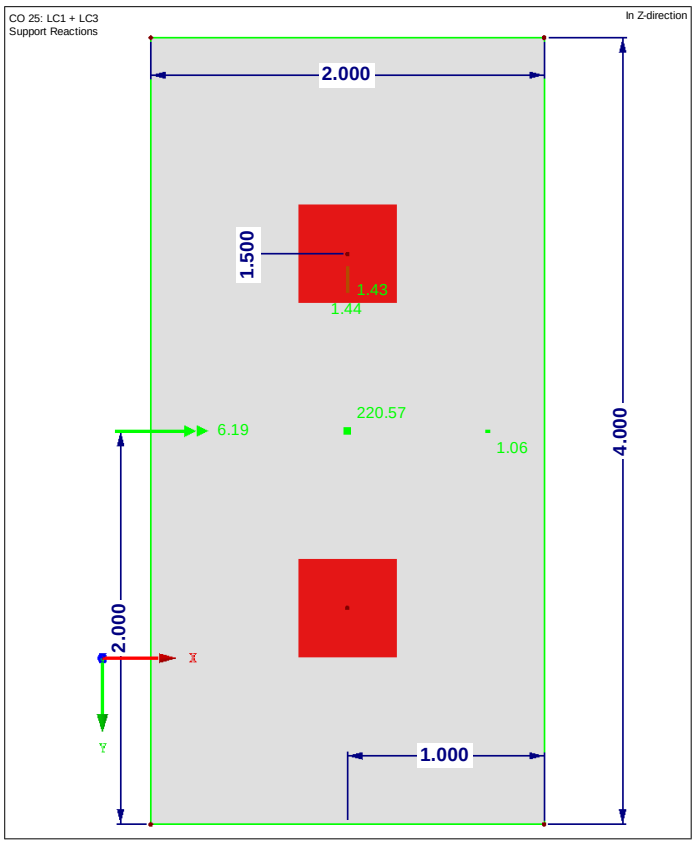
OVERTURN:	Mx=	13.14	kNm	overturn moment				
	Fz=	153	kN	total vertical load				
	e=	0.086	m	eccentricity				
	B/2=	1	m					
	F.O.S=	11.644	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	1.72	kN	lateral force in y				
	Fy=	12.72	kN	lateral force in x				
	Hd=	12.84	kN	sliding force				
	Rd=	61.2	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	4.768	>1.50	Factor of safety against sliding				

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combined footing P39

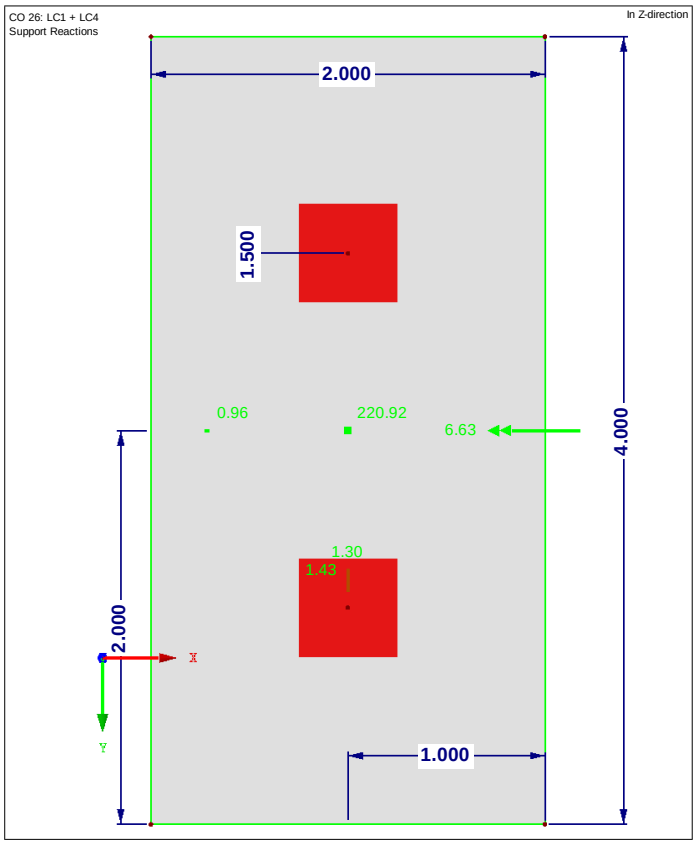


OVERTURN:	Mx=	3.19	kNm	overturn moment
	Fz=	220.57	kN	total vertical load
	e=	0.014	m	eccentricity
	B/2=	2	m	
	F.O.S=	138.288	>1.50	Factor of safety against overturning
SLIDING:	Fx=	1.06	kN	lateral force in y
	Fy=	1.43	kN	lateral force in x
	Hd=	1.78	kN	sliding force
	Rd=	88.228	kN	resistance to sliding =coefficient of friction* total vertical load
	F.O.S=	49.566	>1.50	Factor of safety against sliding

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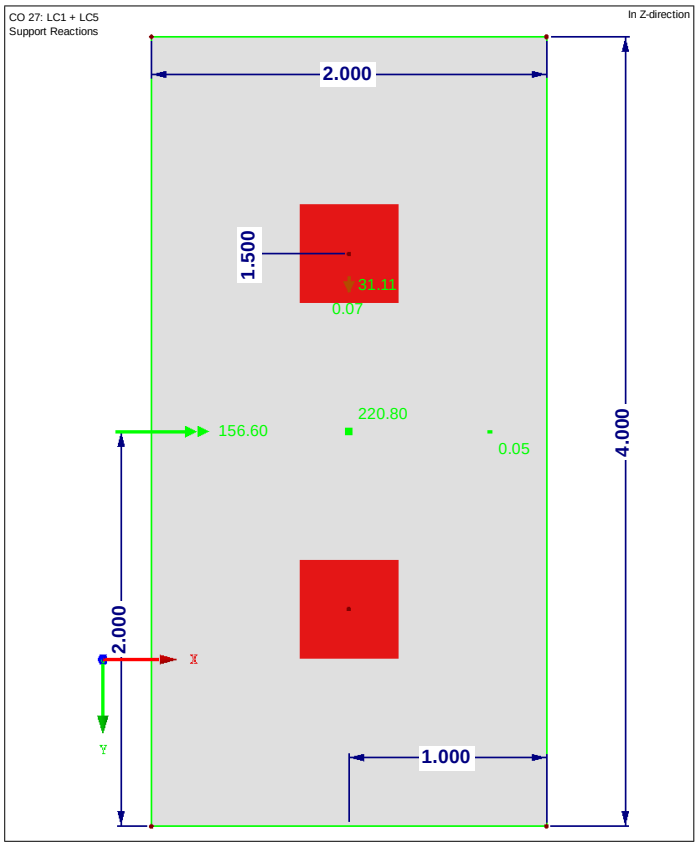


OVERTURN:	Mx=	6.63	kNm	overturn moment				
	Fz=	220.92	kN	total vertical load				
	e=	0.030	m	eccentricity				
	B/2=	2	m					
	F.O.S=	66.643	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	0.96	kN	lateral force in y				
	Fy=	1.43	kN	lateral force in x				
	Hd=	1.72	kN	sliding force				
	Rd=	88.368	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	51.307	>1.50	Factor of safety against sliding				

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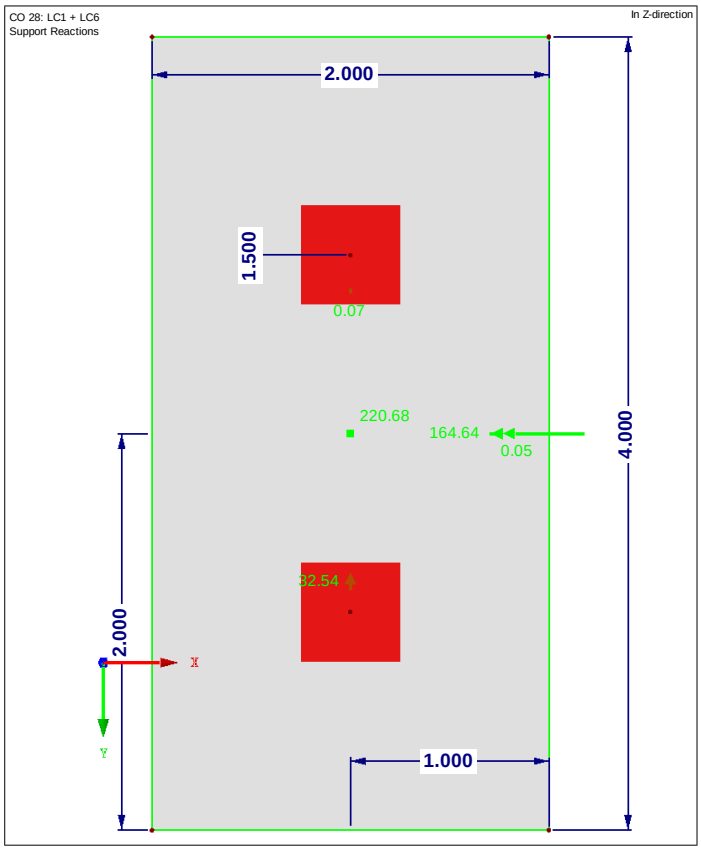


OVERTURN:	Mx=	156.6	kNm	overturn moment				
	Fz=	220.8	kN	total vertical load				
	e=	0.709	m	eccentricity				
	B/2=	2	m					
	F.O.S=	2.820	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	0.05	kN	lateral force in y				
	Fy=	31.11	kN	lateral force in x				
	Hd=	31.11	kN	sliding force				
	Rd=	88.32	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	2.839	>1.50	Factor of safety against sliding				

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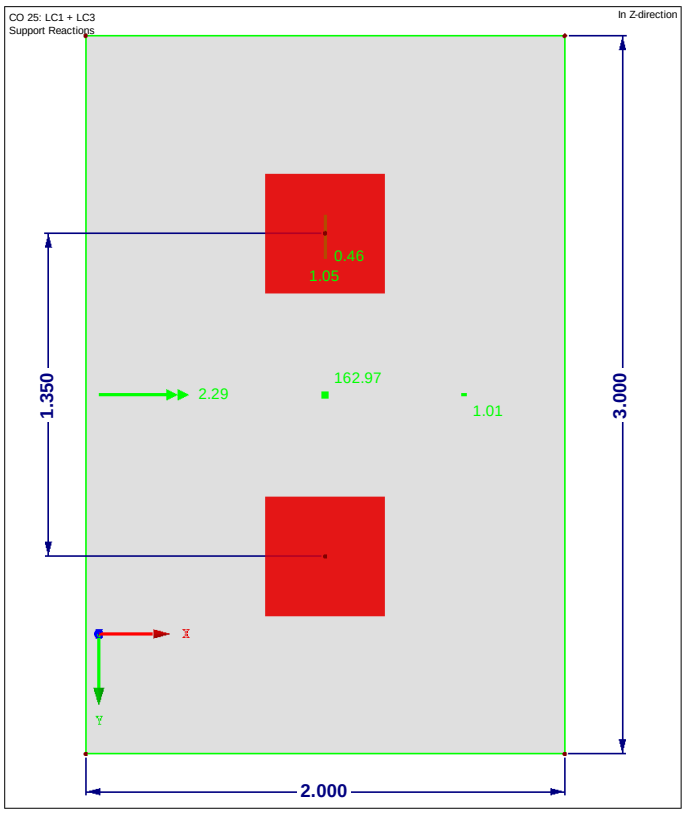
OVERTURN:	Mx=	164.64	kNm	overturn moment					
	Fz=	220.68	kN	total vertical load					
	e=	0.746	m	eccentricity					
	B/2=	2	m						
	F.O.S=	2.681	>1.50	Factor of safety against overturning					
SLIDING:	Fx=	0.05	kN	lateral force in y					
	Fy=	32.54	kN	lateral force in x					
	Hd=	32.54	kN	sliding force					
	Rd=	88.272	kN	resistance to sliding =coefficient of friction* total vertical load					
	F.O.S=	2.713	>1.50	Factor of safety against sliding					

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combined footing P42

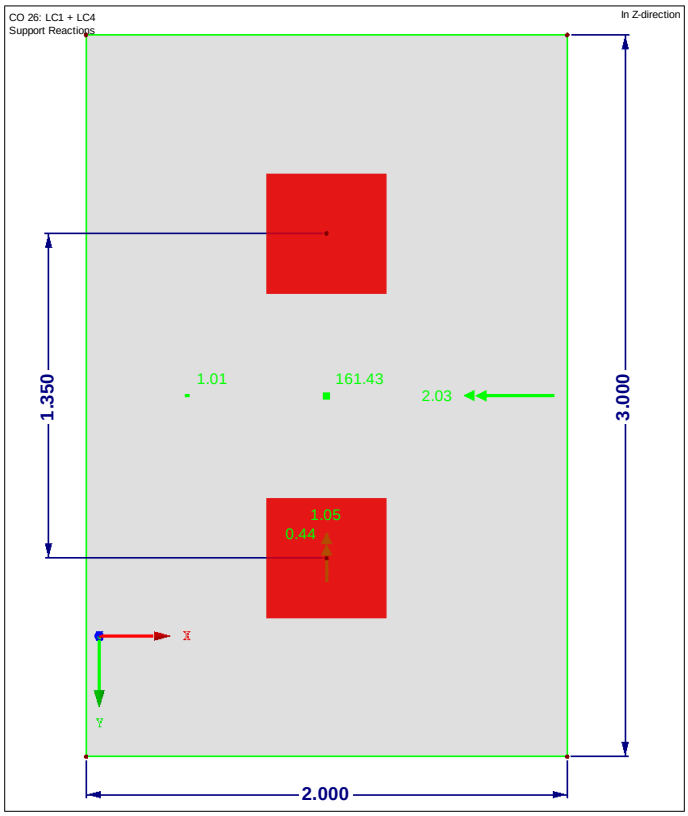


OVERTURN:	Mx=	2.29	kNm	overturn moment
	Fz=	162.97	kN	total vertical load
	e=	0.014	m	eccentricity
	B/2=	1.5	m	
	F.O.S=	106.749	>1.50	Factor of safety against overturning
SLIDING:	Fx=	1.01	kN	lateral force in x
	Fy=	0.46	kN	lateral force in y
	Hd=	1.11	kN	sliding force
	Rd=	65.188	kN	resistance to sliding =coefficient of friction* total gravity load
	F.O.S=	58.737	>1.50	Factor of safety against sliding

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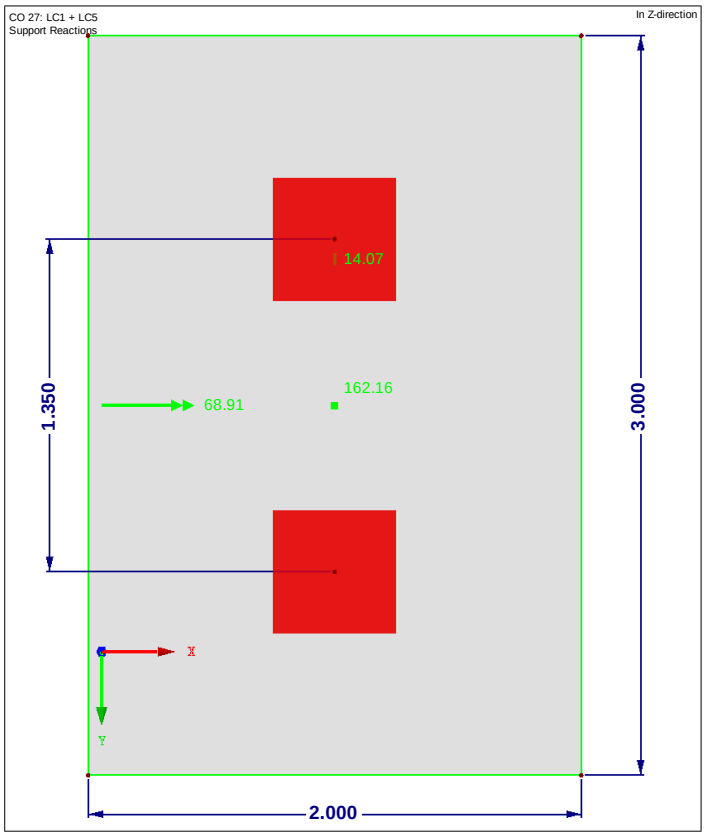


OVERTURN:	Mx=	2.03	kNm	overturn moment				
	Fz=	161.43	kN	total vertical load				
	e=	0.013	m	eccentricity				
	B/2=	1.5	m					
	F.O.S=	119.283	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	1.01	kN	lateral force in x				
	Fy=	0.44	kN	lateral force in y				
	Hd=	1.10	kN	sliding force				
	Rd=	64.572	kN	resistance to sliding =coefficient of friction* total gravity load				
	F.O.S=	58.612	>1.50	Factor of safety against sliding				

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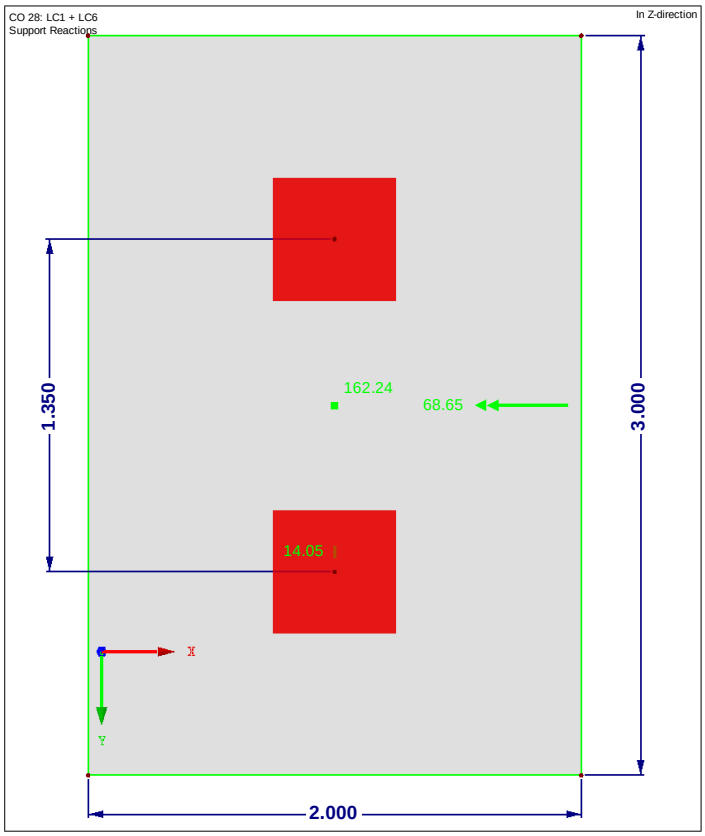


OVERTURN:	Mx=	68.91	kNm	overturn moment				
	Fz=	162.16	kN	total vertical load				
	e=	0.425	m	eccentricity				
	B/2=	1.5	m					
	F.O.S=	3.530	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	0	kN	lateral force in x				
	Fy=	14.07	kN	lateral force in y				
	Hd=	14.07	kN	sliding force				
	Rd=	64.864	kN	resistance to sliding =coefficient of friction* total gravity load				
	F.O.S=	4.610	>1.50	Factor of safety against sliding				

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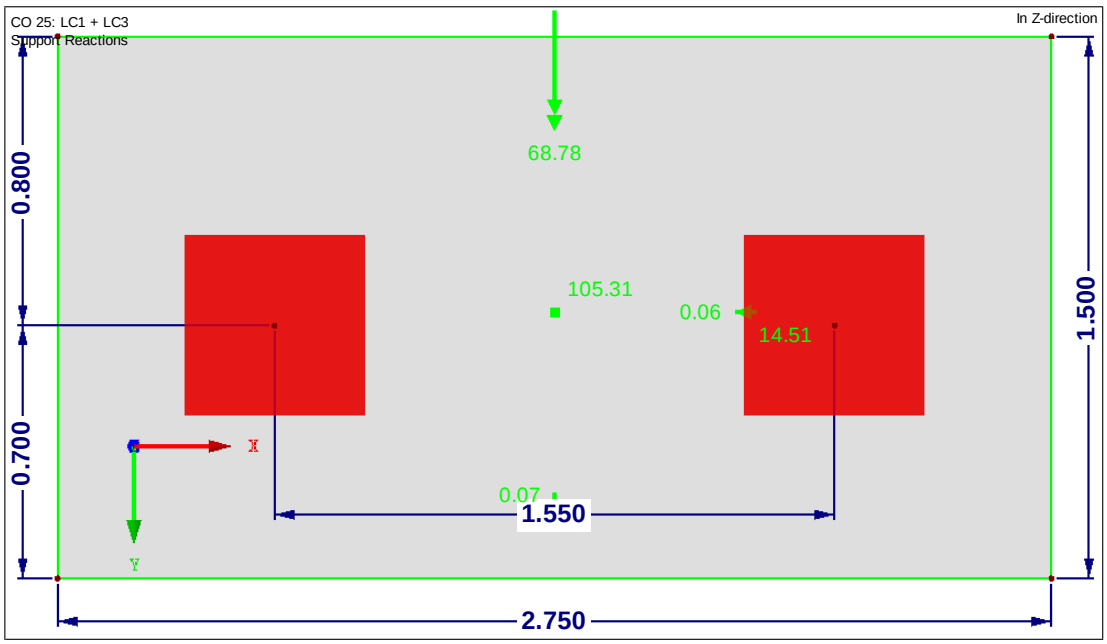
OVERTURN:	Mx=	68.65	kNm	overturn moment				
	Fz=	162.24	kN	total vertical load				
	e=	0.423	m	eccentricity				
	B/2=	1.5	m					
	F.O.S=	3.545	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	0	kN	lateral force in x				
	Fy=	14.05	kN	lateral force in y				
	Hd=	14.05	kN	sliding force				
	Rd=	64.896	kN	resistance to sliding =coefficient of friction* total gravity load				
	F.O.S=	4.619	>1.50	Factor of safety against sliding				

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combined footing P41

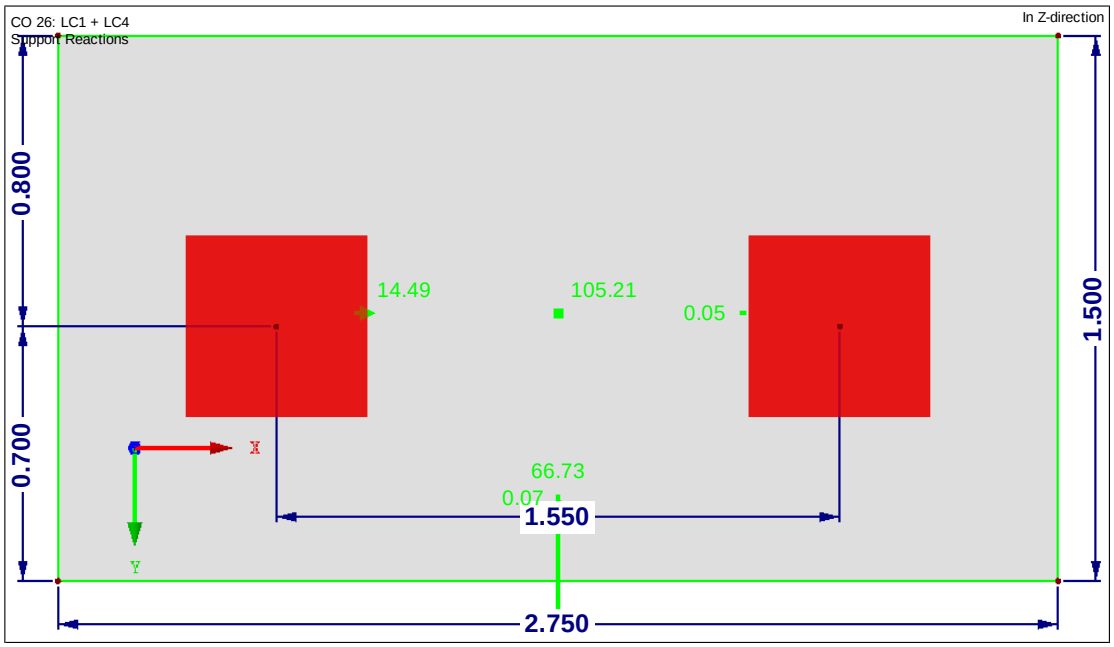


OVERTURN:	My=	68.78	kNm	overturn moment				
	Fz=	105.31	kN	total vertical load				
	e=	0.653	m	eccentricity				
	B/2=	1.375	m					
	F.O.S=	2.105	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	14.51	kN	lateral force in x				
	Fy=	0.07	kN	lateral force in y				
	Hd=	14.51	kN	sliding force				
	Rd=	42.124	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	2.903	>1.50	Factor of safety against sliding				

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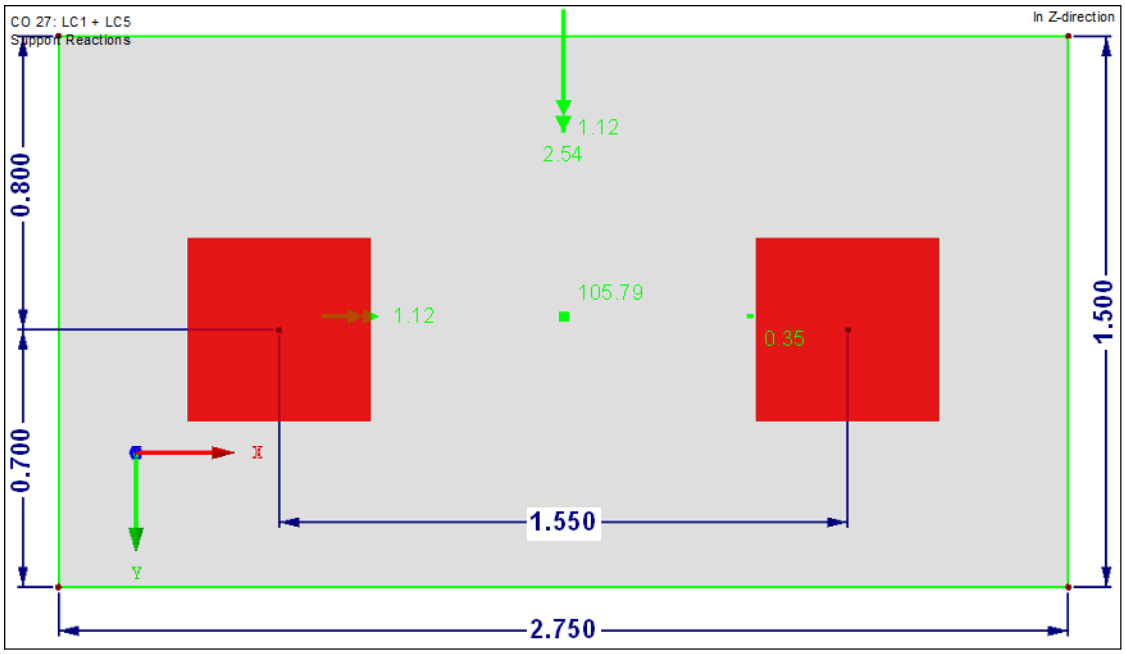


OVERTURN:	My=	66.73	kNm	overturn moment
	Fz=	105.21	kN	total vertical load
	e=	0.634	m	eccentricity
	B/2=	1.375	m	
	F.O.S=	2.168	>1.50	Factor of safety against overturning
SLIDING:	Fx=	14.49	kN	lateral force in x
	Fy=	0.07	kN	lateral force in y
	Hd=	14.49	kN	sliding force
	Rd=	42.084	kN	resistance to sliding =coefficient of friction* total vertical load
	F.O.S=	2.904	>1.50	Factor of safety against sliding

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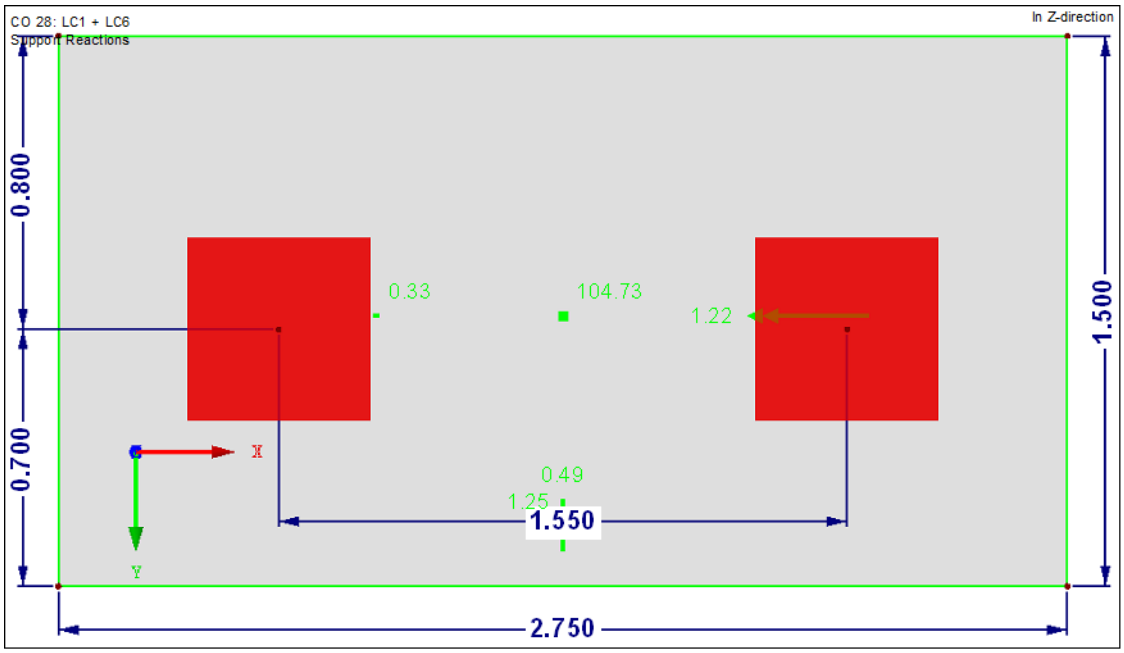


OVERTURN:	Mx=	2.54	kNm	overturn moment				
	Fz=	105.79	kN	total vertical load				
	e=	0.024	m	eccentricity				
	B/2=	0.763	m					
	F.O.S=	31.779	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	0.35	kN	lateral force in x				
	Fy=	1.12	kN	lateral force in y				
	Hd=	1.17	kN	sliding force				
	Rd=	42.316	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	36.062	>1.50	Factor of safety against sliding				

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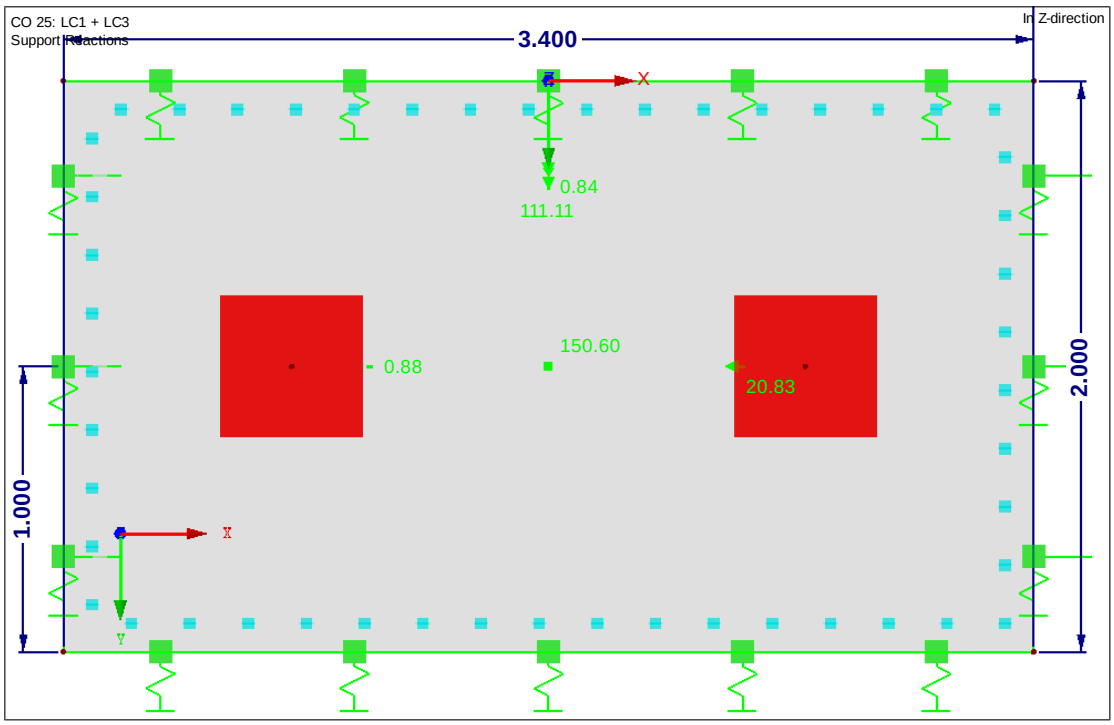
OVERTURN:	Mx=	1.22	kNm	overturn moment				
	Fz=	104.73	kN	total vertical load				
	e=	0.012	m	eccentricity				
	B/2=	0.763	m					
	F.O.S=	65.499	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	0.33	kN	lateral force in x				
	Fy=	1.25	kN	lateral force in y				
	Hd=	1.29	kN	sliding force				
	Rd=	41.892	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	32.403	>1.50	Factor of safety against sliding				

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combined footing P46

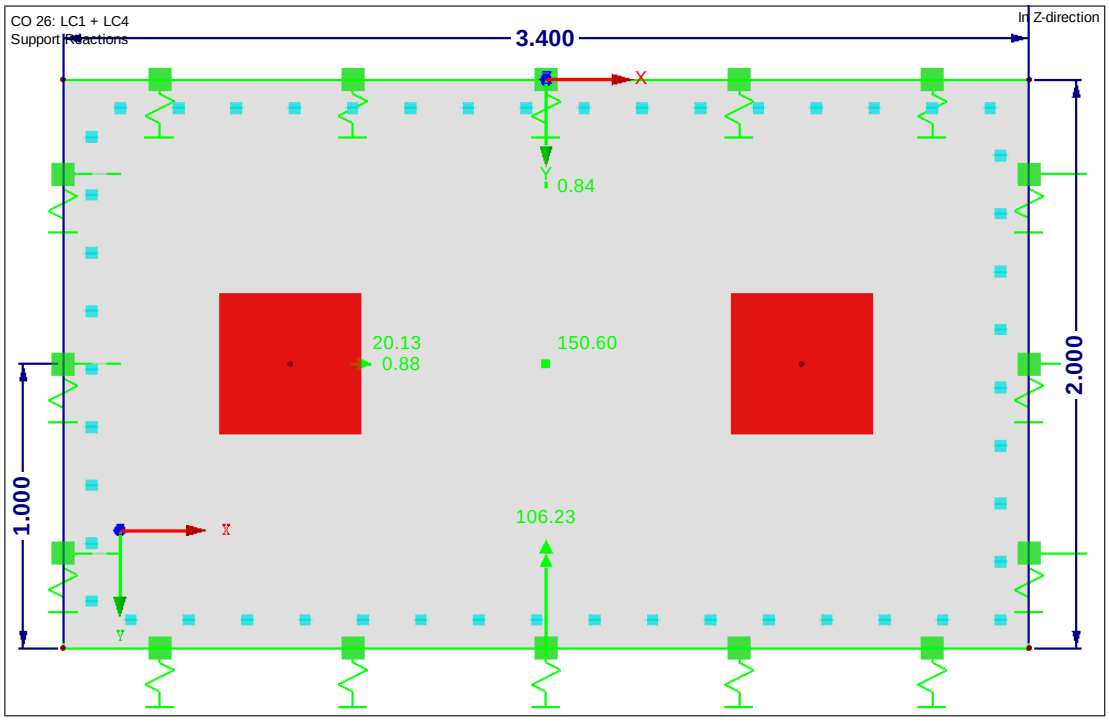


OVERTURN:	Mx=	111.11	kNm	overturn moment				
	Fz=	150.6	kN	total vertical load				
	e=	0.738	m	eccentricity				
	B/2=	1.7	m					
	F.O.S=	2.304	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	20.83	kN	lateral force in y				
	Fy=	0.84	kN	lateral force in x				
	Hd=	20.85	kN	sliding force				
	Rd=	60.24	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	2.890	>1.50	Factor of safety against sliding				

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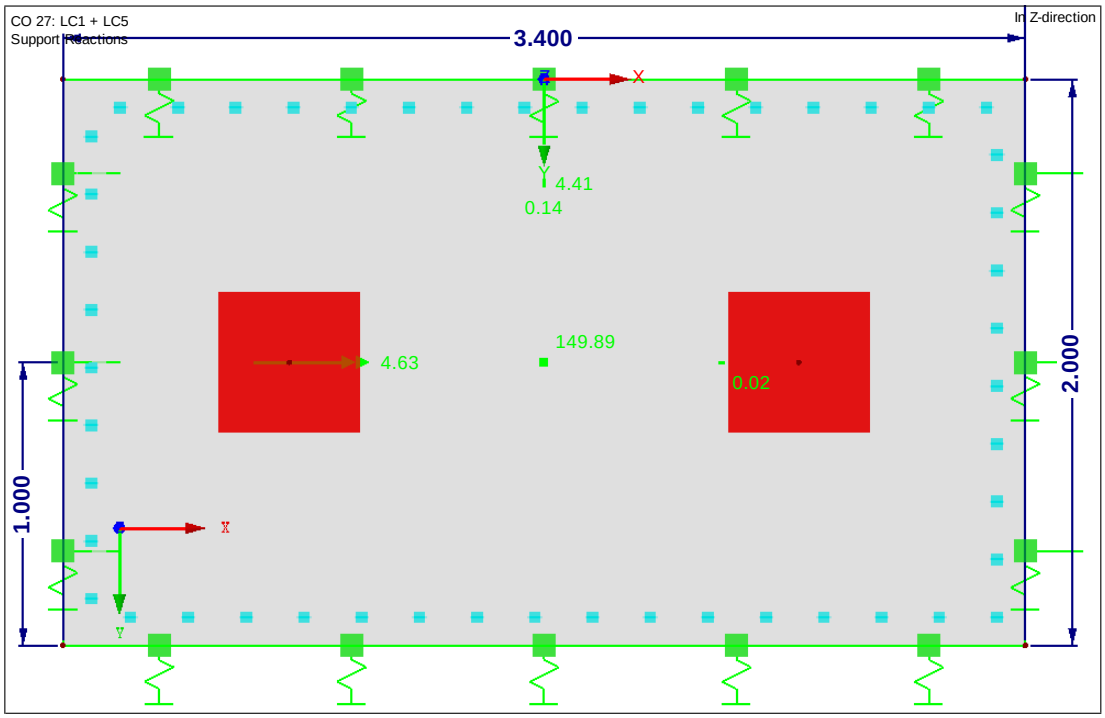


OVERTURN:	Mx=	106.23	kNm	overturn moment				
	Fz=	150.6	kN	total vertical load				
	e=	0.705	m	eccentricity				
	B/2=	1.7	m					
	F.O.S=	2.410	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	20.13	kN	lateral force in y				
	Fy=	0.84	kN	lateral force in x				
	Hd=	20.15	kN	sliding force				
	Rd=	60.24	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	2.990	>1.50	Factor of safety against sliding				

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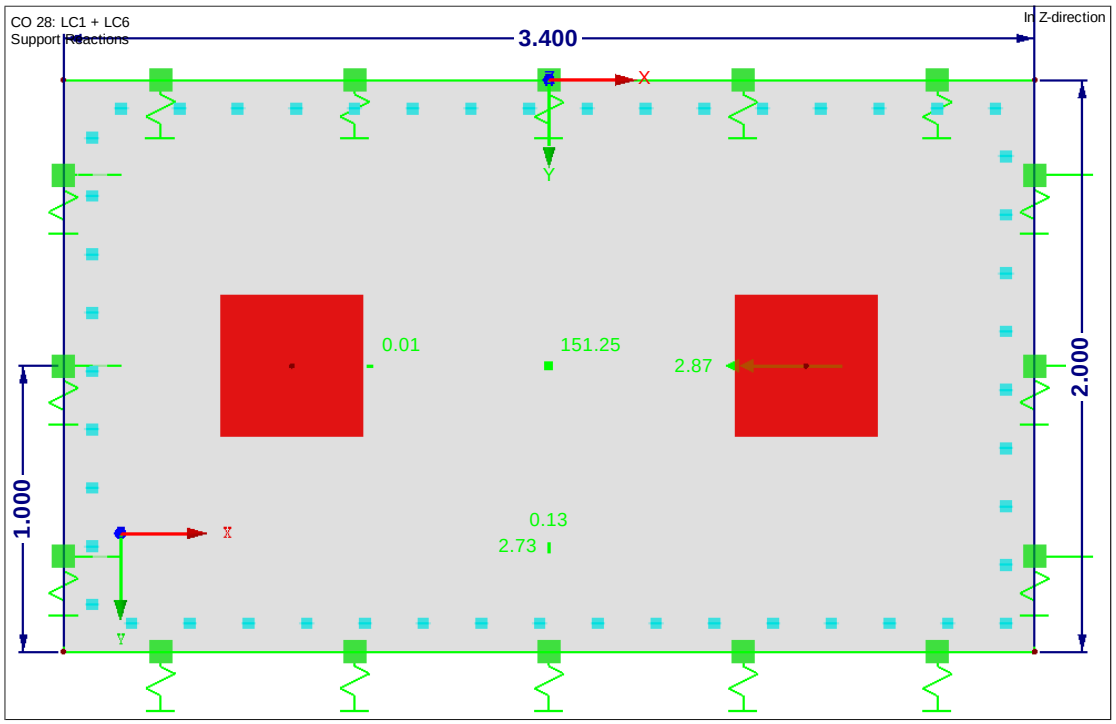


OVERTURN:	Mx=	4.63	kNm	overturn moment				
	Fz=	149.99	kN	total vertical load				
	e=	0.031	m	eccentricity				
	B/2=	1	m					
	F.O.S=	32.395	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	0.02	kN	lateral force in y				
	Fy=	4.41	kN	lateral force in x				
	Hd=	4.41	kN	sliding force				
	Rd=	59.996	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	13.604	>1.50	Factor of safety against sliding				

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OVERTURN:	Mx=	2.87	kNm	overturn moment				
	Fz=	151.25	kN	total vertical load				
	e=	0.019	m	eccentricity				
	B/2=	1	m					
	F.O.S=	52.700	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	0.01	kN	lateral force in y				
	Fy=	2.73	kN	lateral force in x				
	Hd=	2.73	kN	sliding force				
	Rd=	60.5	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	22.161	>1.50	Factor of safety against sliding				

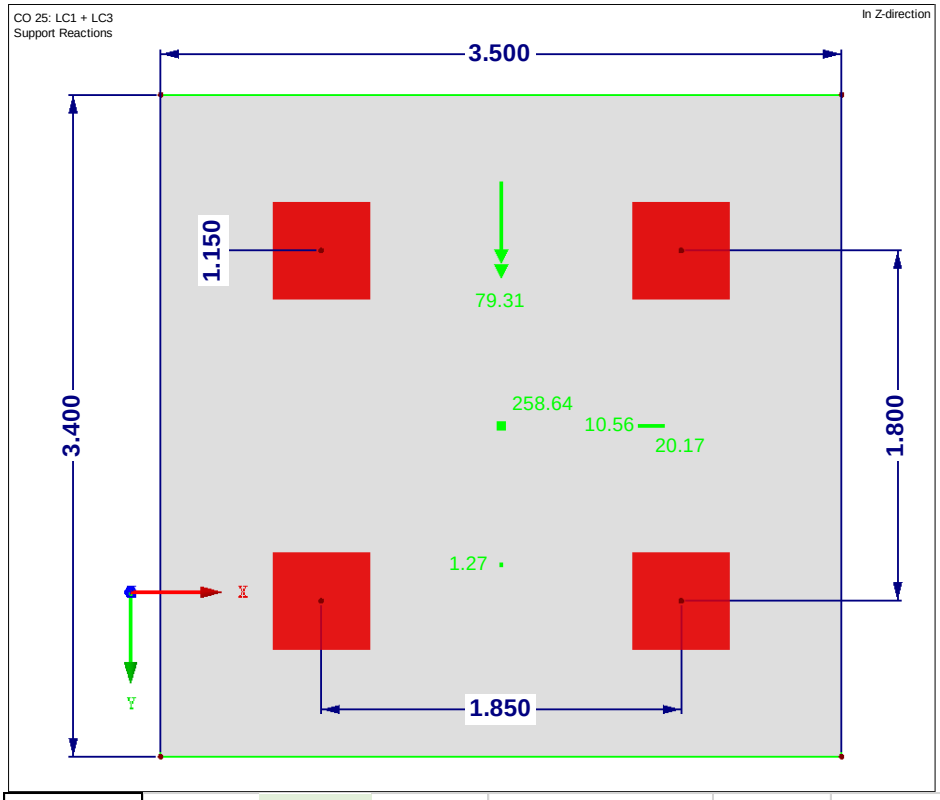
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6.4. Stability check for 4-pedestal foundation P37, P38, P40, P43, P44, P45

Combined footing P37

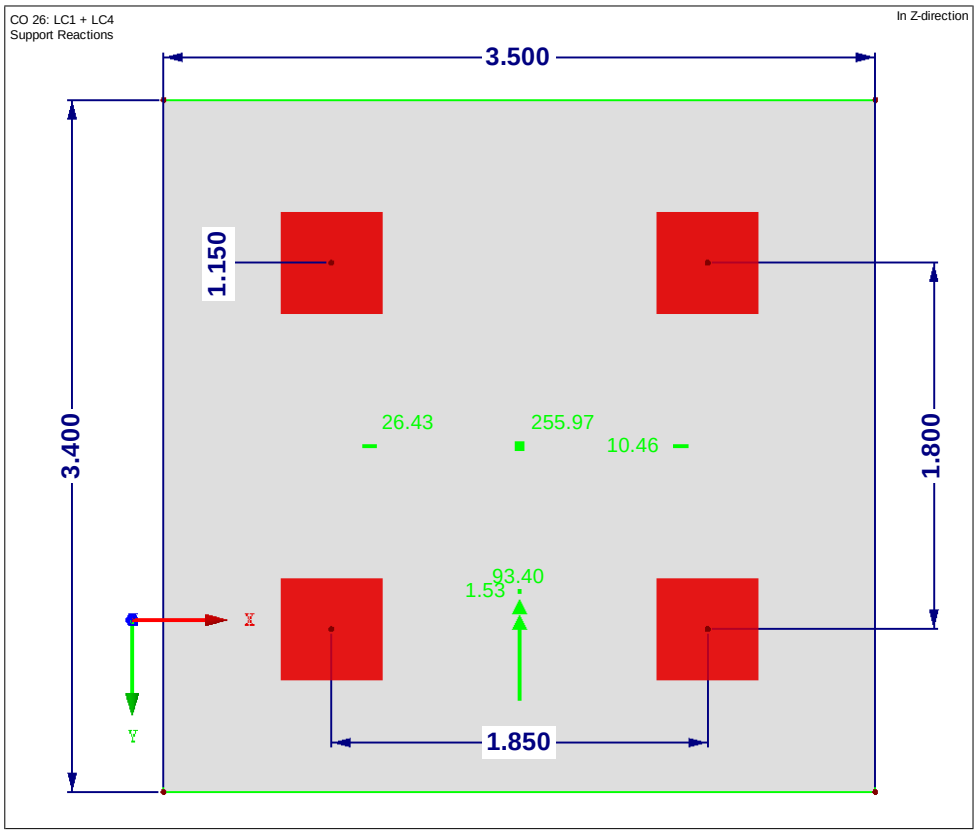


OVERTURN:	Mx=	79.31	kNm	overturn moment				
	Fz=	258.64	kN	total vertical load				
	e=	0.307	m	eccentricity				
	B/2=	1.75	m					
	F.O.S=	5.707	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	20.17	kN	lateral force in y				
	Fy=	1.27	kN	lateral force in x				
	Hd=	20.21	kN	sliding force				
	Rd=	103.456	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	5.119	>1.50	Factor of safety against sliding				

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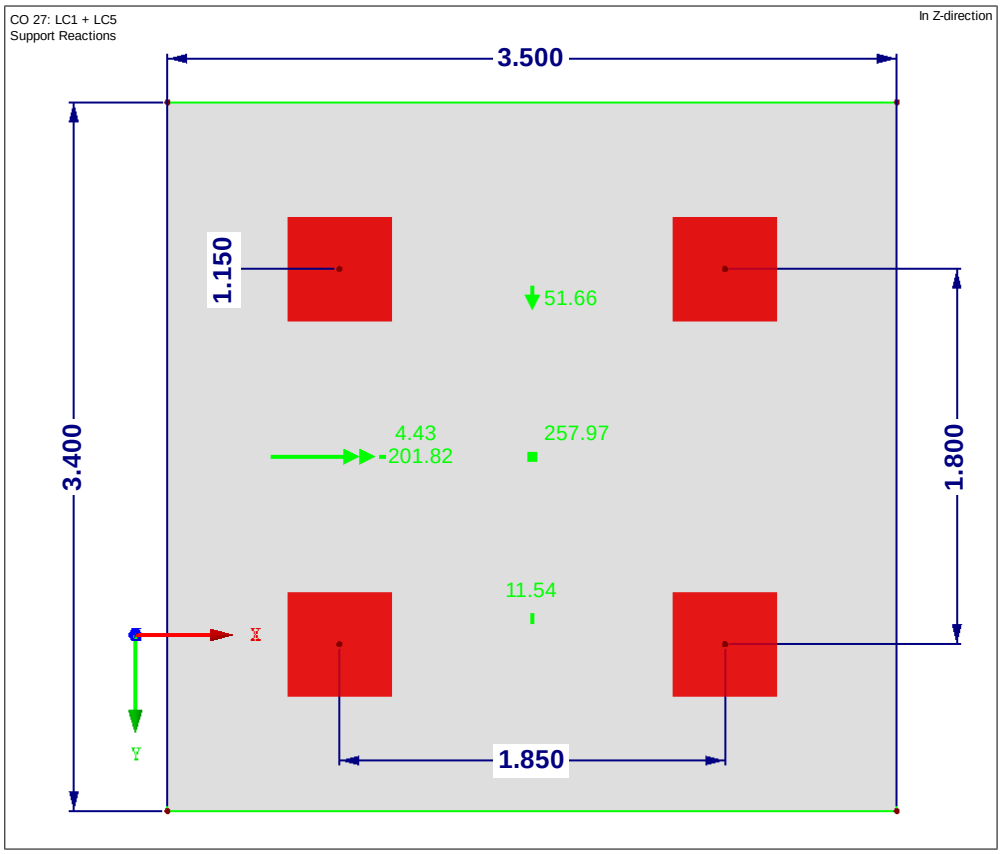


OVERTURN:	Mx=	93.4	kNm	overturn moment					
	Fz=	255.97	kN	total vertical load					
	e=	0.365	m	eccentricity					
	B/2=	1.75	m						
	F.O.S=	4.796	>1.50	Factor of safety against overturning					
SLIDING:	Fx=	26.43	kN	lateral force in y					
	Fy=	1.53	kN	lateral force in x					
	Hd=	26.47	kN	sliding force					
	Rd=	102.388	kN	resistance to sliding =coefficient of friction* total vertical load					
	F.O.S=	3.867	>1.50	Factor of safety against sliding					

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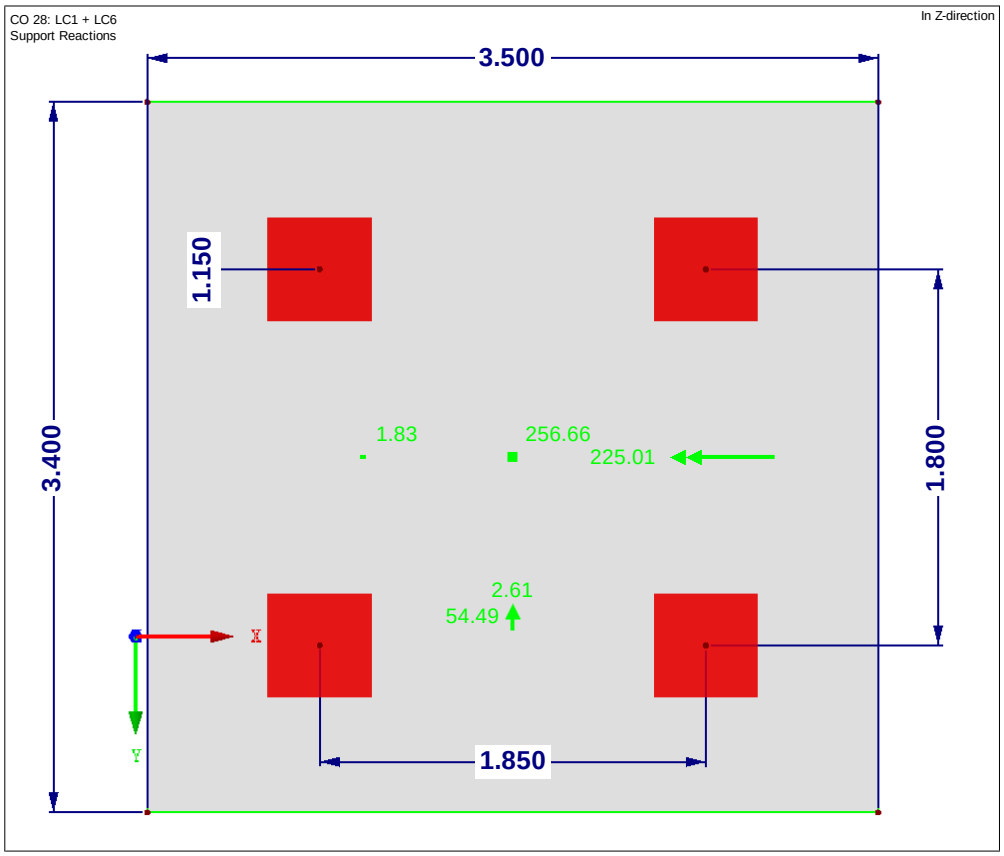


OVERTURN:	Mx=	201.82	kNm	overturn moment				
	Fz=	257.97	kN	total vertical load				
	e=	0.782	m	eccentricity				
	B/2=	1.7	m					
	F.O.S=	2.173	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	4.43	kN	lateral force in y				
	Fy=	51.66	kN	lateral force in x				
	Hd=	51.85	kN	sliding force				
	Rd=	103.188	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	1.990	>1.50	Factor of safety against sliding				

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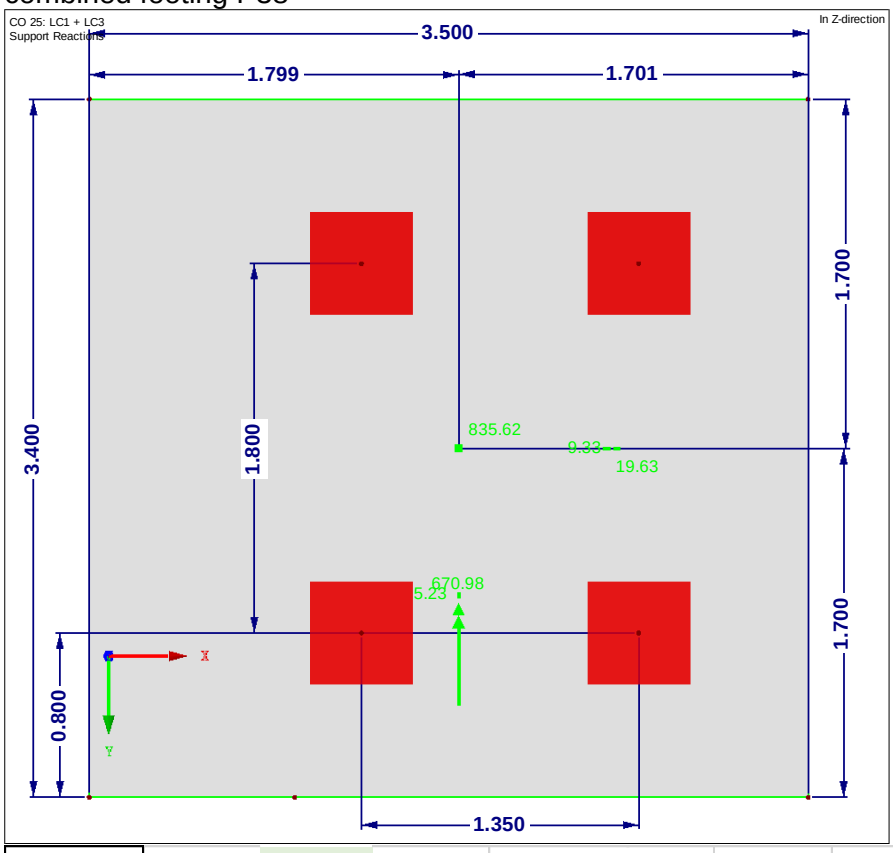
OVERTURN:	Mx=	225.01	kNm	overturn moment				
	Fz=	256.66	kN	total vertical load				
	e=	0.877	m	eccentricity				
	B/2=	1.7	m					
	F.O.S=	1.939	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	1.83	kN	lateral force in y				
	Fy=	54.49	kN	lateral force in x				
	Hd=	54.52	kN	sliding force				
	Rd=	102.664	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	1.883	>1.50	Factor of safety against sliding				

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combined footing P38

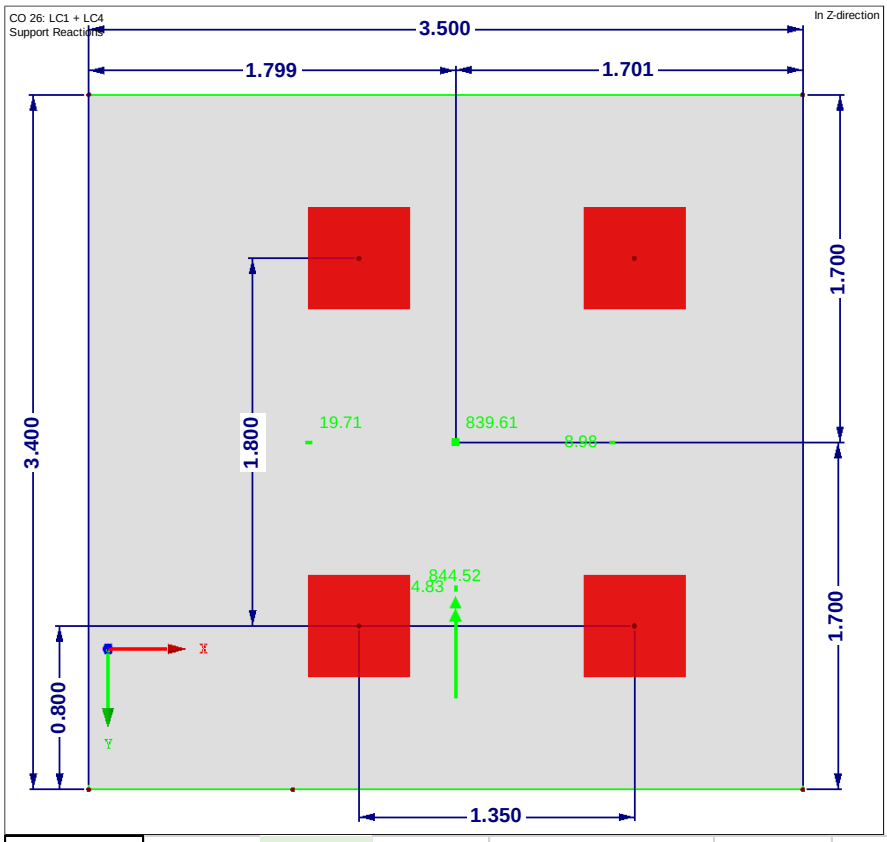


OVERTURN:	Mx=	670.98	kNm	overturn moment					
	Fz=	835.62	kN	total vertical load					
	e=	0.803	m	eccentricity					
	B/2=	1.7	m						
	F.O.S=	2.117	>1.50	Factor of safety against overturning					
SLIDING:	Fx=	19.63	kN	lateral force in x					
	Fy=	5.23	kN	lateral force in y					
	Hd=	20.31	kN	sliding force					
	Rd=	334.248	kN	resistance to sliding =coefficient of friction* total vertical load					
	F.O.S=	16.453	>1.50	Factor of safety against sliding					

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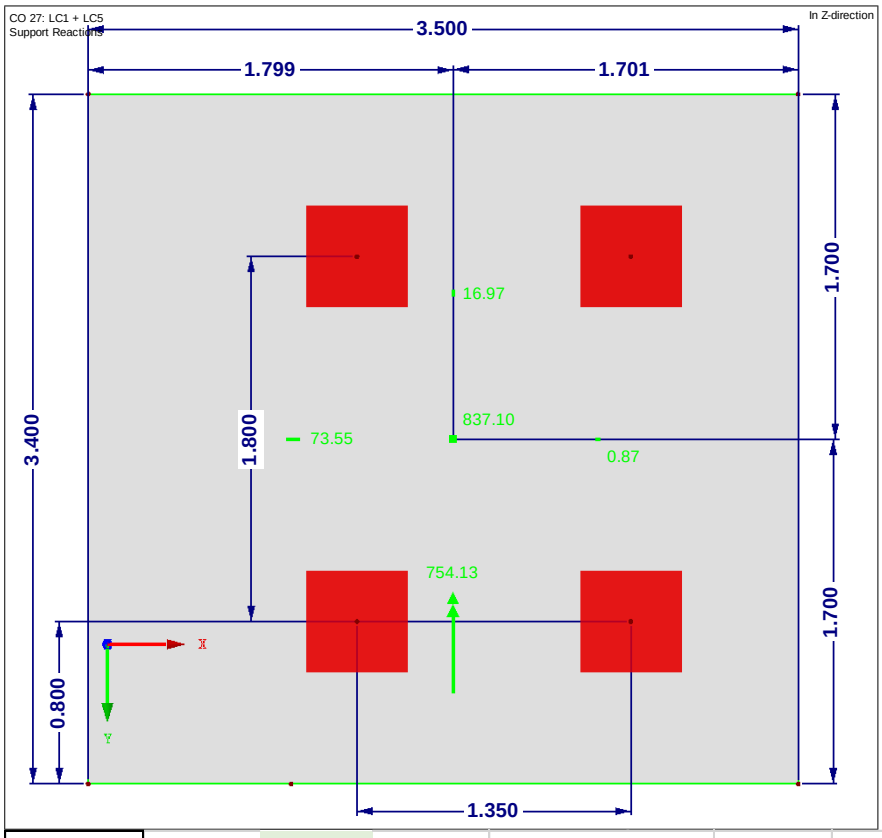


OVERTURN:	Mx=	844.52	kNm	overturn moment				
	Fz=	839.61	kN	total vertical load				
	e=	1.006	m	eccentricity				
	B/2=	1.7	m					
	F.O.S=	1.690	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	19.71	kN	lateral force in x				
	Fy=	4.83	kN	lateral force in y				
	Hd=	20.29	kN	sliding force				
	Rd=	335.844	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	16.550	>1.50	Factor of safety against sliding				

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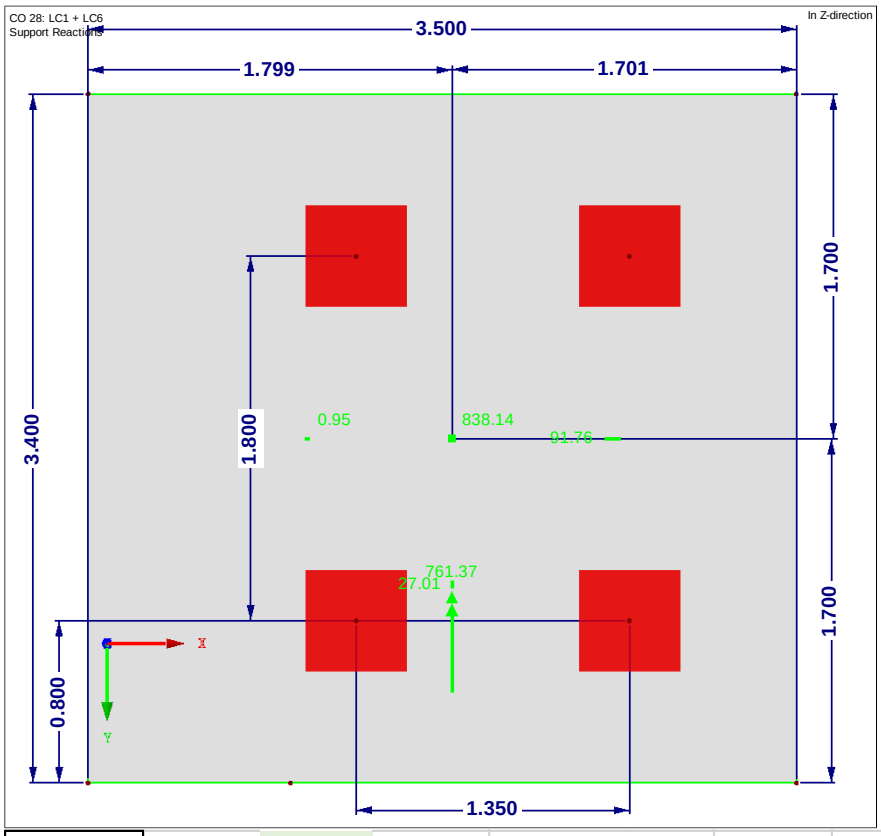


OVERTURN:	Mx=	754.13	kNm	overturn moment				
	Fz=	837.1	kN	total vertical load				
	e=	0.901	m	eccentricity				
	B/2=	1.7	m					
	F.O.S=	1.887	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	0.87	kN	lateral force in x				
	Fy=	16.97	kN	lateral force in y				
	Hd=	16.99	kN	sliding force				
	Rd=	334.84	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	19.705	>1.50	Factor of safety against sliding				

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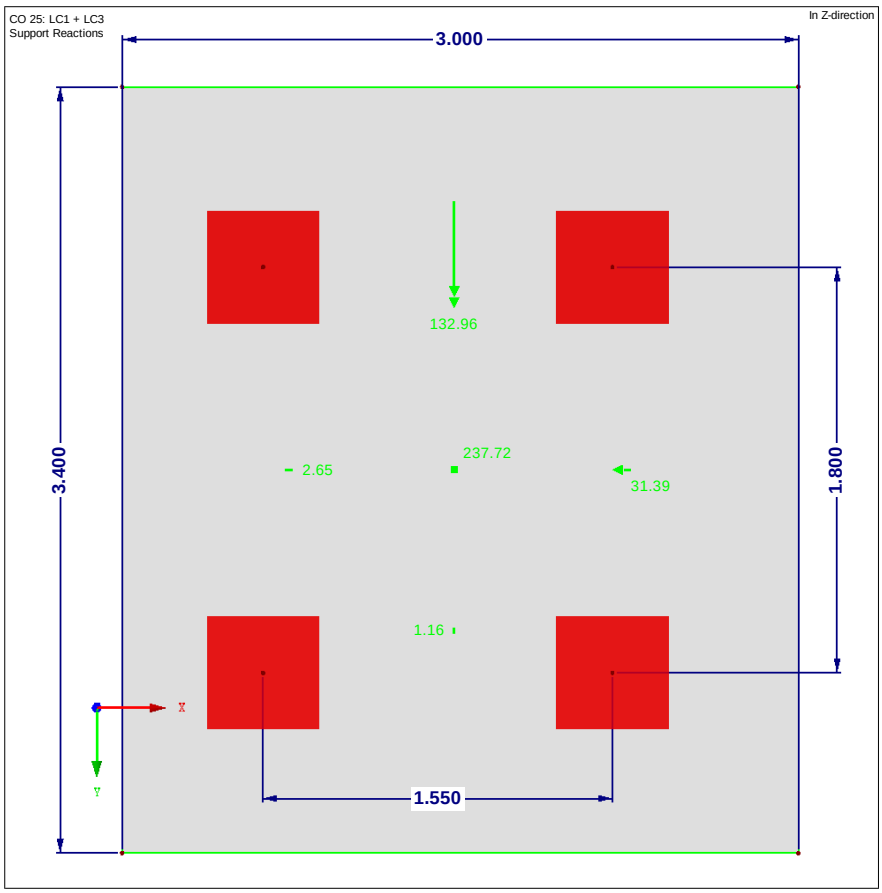
OVERTURN:	Mx=	761.37	kNm	overturn moment				
	Fz=	838.14	kN	total vertical load				
	e=	0.908	m	eccentricity				
	B/2=	1.7	m					
	F.O.S=	1.871	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	0.95	kN	lateral force in x				
	Fy=	27.01	kN	lateral force in y				
	Hd=	27.03	kN	sliding force				
	Rd=	335.256	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	12.405	>1.50	Factor of safety against sliding				

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Combined footing P40

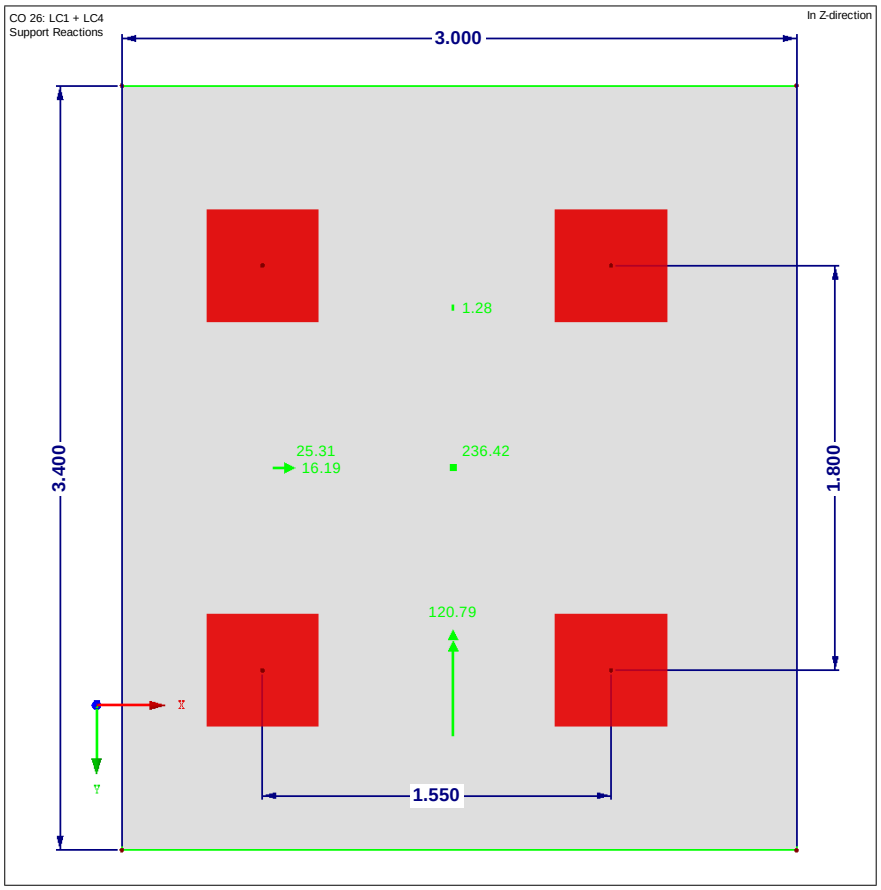


OVERTURN:	Mx=	132.96	kNm	overturn moment				
	Fz=	237.72	kN	total vertical load				
	e=	0.559	m	eccentricity				
	B/2=	1.5	m					
	F.O.S=	2.682	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	31.39	kN	lateral force in y				
	Fy=	1.16	kN	lateral force in x				
	Hd=	31.41	kN	sliding force				
	Rd=	95.088	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	3.027	>1.50	Factor of safety against sliding				

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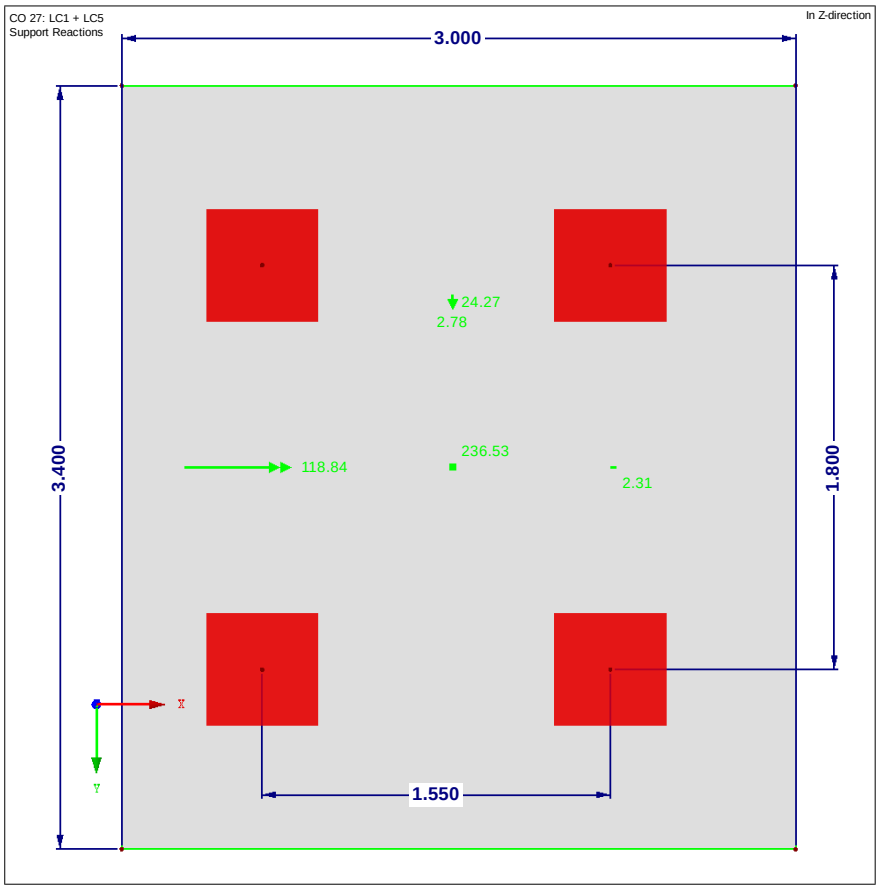


OVERTURN:	Mx=	120.79	kNm	overturn moment				
	Fz=	236.42	kN	total vertical load				
	e=	0.511	m	eccentricity				
	B/2=	1.5	m					
	F.O.S=	2.936	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	25.31	kN	lateral force in y				
	Fy=	1.28	kN	lateral force in x				
	Hd=	25.34	kN	sliding force				
	Rd=	94.568	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	3.732	>1.50	Factor of safety against sliding				

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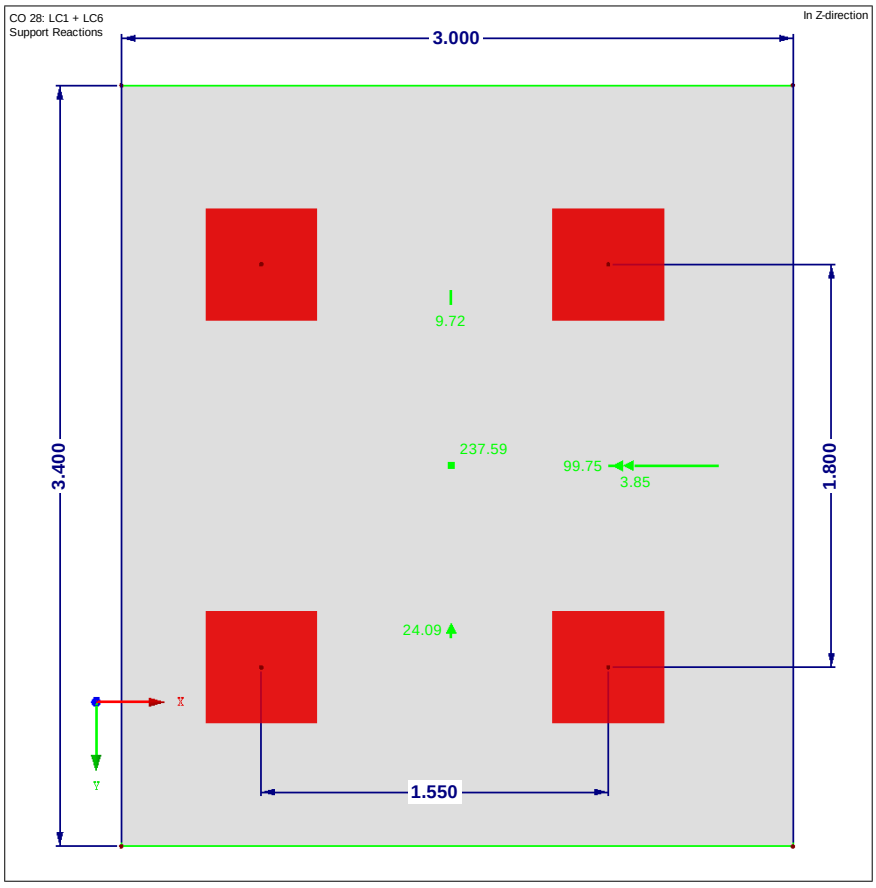


OVERTURN:	Mx=	118.84	kNm	overturn moment					
	Fz=	236.53	kN	total vertical load					
	e=	0.502	m	eccentricity					
	B/2=	1.7	m						
	F.O.S=	3.384	>1.50	Factor of safety against overturning					
SLIDING:	Fx=	2.31	kN	lateral force in y					
	Fy=	24.27	kN	lateral force in x					
	Hd=	24.38	kN	sliding force					
	Rd=	94.612	kN	resistance to sliding =coefficient of friction* total vertical load					
	F.O.S=	3.881	>1.50	Factor of safety against sliding					

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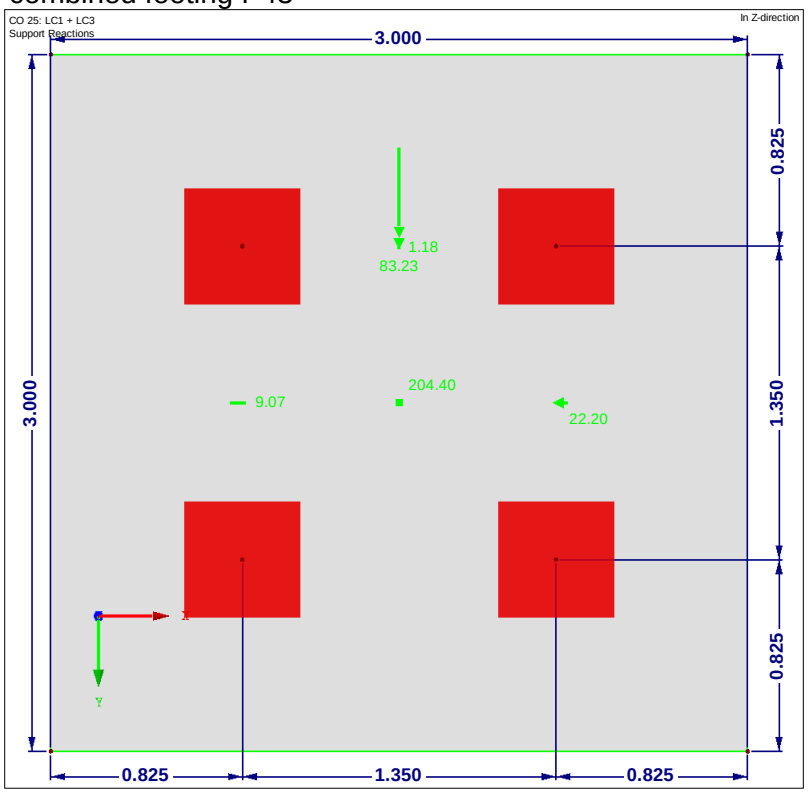
OVERTURN:	Mx=	99.75	kNm	overturn moment				
	Fz=	237.59	kN	total vertical load				
	e=	0.420	m	eccentricity				
	B/2=	1.7	m					
	F.O.S=	4.049	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	3.85	kN	lateral force in y				
	Fy=	24.09	kN	lateral force in x				
	Hd=	24.40	kN	sliding force				
	Rd=	95.036	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	3.896	>1.50	Factor of safety against sliding				

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combined footing P43

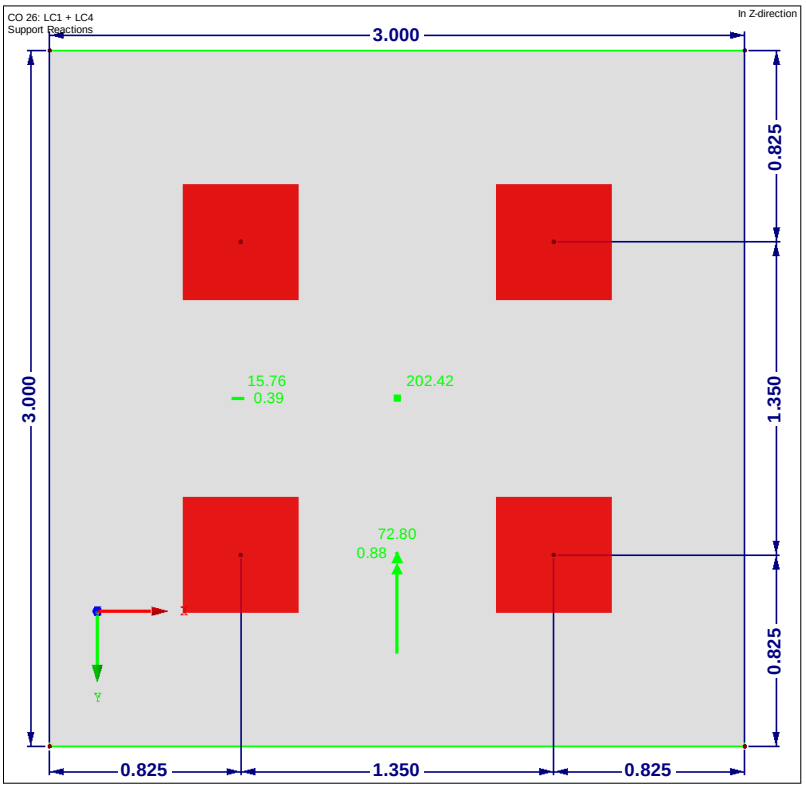


OVERTURN:	Mx=	83.23	kNm	overturn moment				
	Fz=	104.4	kN	total vertical load				
	e=	0.797	m	eccentricity				
	B/2=	1.5	m					
	F.O.S=	1.882	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	22.2	kN	lateral force in x				
	Fy=	1.18	kN	lateral force in y				
	Hd=	22.23	kN	sliding force				
	Rd=	41.76	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	1.878	>1.50	Factor of safety against sliding				

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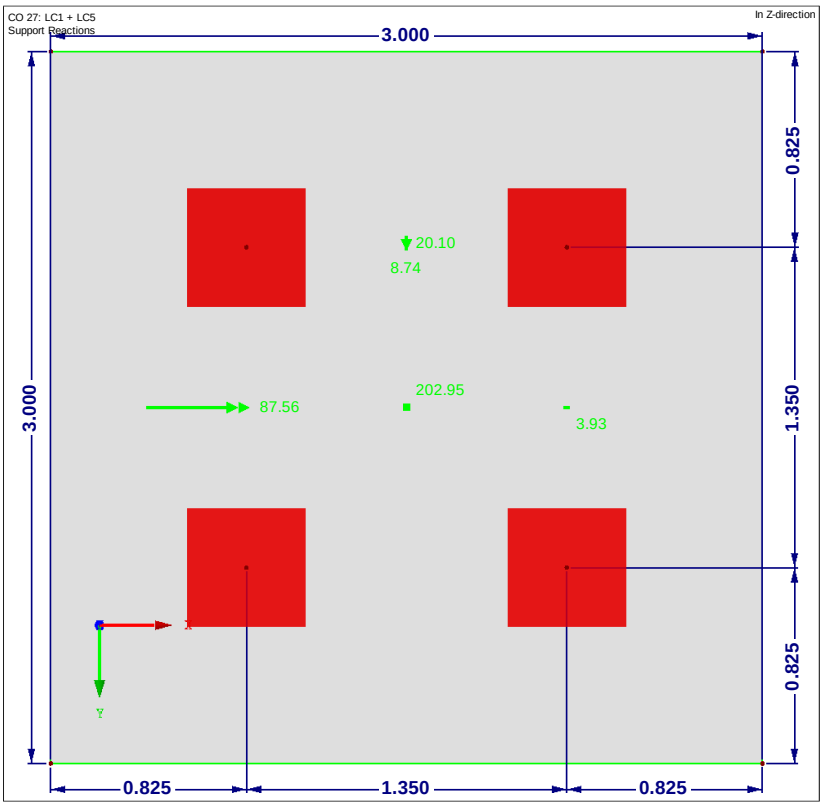


OVERTURN:	Mx=	72.8	kNm	overturn moment				
	Fz=	202.42	kN	total vertical load				
	e=	0.360	m	eccentricity				
	B/2=	1.5	m					
	F.O.S=	4.171	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	15.76	kN	lateral force in x				
	Fy=	0.88	kN	lateral force in y				
	Hd=	15.78	kN	sliding force				
	Rd=	80.968	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	5.130	>1.50	Factor of safety against sliding				

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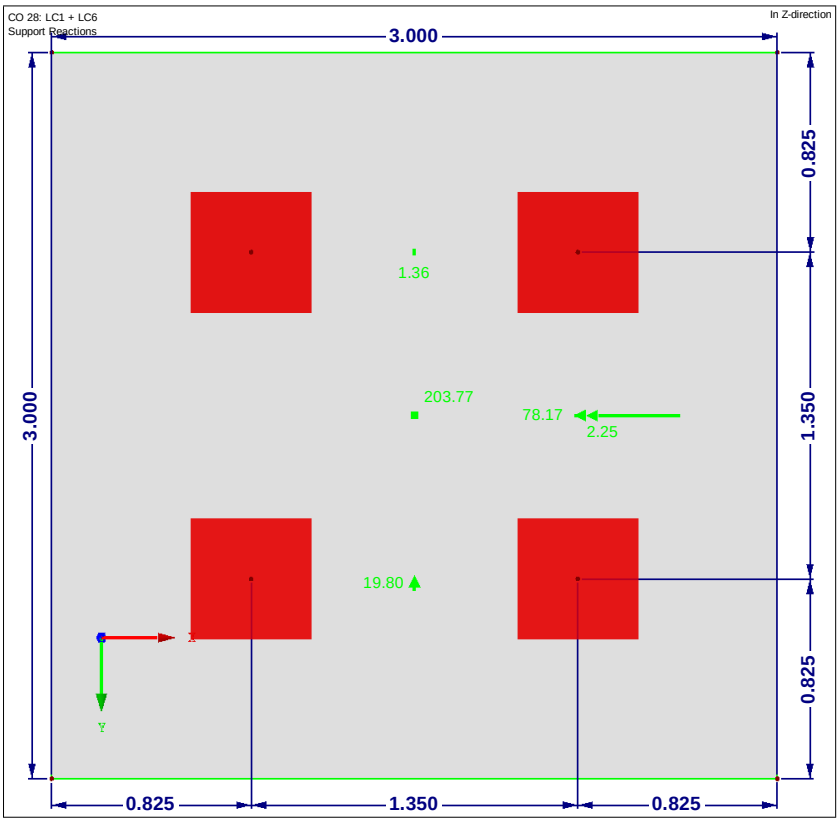


OVERTURN:	Mx=	87.56	kNm	overturn moment				
	Fz=	202.95	kN	total vertical load				
	e=	0.431	m	eccentricity				
	B/2=	1.5	m					
	F.O.S=	3.477	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	3.93	kN	lateral force in x				
	Fy=	20.1	kN	lateral force in y				
	Hd=	20.48	kN	sliding force				
	Rd=	81.18	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	3.964	>1.50	Factor of safety against sliding				

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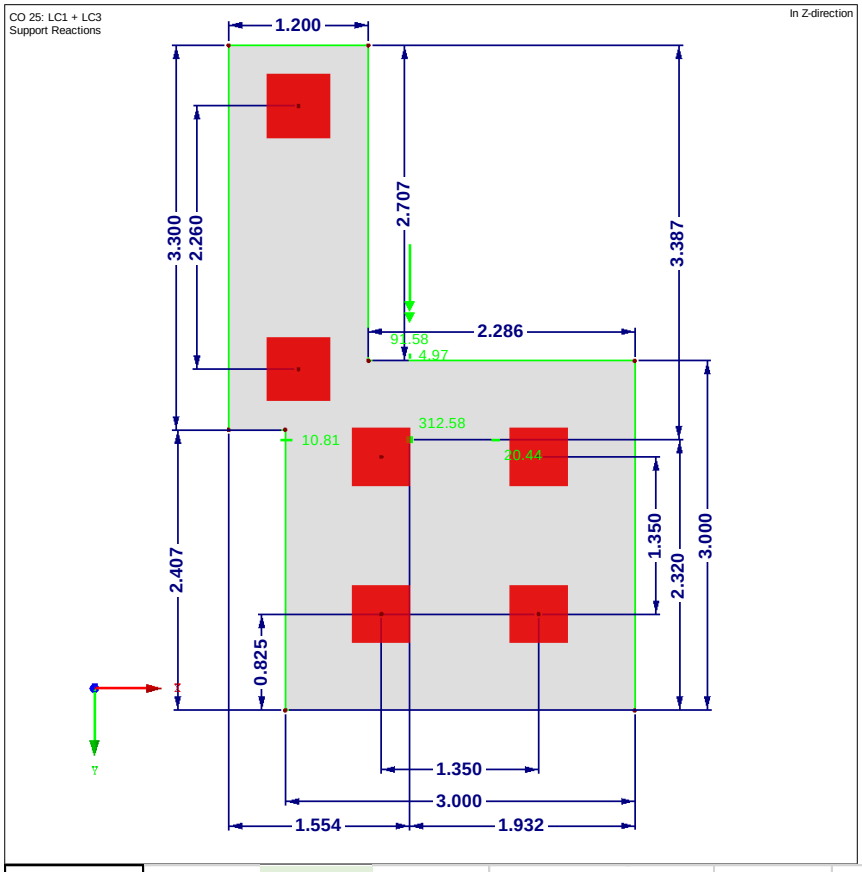
OVERTURN:	Mx=	78.17	kNm	overturn moment				
	Fz=	203.77	kN	total vertical load				
	e=	0.384	m	eccentricity				
	B/2=	1.5	m					
	F.O.S=	3.910	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	2.25	kN	lateral force in x				
	Fy=	19.8	kN	lateral force in y				
	Hd=	19.93	kN	sliding force				
	Rd=	81.508	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	4.090	>1.50	Factor of safety against sliding				

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combined footing P44

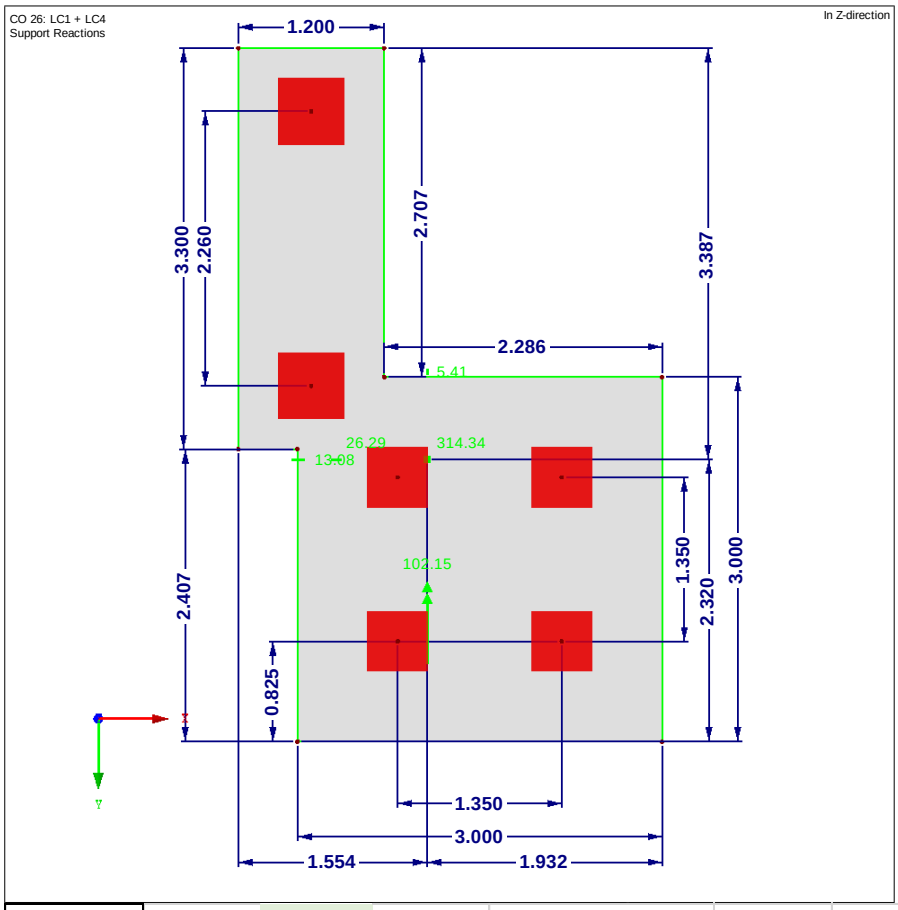


OVERTURN:	Mx=	91.58	kNm	overturn moment				
	Fz=	312.58	kN	total vertical load				
	e=	0.293	m	eccentricity				
	B/2=	1.554	m					
	F.O.S=	5.304	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	20.44	kN	lateral force in y				
	Fy=	4.97	kN	lateral force in x				
	Hd=	21.04	kN	sliding force				
	Rd=	125.032	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	5.944	>1.50	Factor of safety against sliding				

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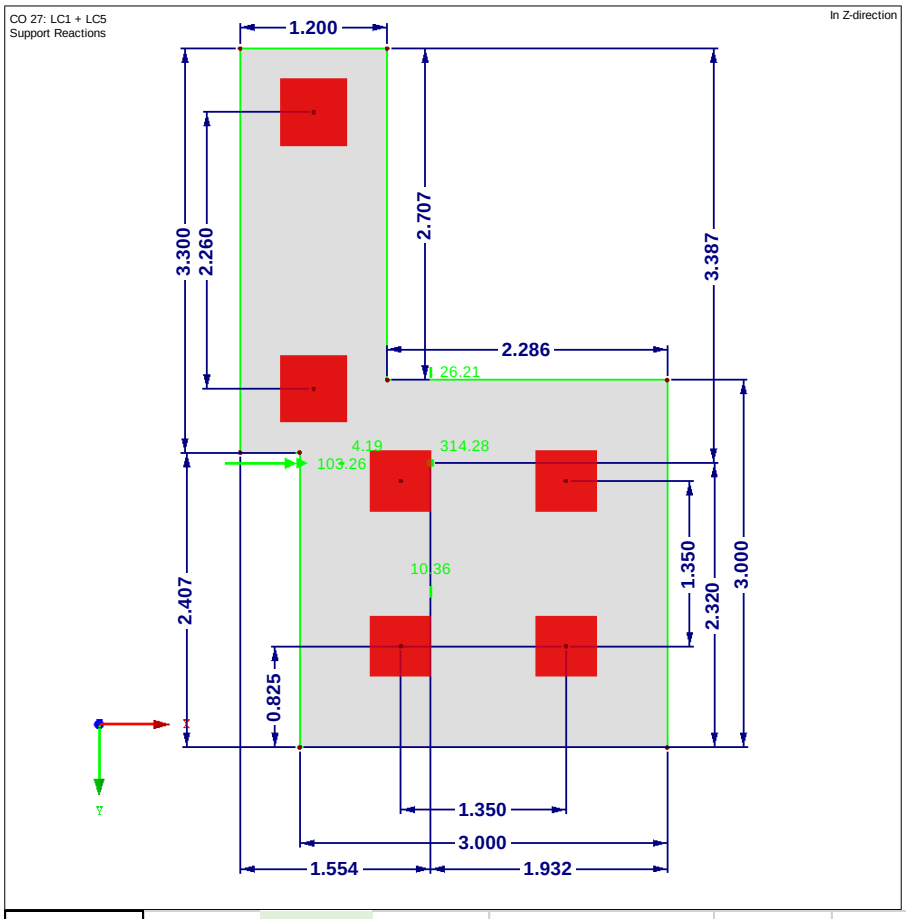


OVERTURN:	Mx=	102.15	kNm	overturn moment					
	Fz=	314.34	kN	total vertical load					
	e=	0.325	m	eccentricity					
	B/2=	1.554	m						
	F.O.S=	4.782	>1.50	Factor of safety against overturning					
SLIDING:	Fx=	26.29	kN	lateral force in y					
	Fy=	5.41	kN	lateral force in x					
	Hd=	26.84	kN	sliding force					
	Rd=	125.736	kN	resistance to sliding =coefficient of friction* total vertical load					
	F.O.S=	4.684	>1.50	Factor of safety against sliding					

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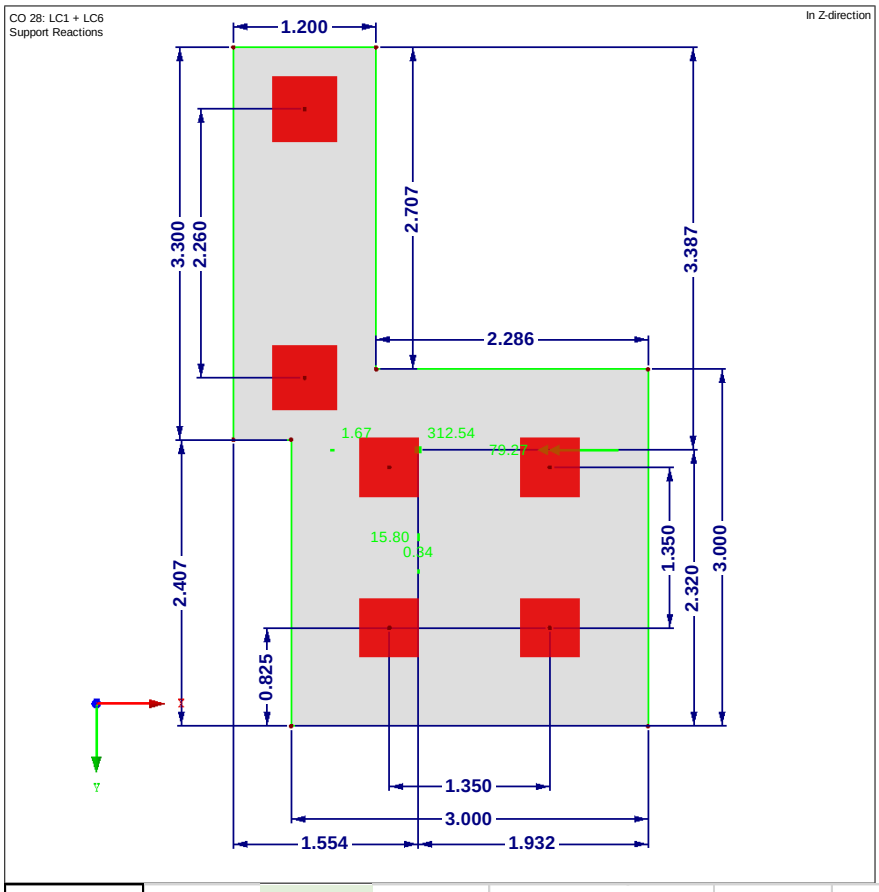


OVERTURN:	Mx=	103.26	kNm	overturn moment
	Fz=	314.28	kN	total vertical load
	e=	0.329	m	eccentricity
	B/2=	2.32	m	
	F.O.S=	7.061	>1.50	Factor of safety against overturning
SLIDING:	Fx=	4.19	kN	lateral force in y
	Fy=	26.21	kN	lateral force in x
	Hd=	26.54	kN	sliding force
	Rd=	125.712	kN	resistance to sliding =coefficient of friction* total vertical load
	F.O.S=	4.736	>1.50	Factor of safety against sliding

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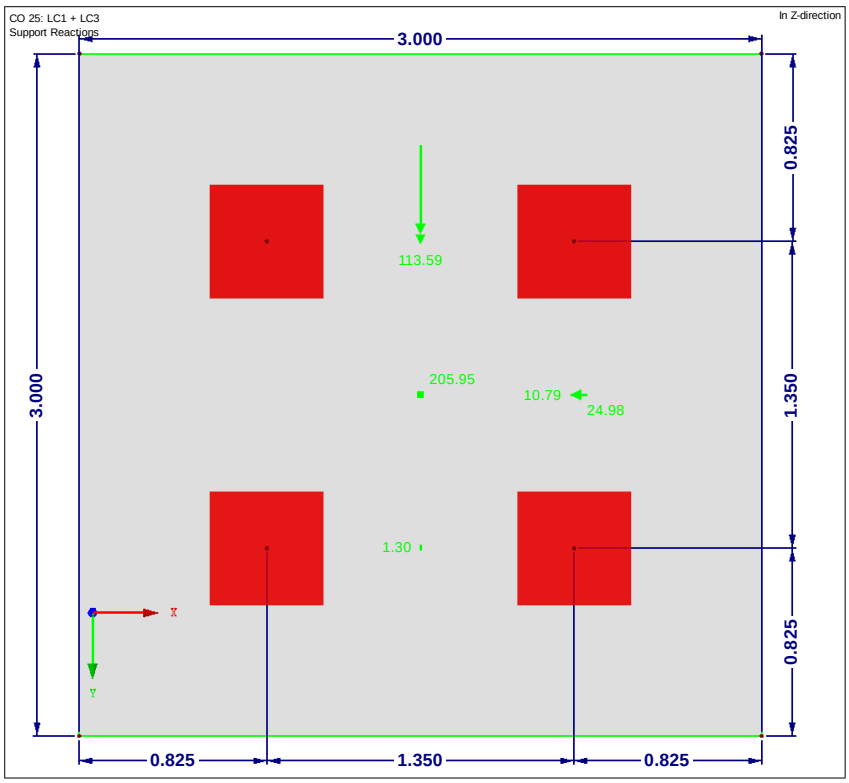
OVERTURN:	Mx=	79.27	kNm	overturn moment
	Fz=	312.54	kN	total vertical load
	e=	0.254	m	eccentricity
	B/2=	2.32	m	
	F.O.S=	9.147	>1.50	Factor of safety against overturning
SLIDING:	Fx=	1.67	kN	lateral force in y
	Fy=	15.8	kN	lateral force in x
	Hd=	15.89	kN	sliding force
	Rd=	125.016	kN	resistance to sliding =coefficient of friction* total vertical load
	F.O.S=	7.869	>1.50	Factor of safety against sliding

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combined footing P45

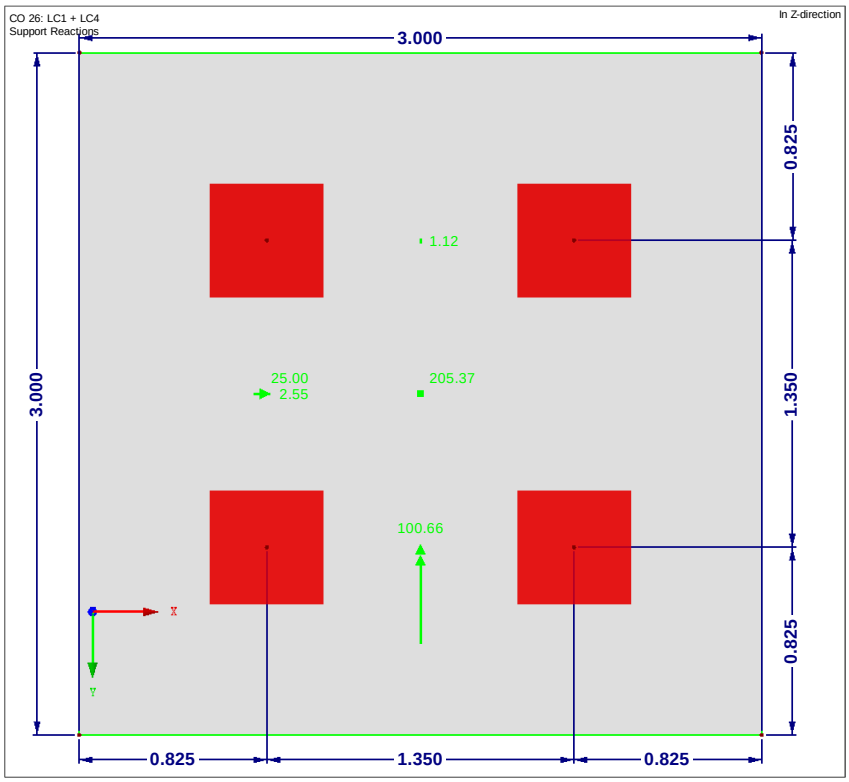


OVERTURN:	Mx=	113.59	kNm	overturn moment				
	Fz=	205.95	kN	total vertical load				
	e=	0.552	m	eccentricity				
	B/2=	1.5	m					
	F.O.S=	2.720	>1.50	Factor of safety against overturning				
SLIDING:	Fx=	24.98	kN	lateral force in x				
	Fy=	1.3	kN	lateral force in y				
	Hd=	25.01	kN	sliding force				
	Rd=	82.38	kN	resistance to sliding =coefficient of friction* total vertical load				
	F.O.S=	3.293	>1.50	Factor of safety against sliding				

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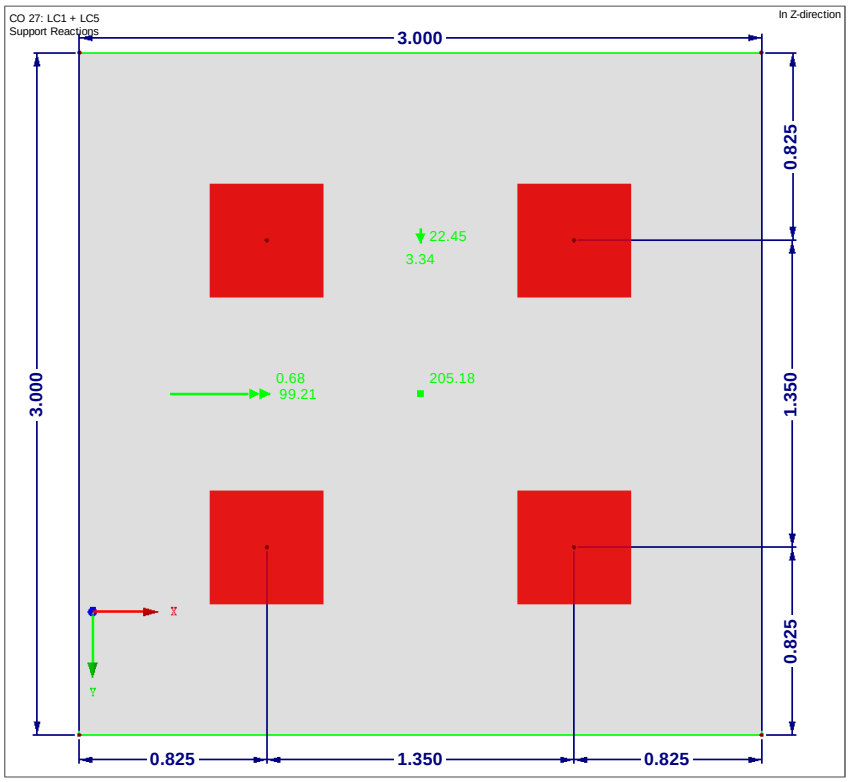


OVERTURN:	Mx=	100.66	kNm	overturn moment					
	Fz=	205.37	kN	total vertical load					
	e=	0.490	m	eccentricity					
	B/2=	1.5	m						
	F.O.S=	3.060	>1.50	Factor of safety against overturning					
SLIDING:	Fx=	25	kN	lateral force in x					
	Fy=	1.12	kN	lateral force in y					
	Hd=	25.03	kN	sliding force					
	Rd=	82.148	kN	resistance to sliding =coefficient of friction* total vertical load					
	F.O.S=	3.283	>1.50	Factor of safety against sliding					

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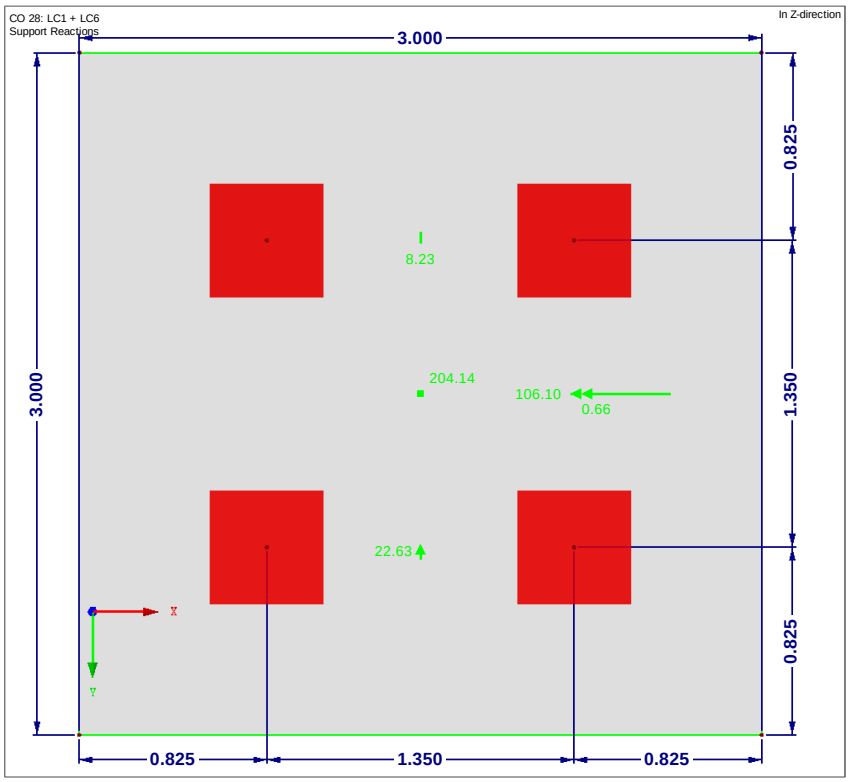


OVERTURN:	Mx=	99.21	kNm	overturn moment					
	Fz=	205.18	kN	total vertical load					
	e=	0.484	m	eccentricity					
	B/2=	1.5	m						
	F.O.S=	3.102	>1.50	Factor of safety against overturning					
SLIDING:	Fx=	0.68	kN	lateral force in x					
	Fy=	22.45	kN	lateral force in y					
	Hd=	22.46	kN	sliding force					
	Rd=	82.072	kN	resistance to sliding =coefficient of friction* total vertical load					
	F.O.S=	3.654	>1.50	Factor of safety against sliding					

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OVERTURN:	Mx=	106.1	kNm	overturn moment					
	Fz=	204.14	kN	total vertical load					
	e=	0.520	m	eccentricity					
	B/2=	1.5	m						
	F.O.S=	2.886	>1.50	Factor of safety against overturning					
SLIDING:	Fx=	0.66	kN	lateral force in x					
	Fy=	22.63	kN	lateral force in y					
	Hd=	22.64	kN	sliding force					
	Rd=	81.656	kN	resistance to sliding =coefficient of friction* total vertical load					
	F.O.S=	3.607	>1.50	Factor of safety against sliding					

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6.5. Pedestal design

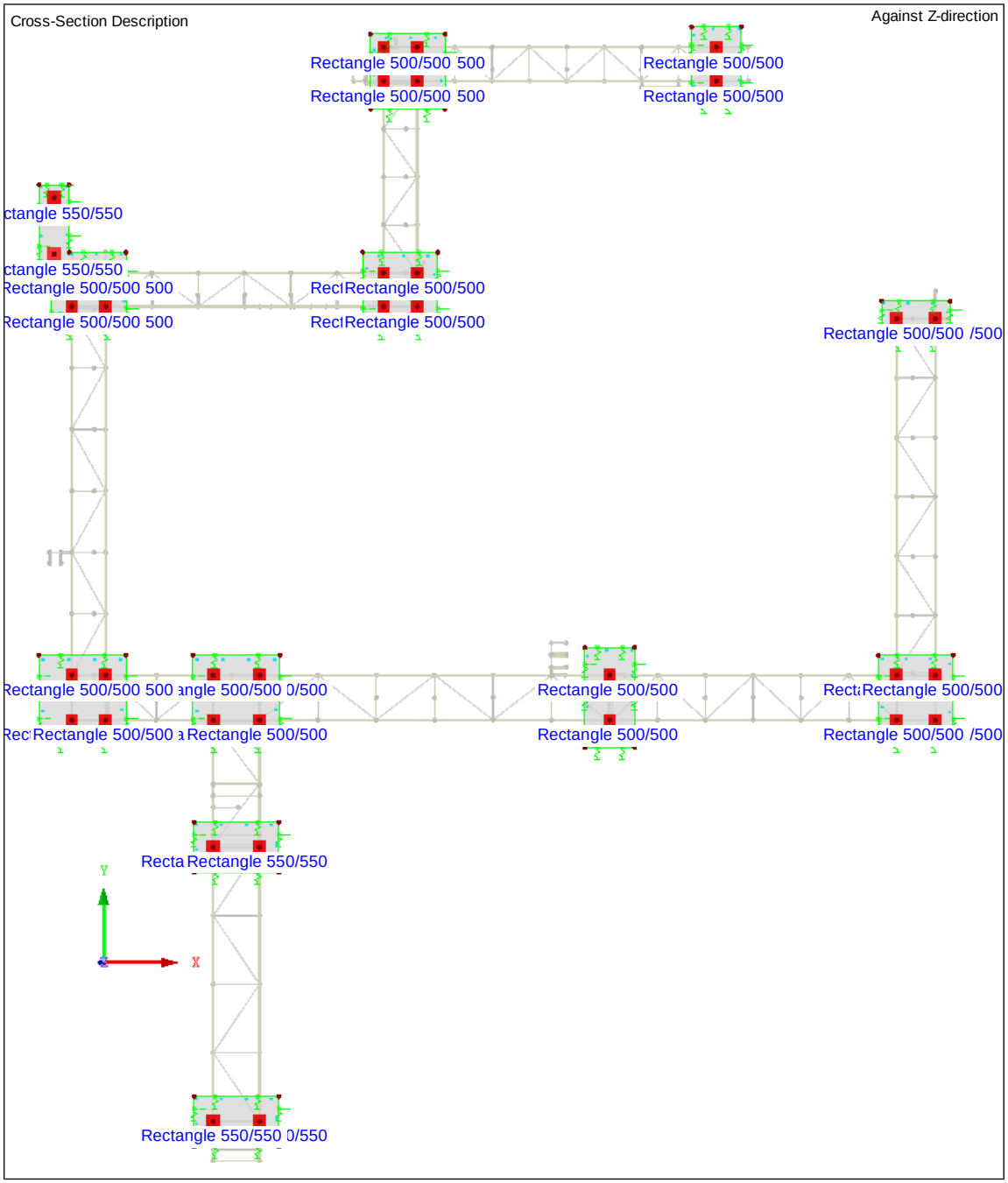


Figure 18 Column layout and dimension

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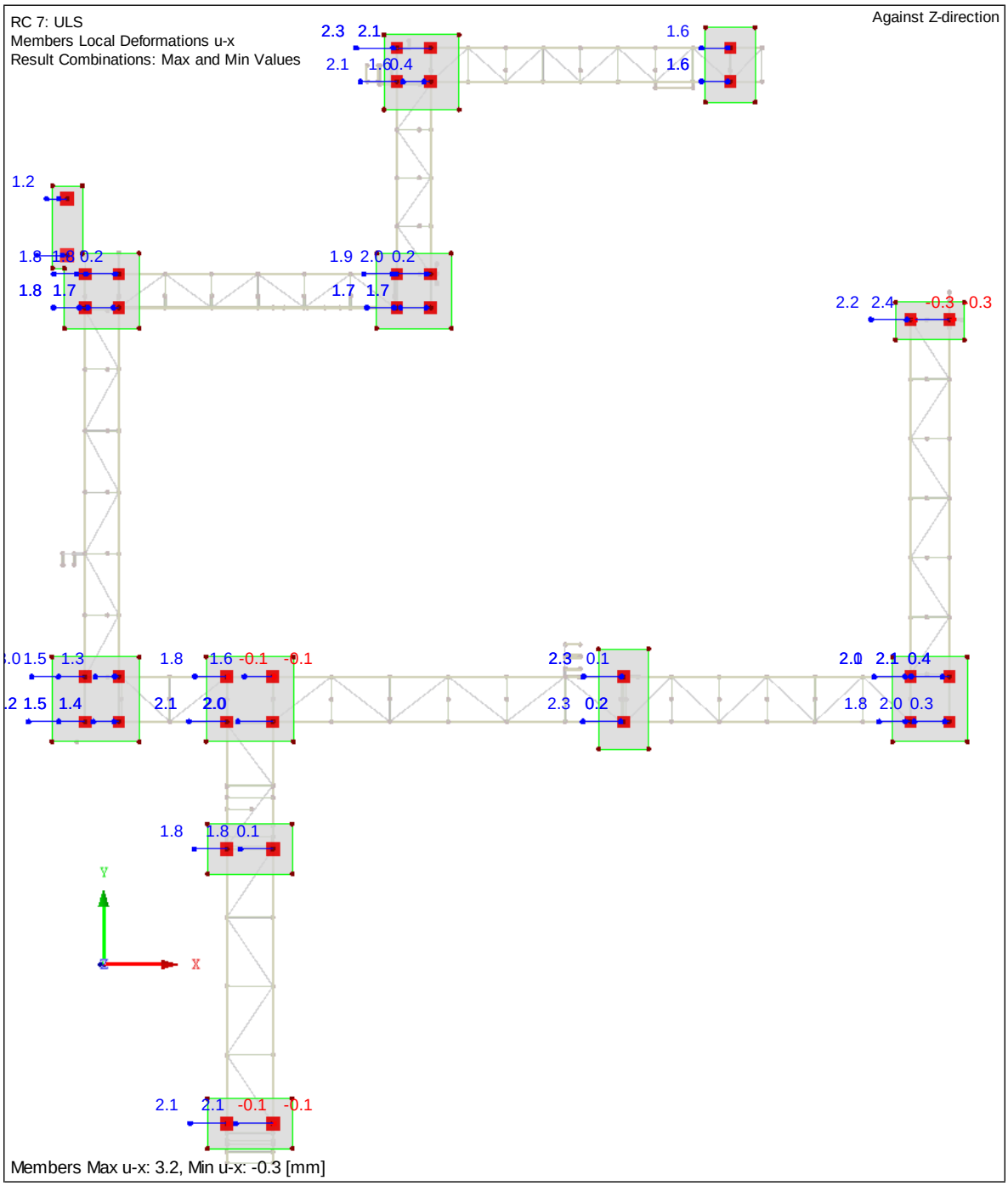


Figure 19 Deformation ux

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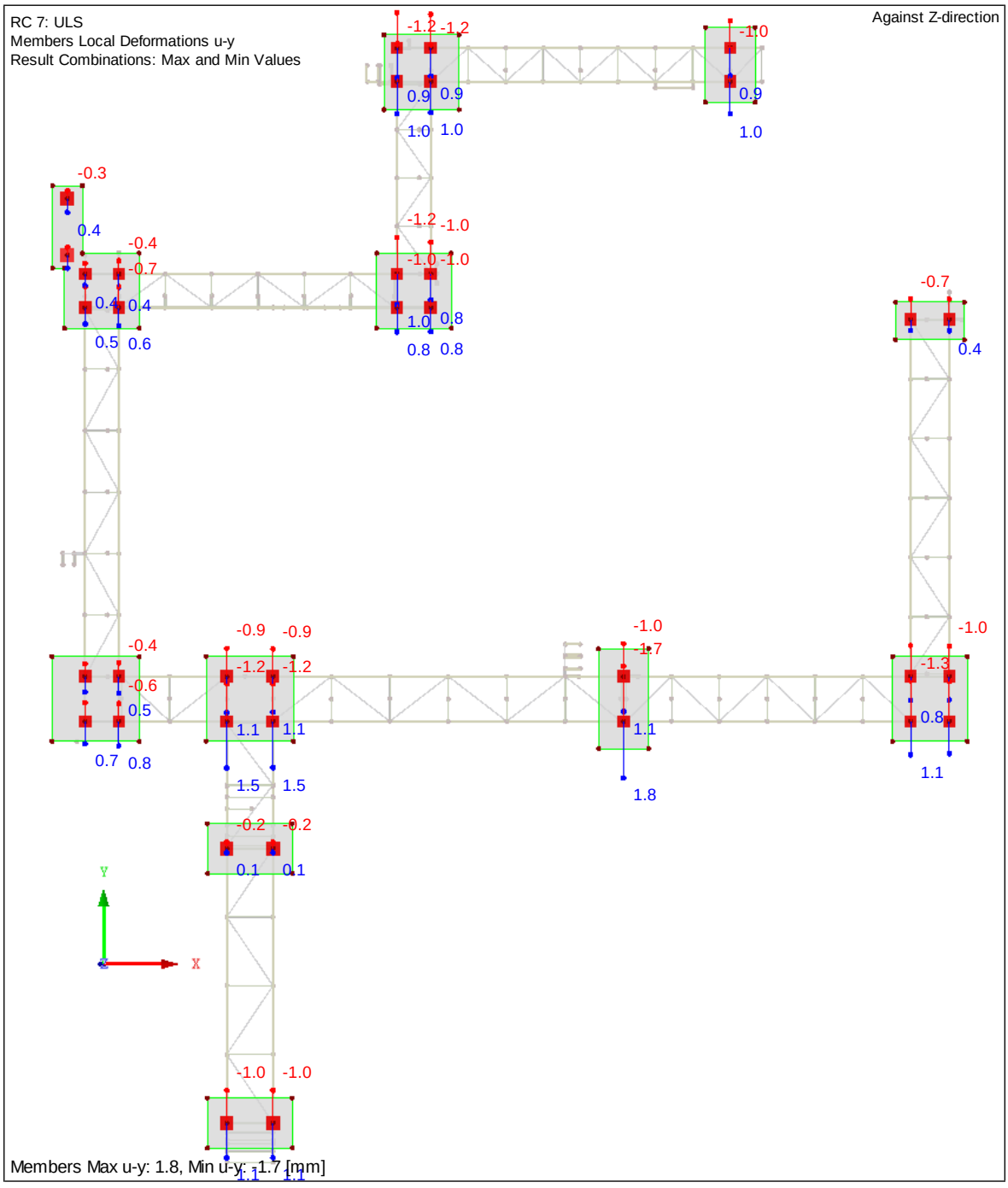


Figure 20 Deformation uy

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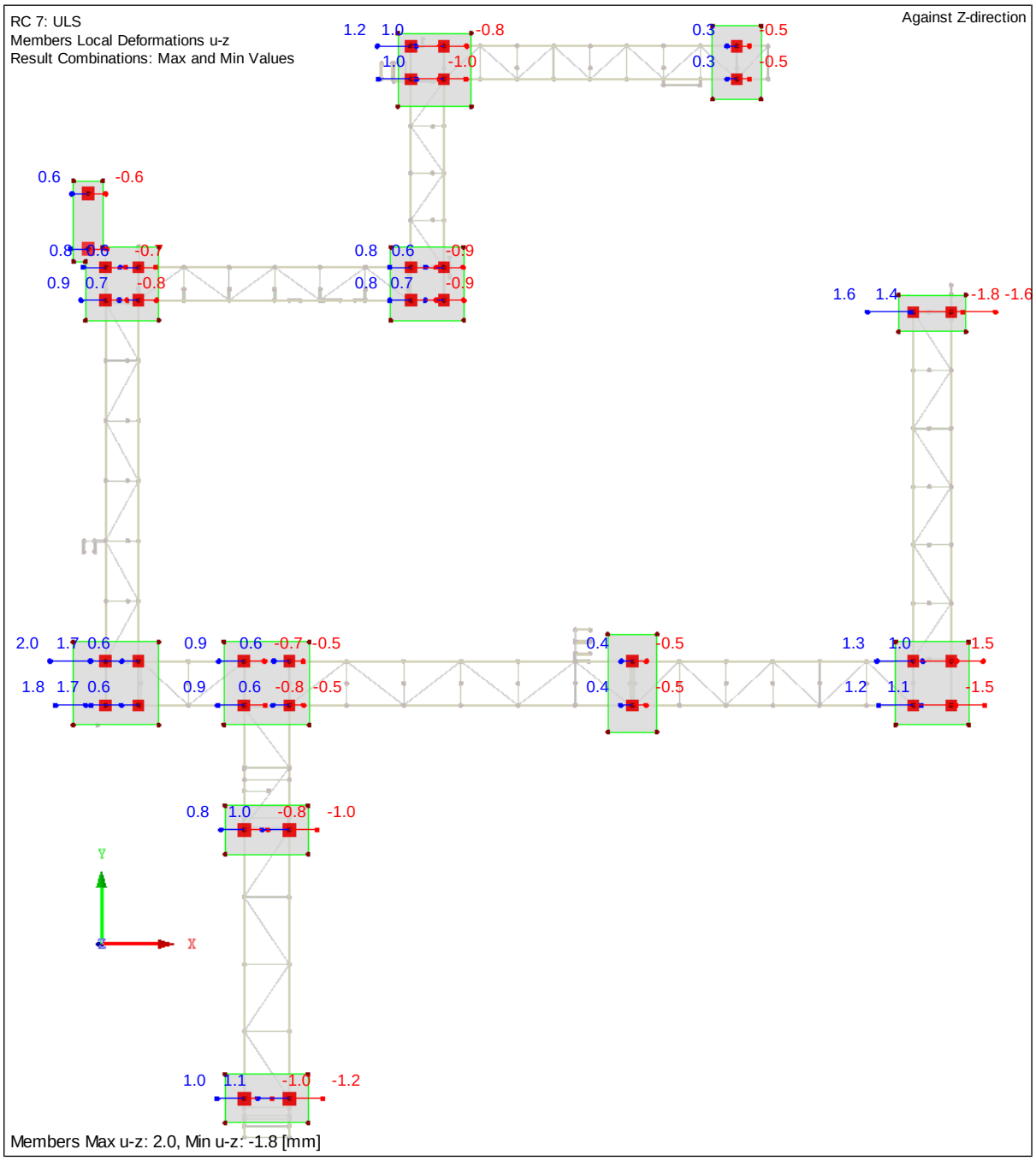


Figure 21 Deformation uz

C01

$u < u_{max}$ = 25mm (according to 6.3.1MD-101-SMEM-EG-ST-CRT-00001), ok!

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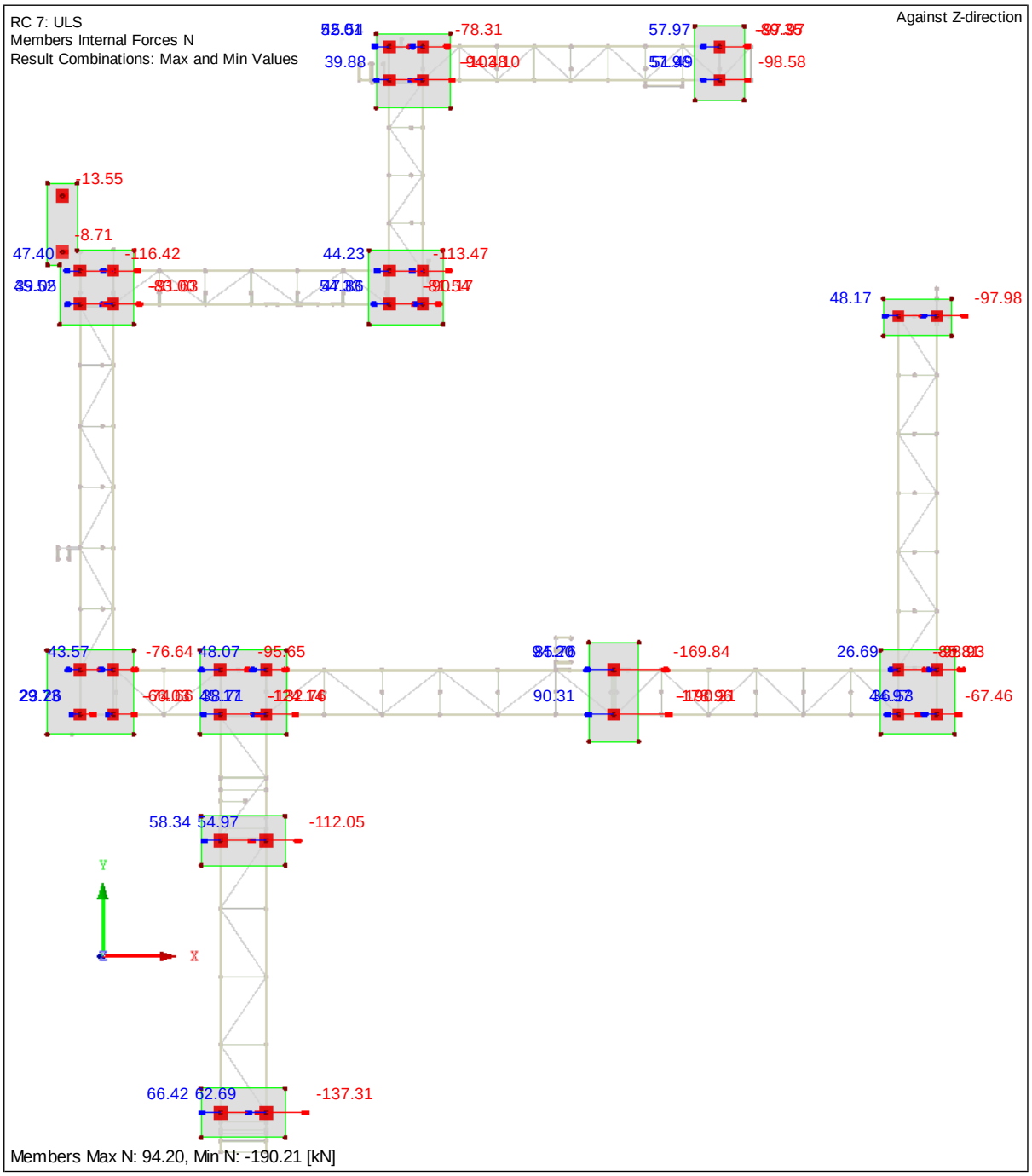


Figure 22 Max. axial force $N_{Ed,max}$ for columns

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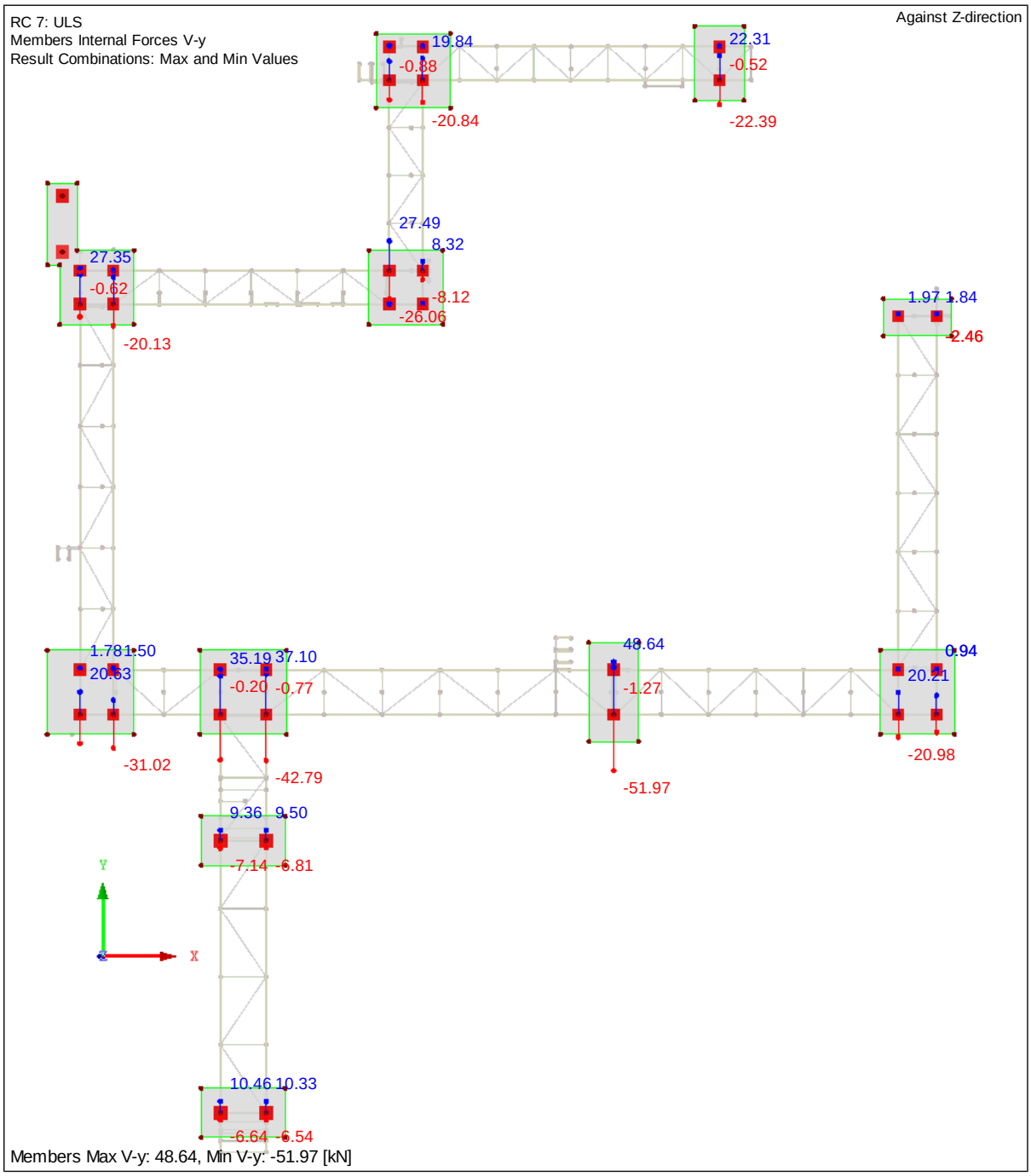


Figure 23 Max. shear force Vy,max for columns

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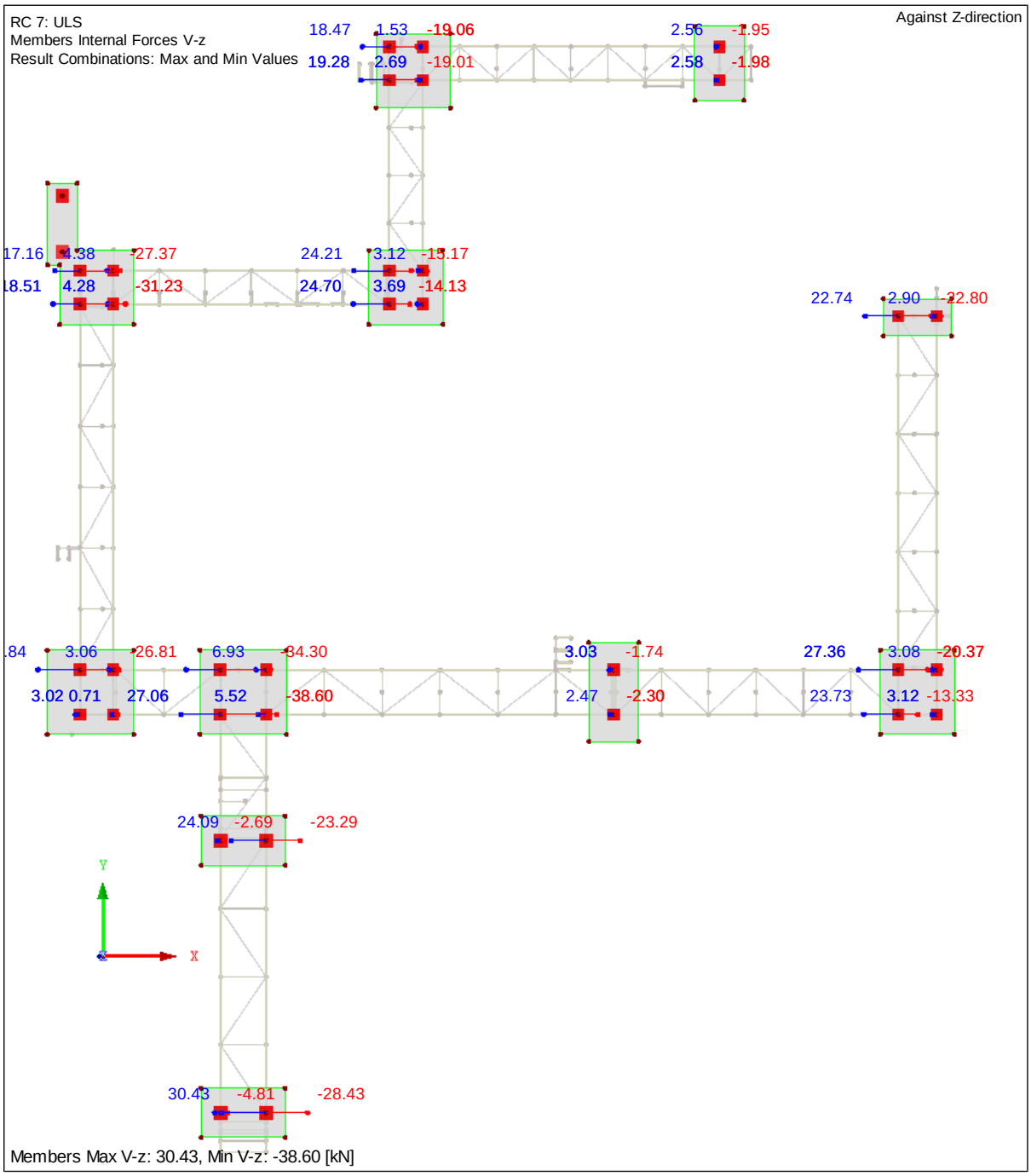


Figure 24 Max. shear force Vz,max for columns

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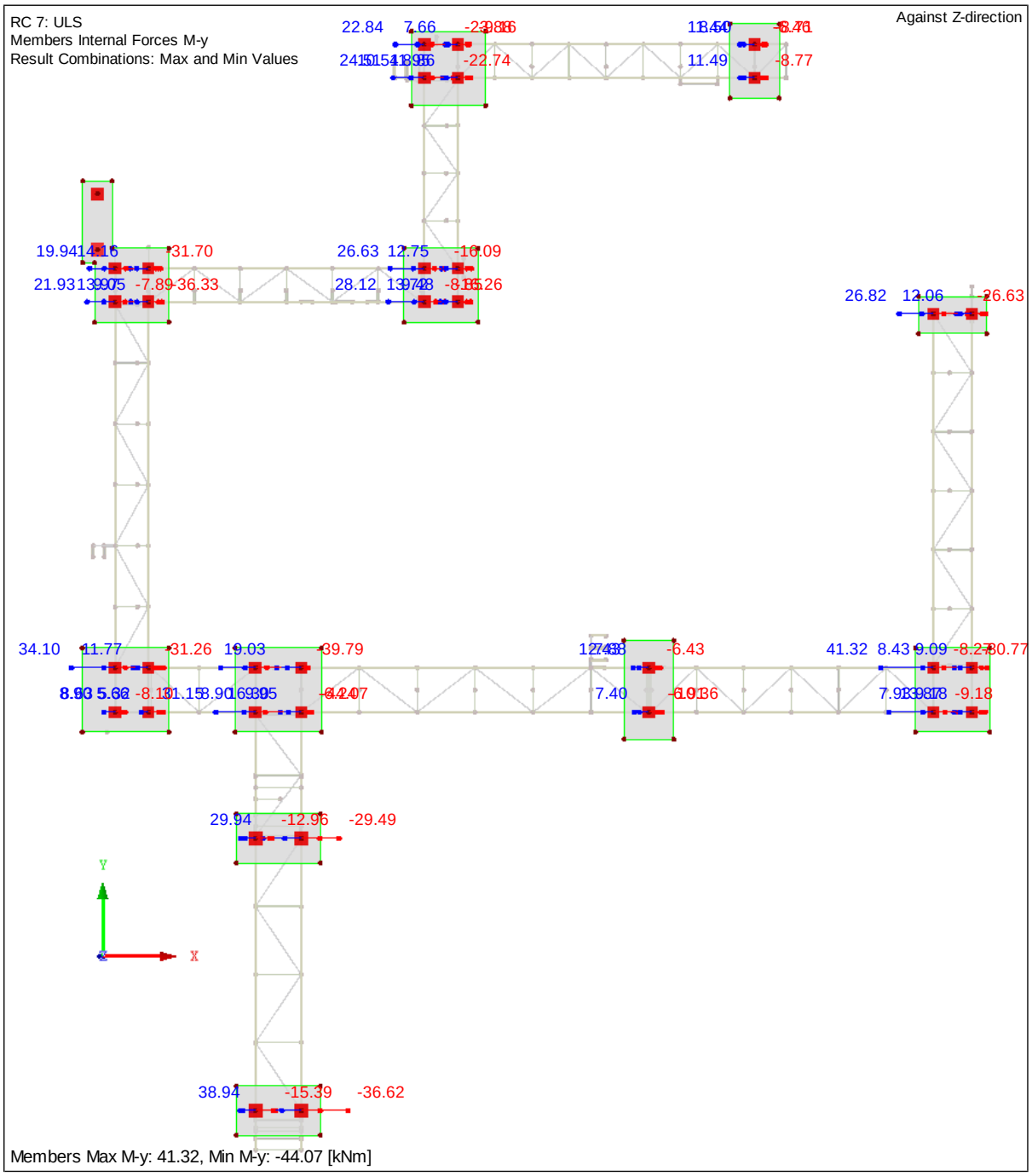


Figure 25 Max. moment My,max for columns

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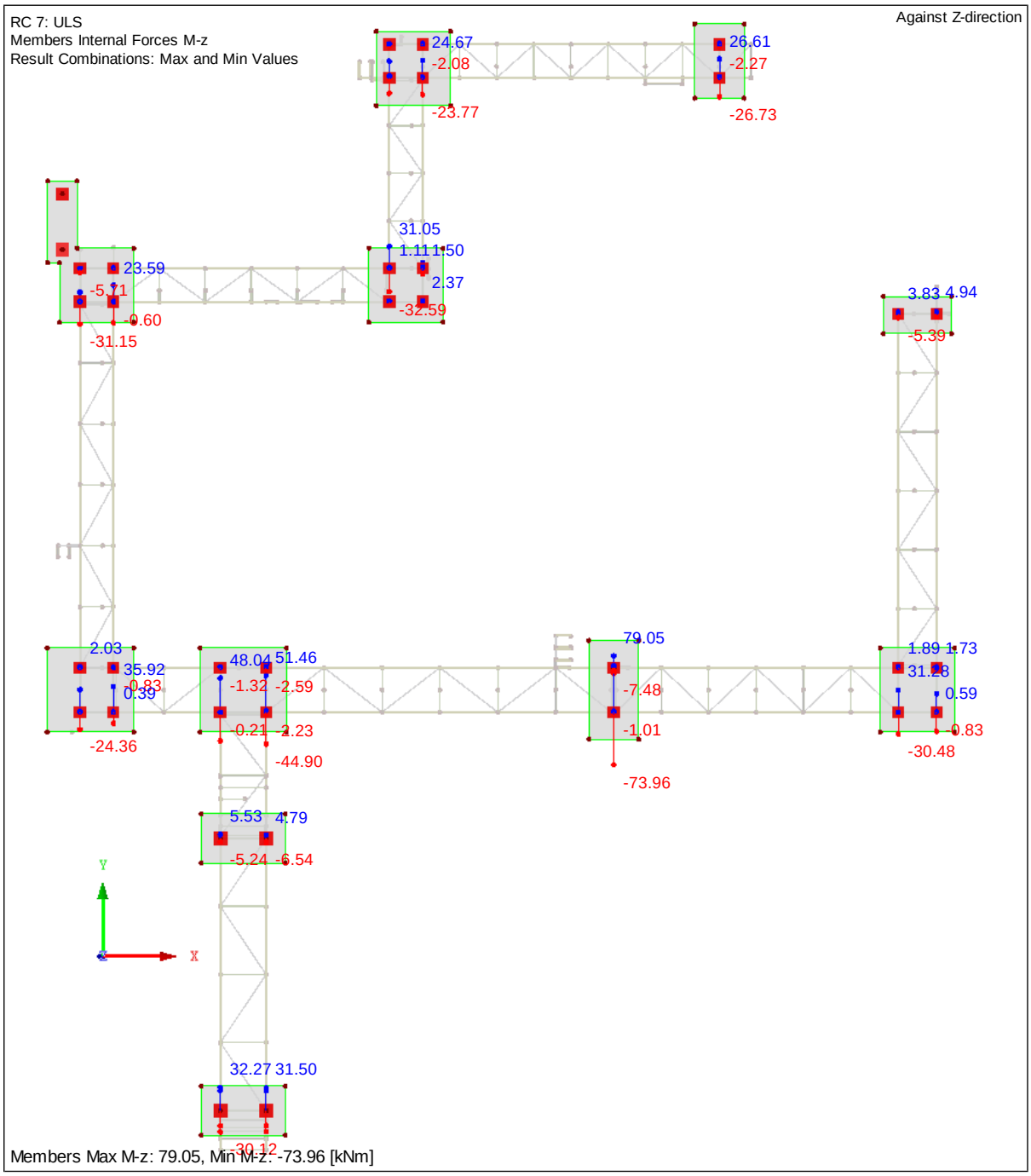


Figure 26 Max. moment Mz,max for columns

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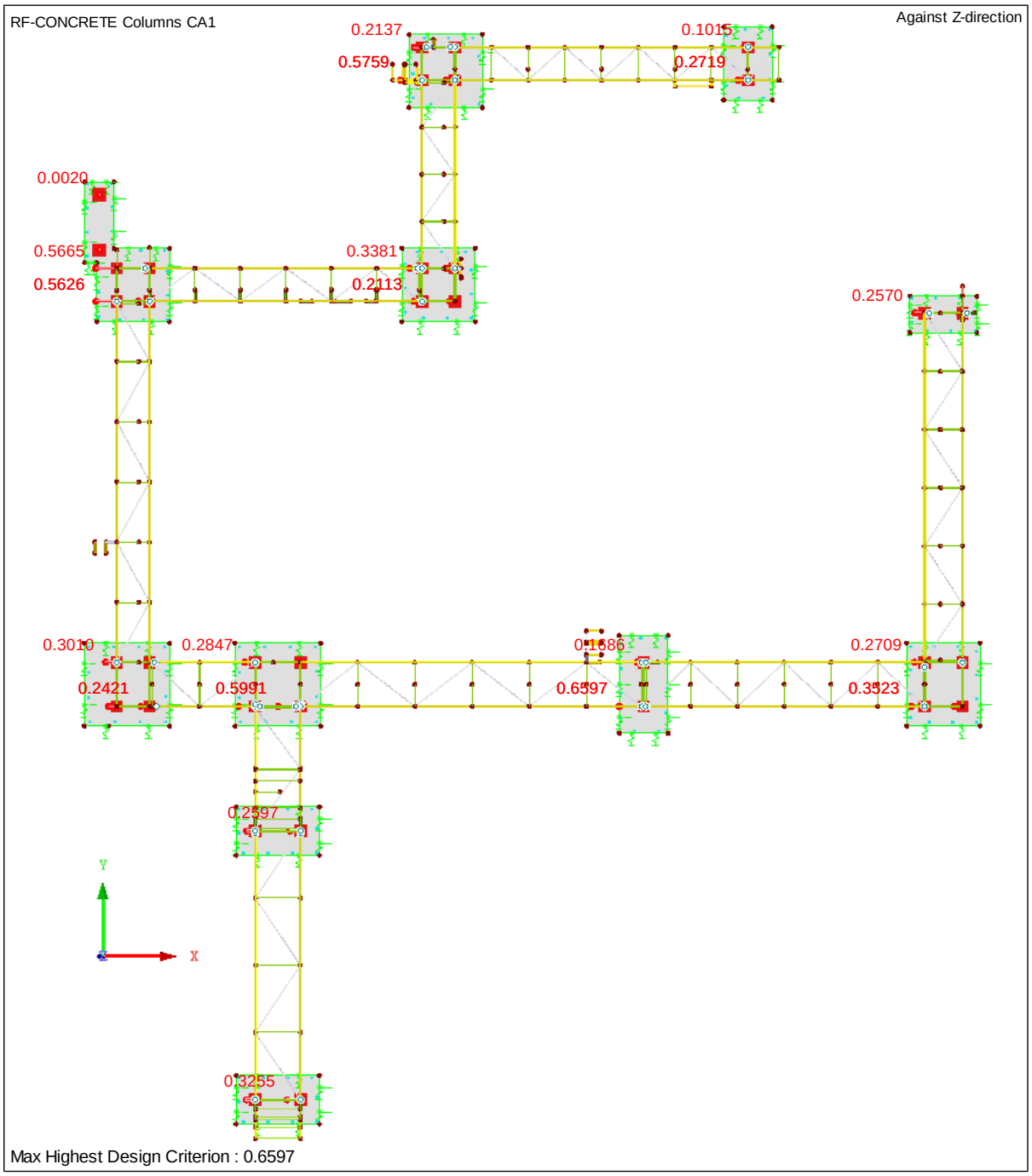


Figure 27 Highest design criterion

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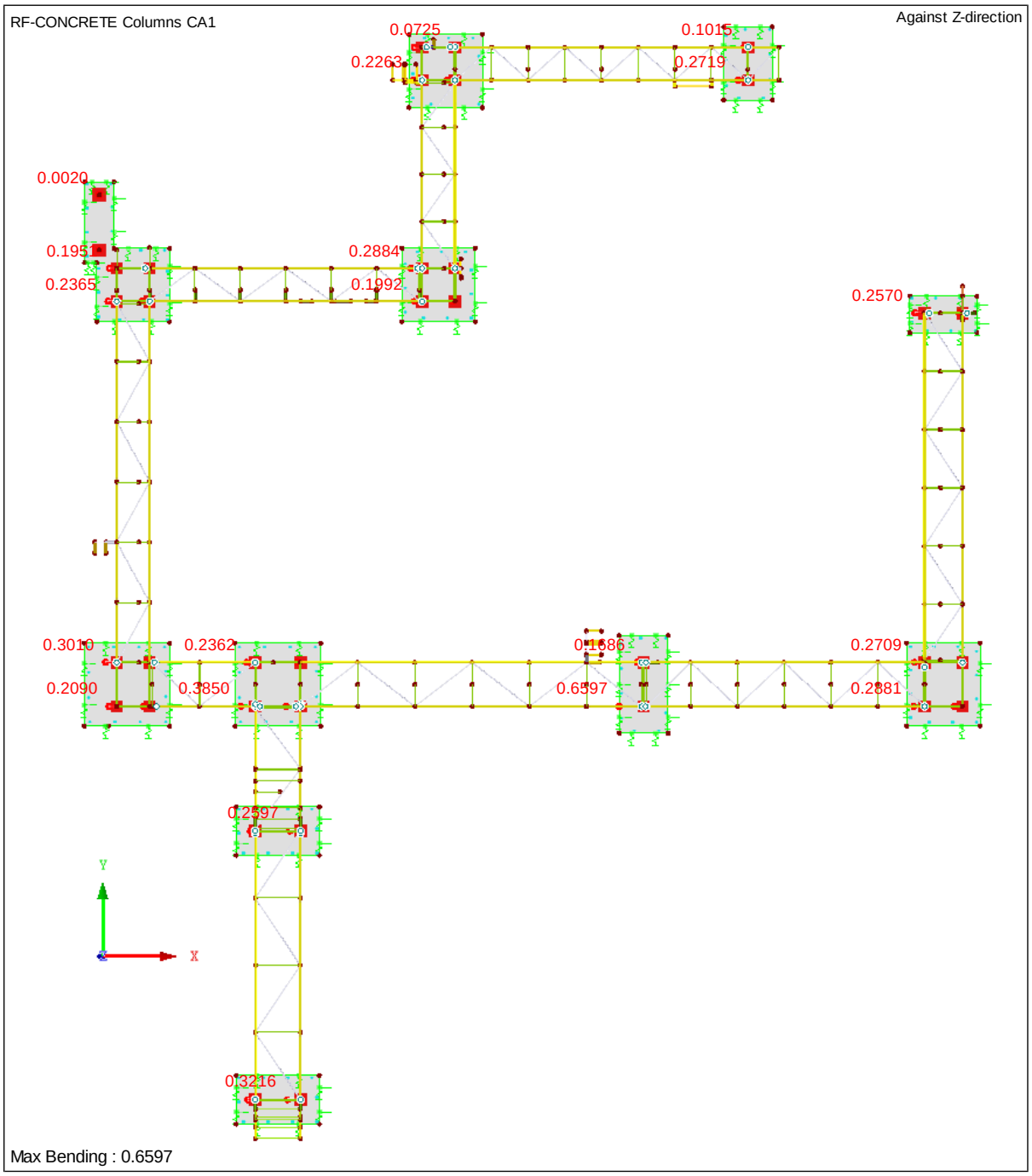


Figure 28 Utility ratio for bending

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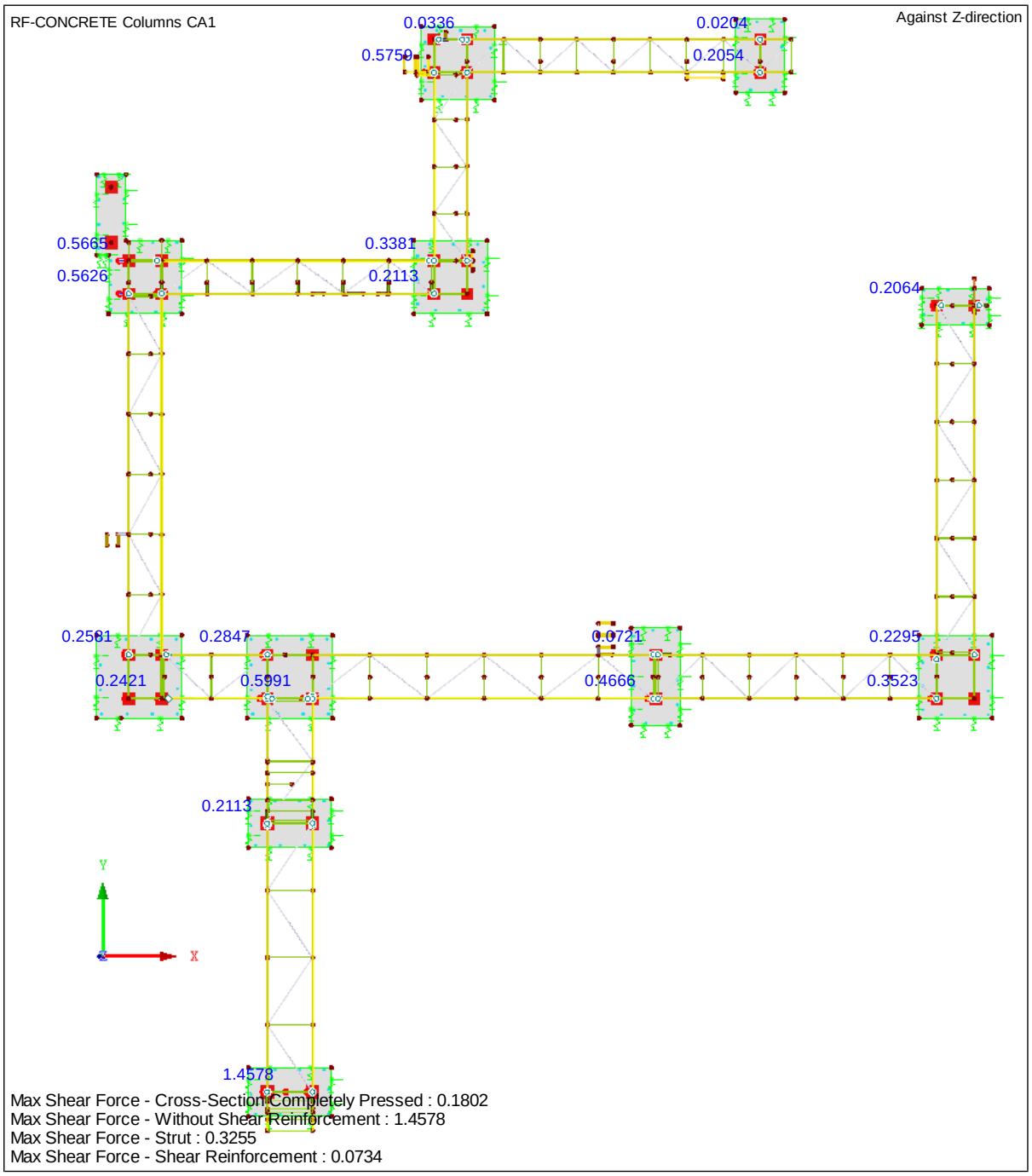


Figure 29 Utility ratio for shear

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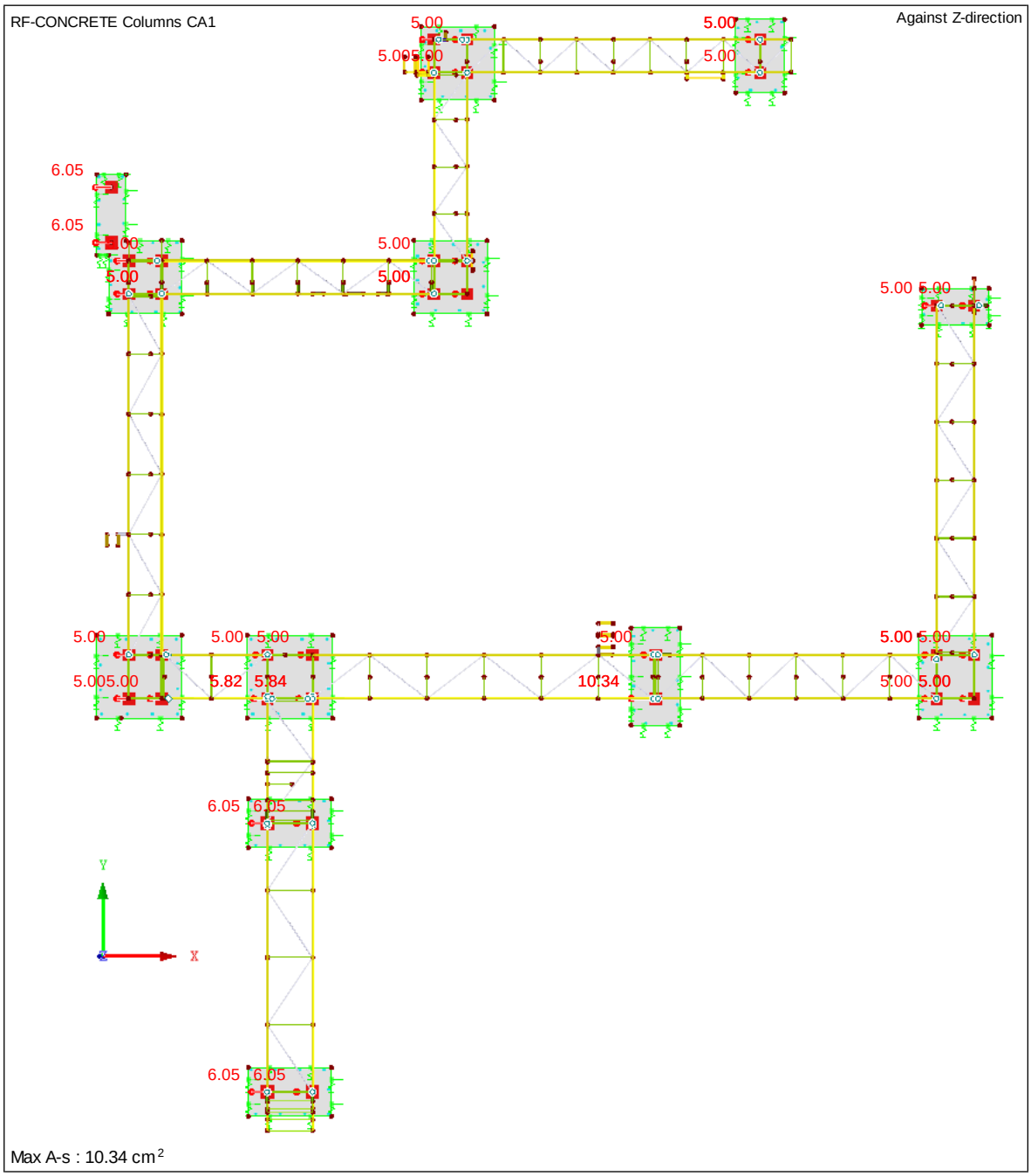


Figure 30 Required longitudinal reinforcement area As

Provide longitudinal reinforcement 8T16, A_{sp} = 8* π*(16mm/2)^2 = 16.08cm^2/m

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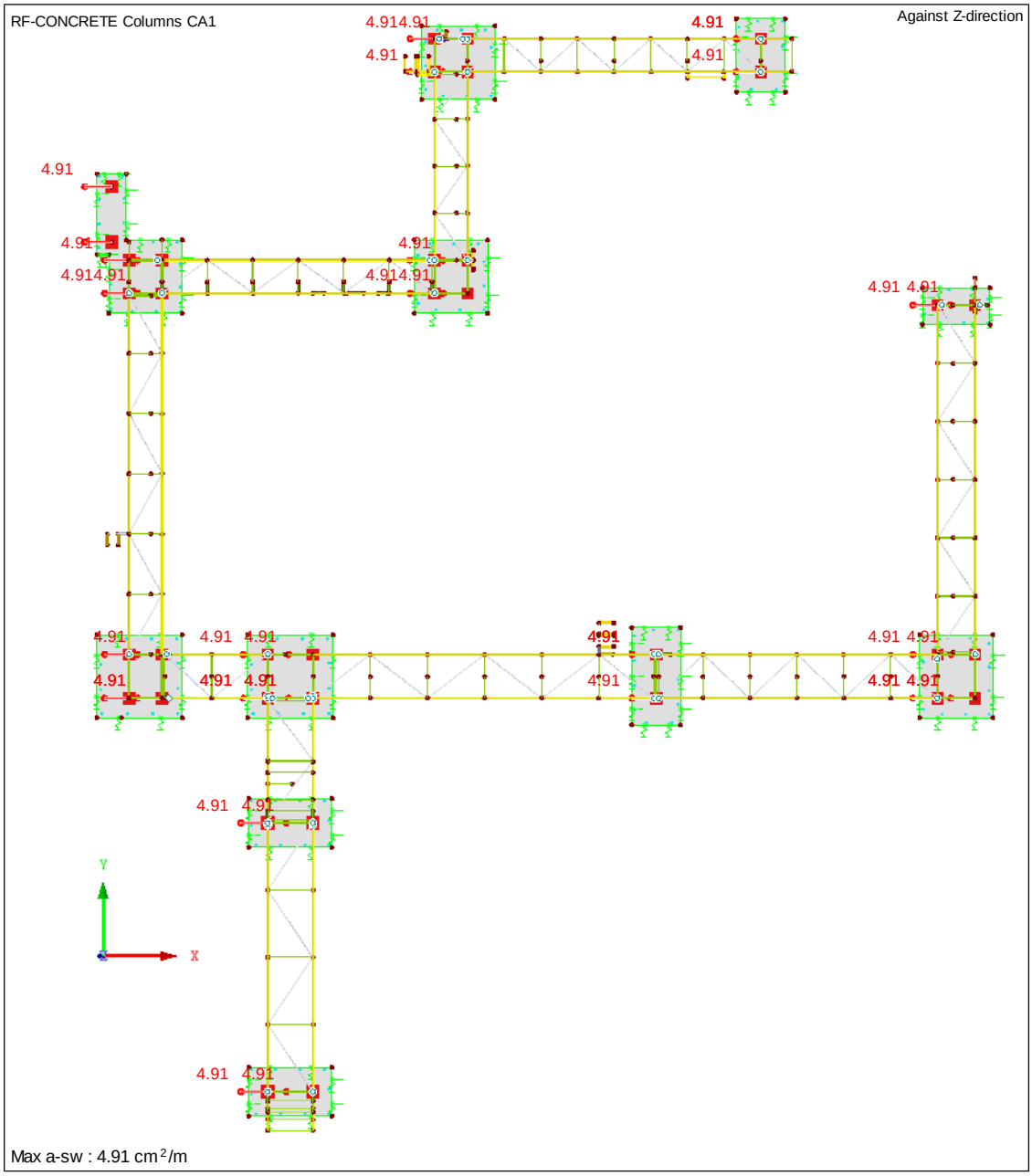


Figure 31 Required shear reinforcement area a_{sw}

Provide shear reinforcement T10/200

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6.6. pad design

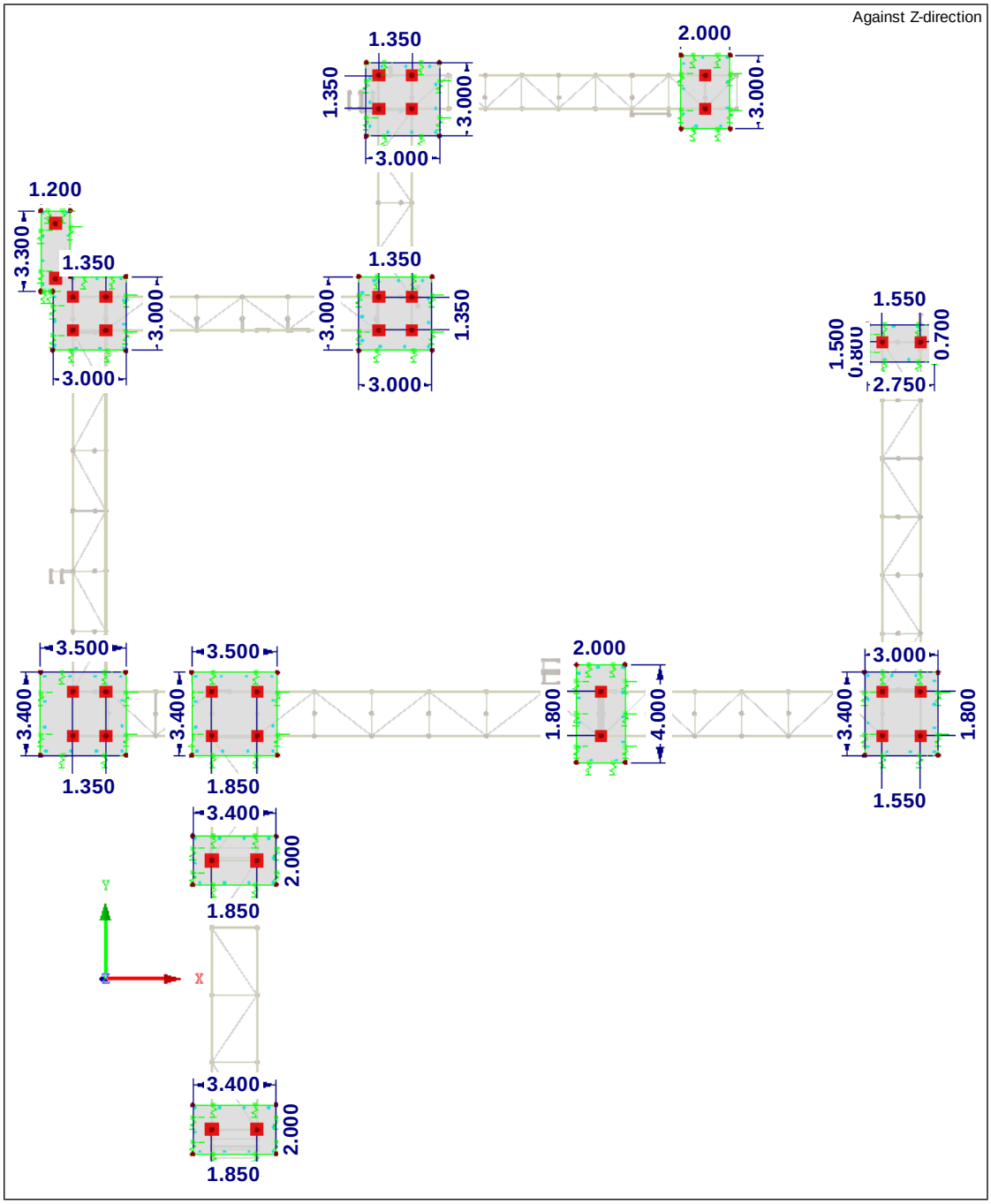


Figure 32 Footing layout and dimension, t=400mm

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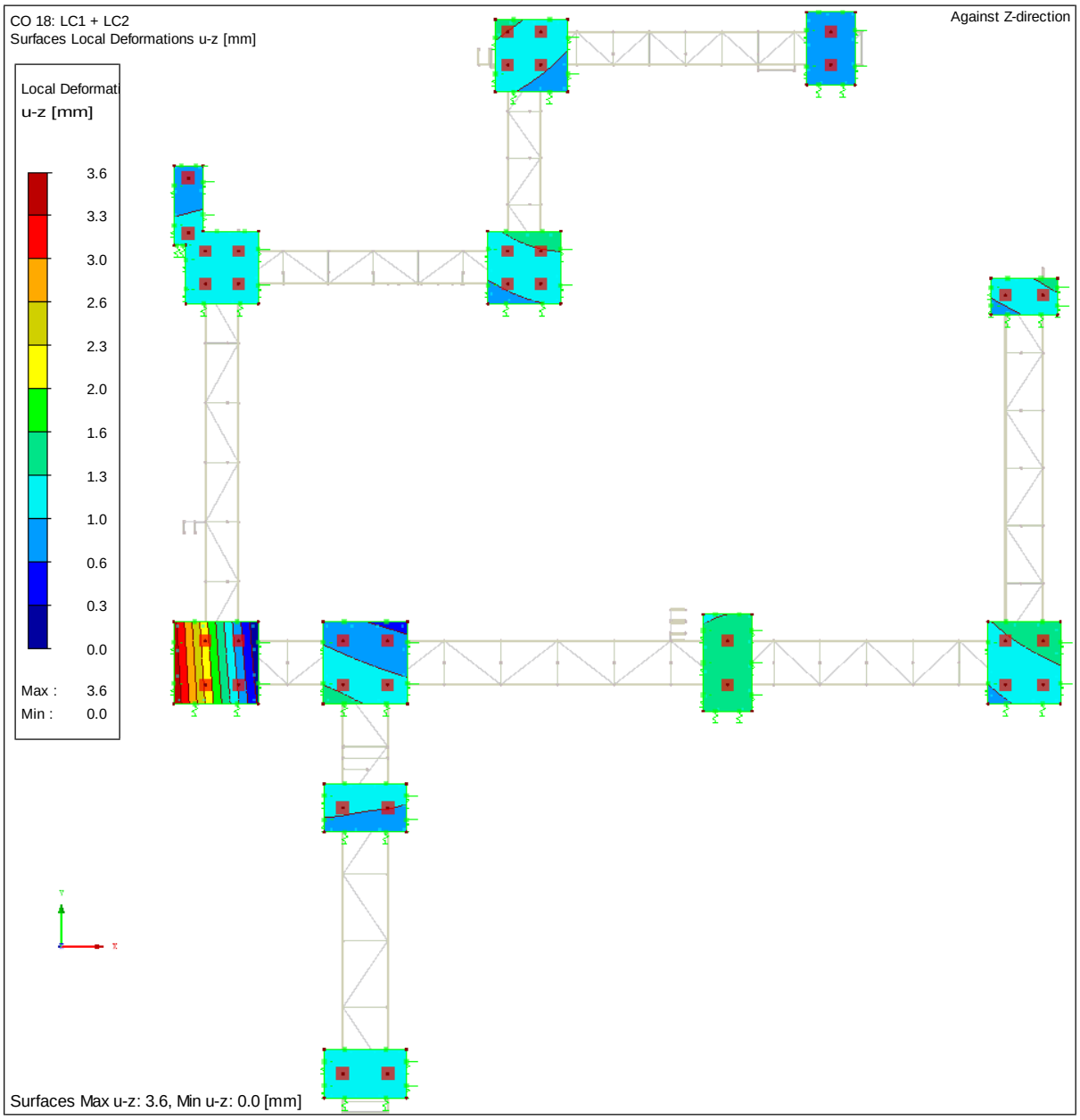


Figure 33 Vertical displacement

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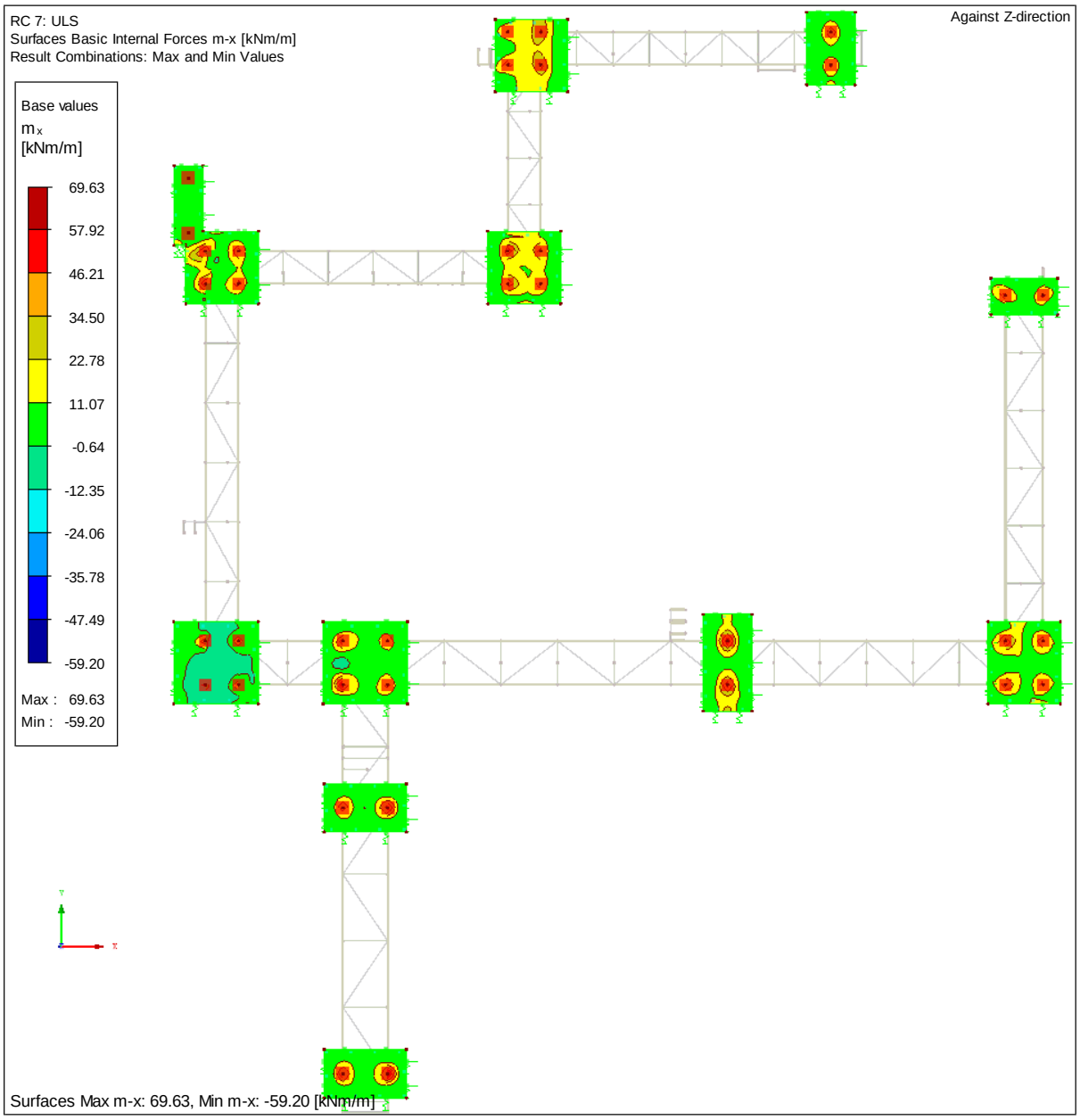


Figure 34 Bending moment Mx

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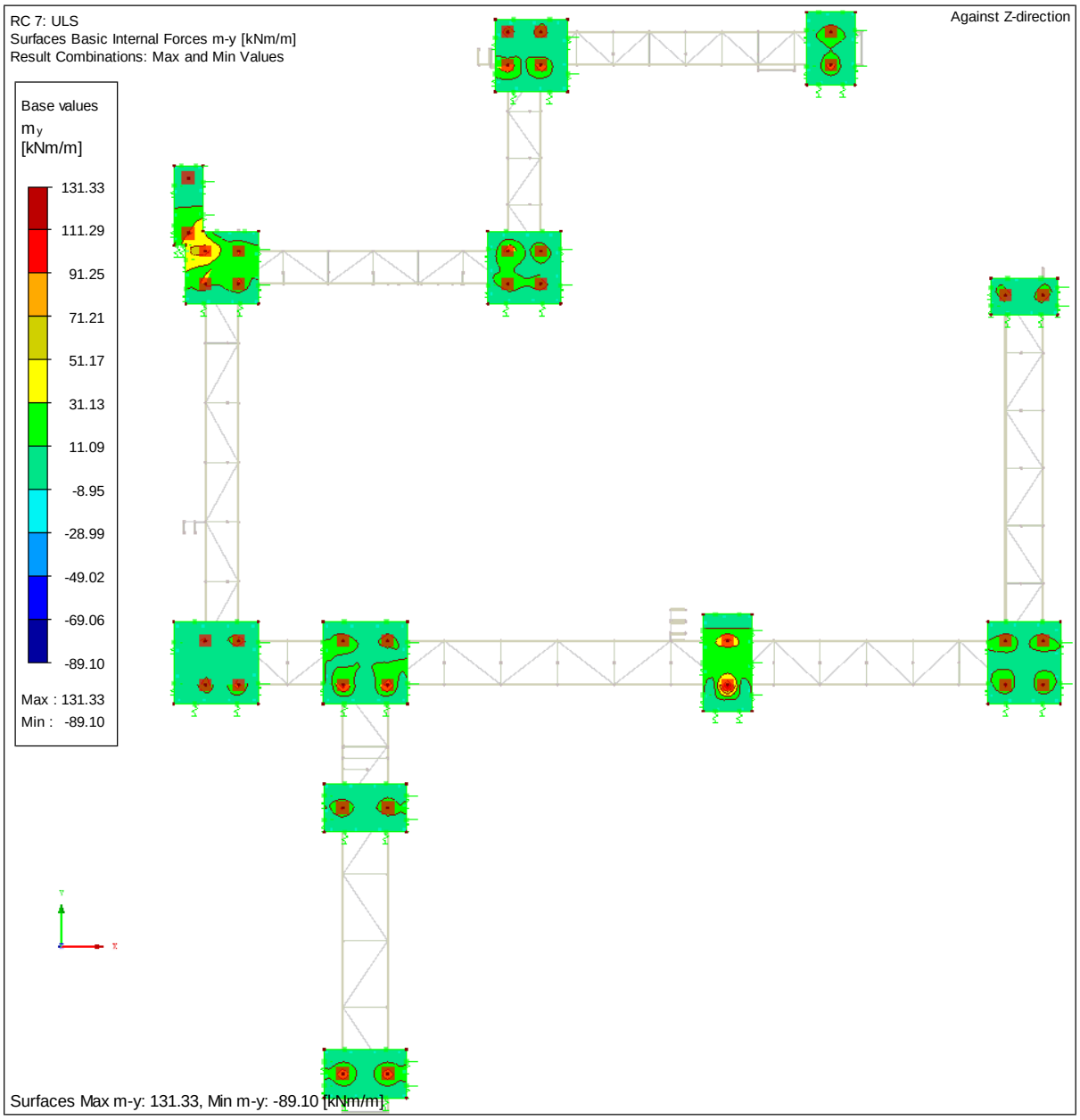


Figure 35 Bending moment My

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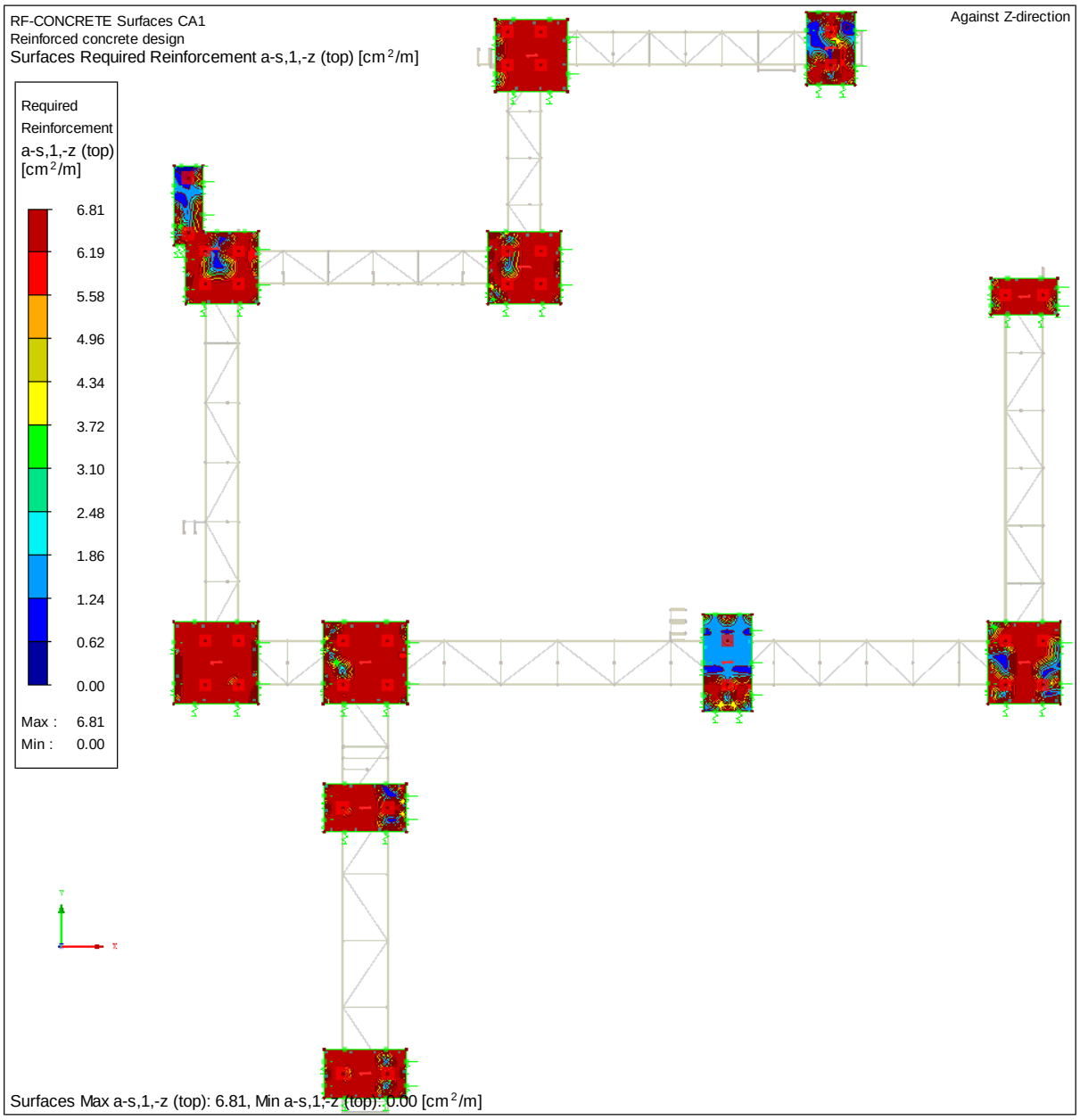


Figure 36 Required reinforcement As,1, top

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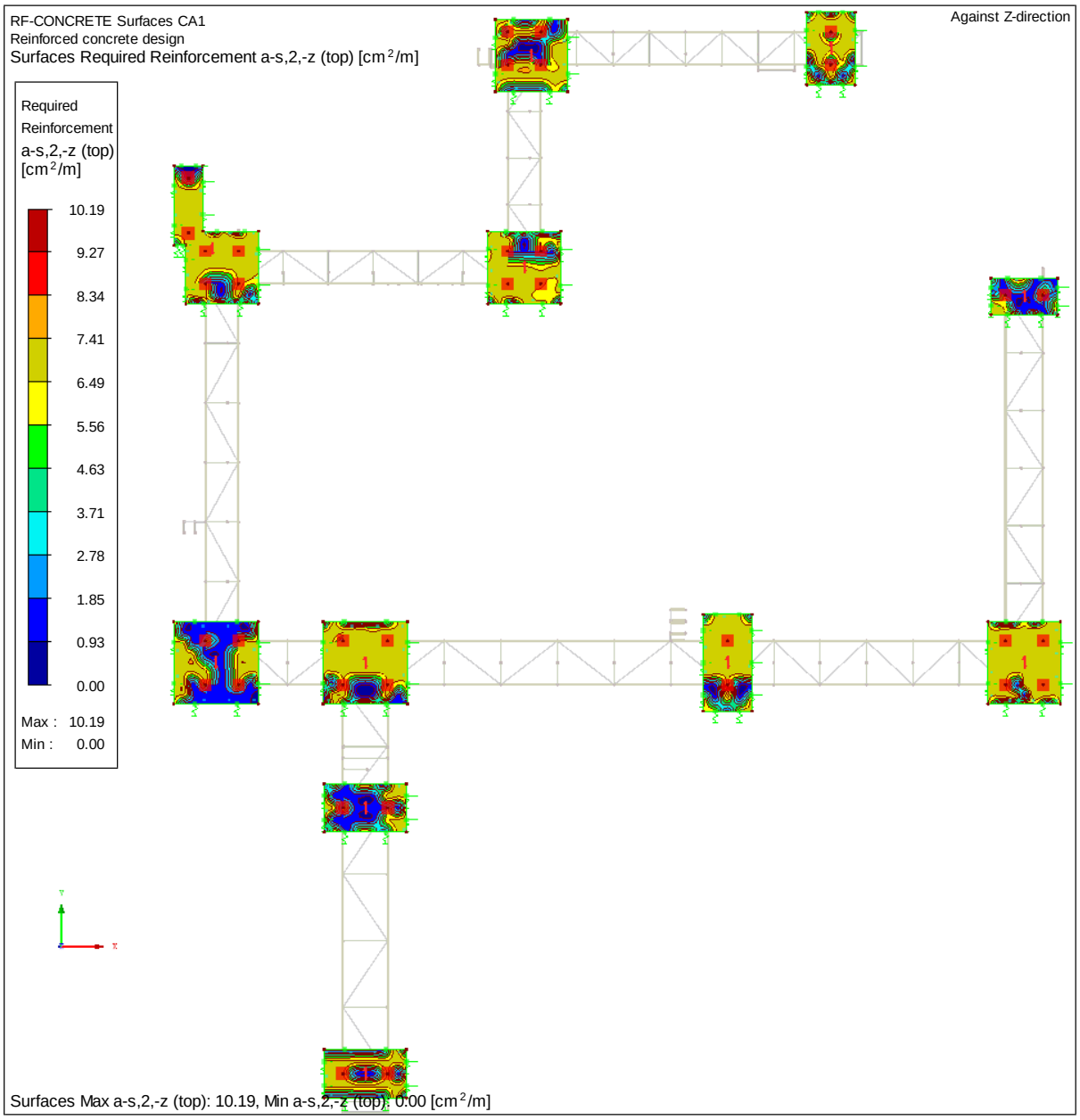


Figure 37 Required reinforcement As,2, top

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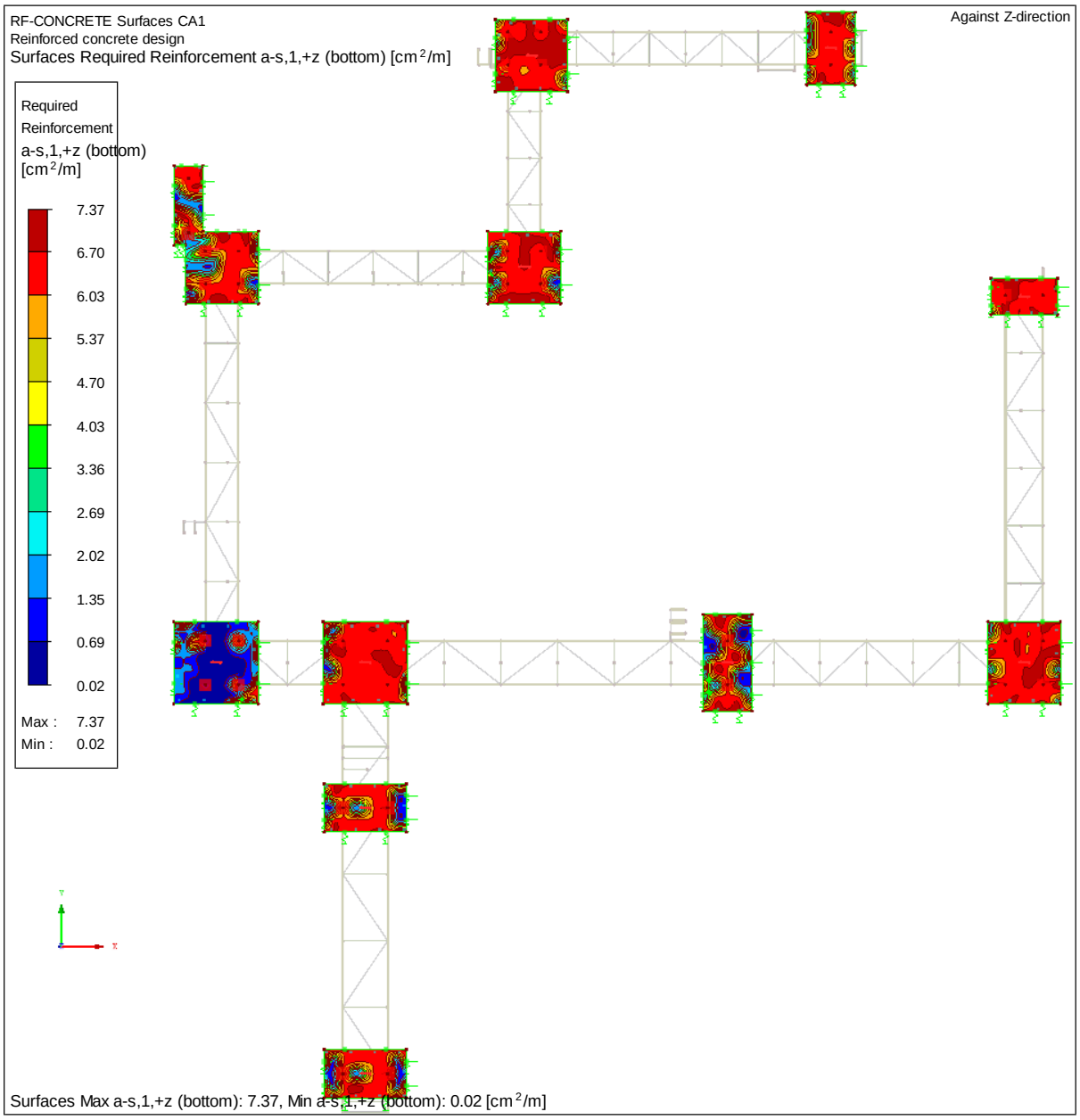


Figure 38 Required reinforcement As,1, bottom

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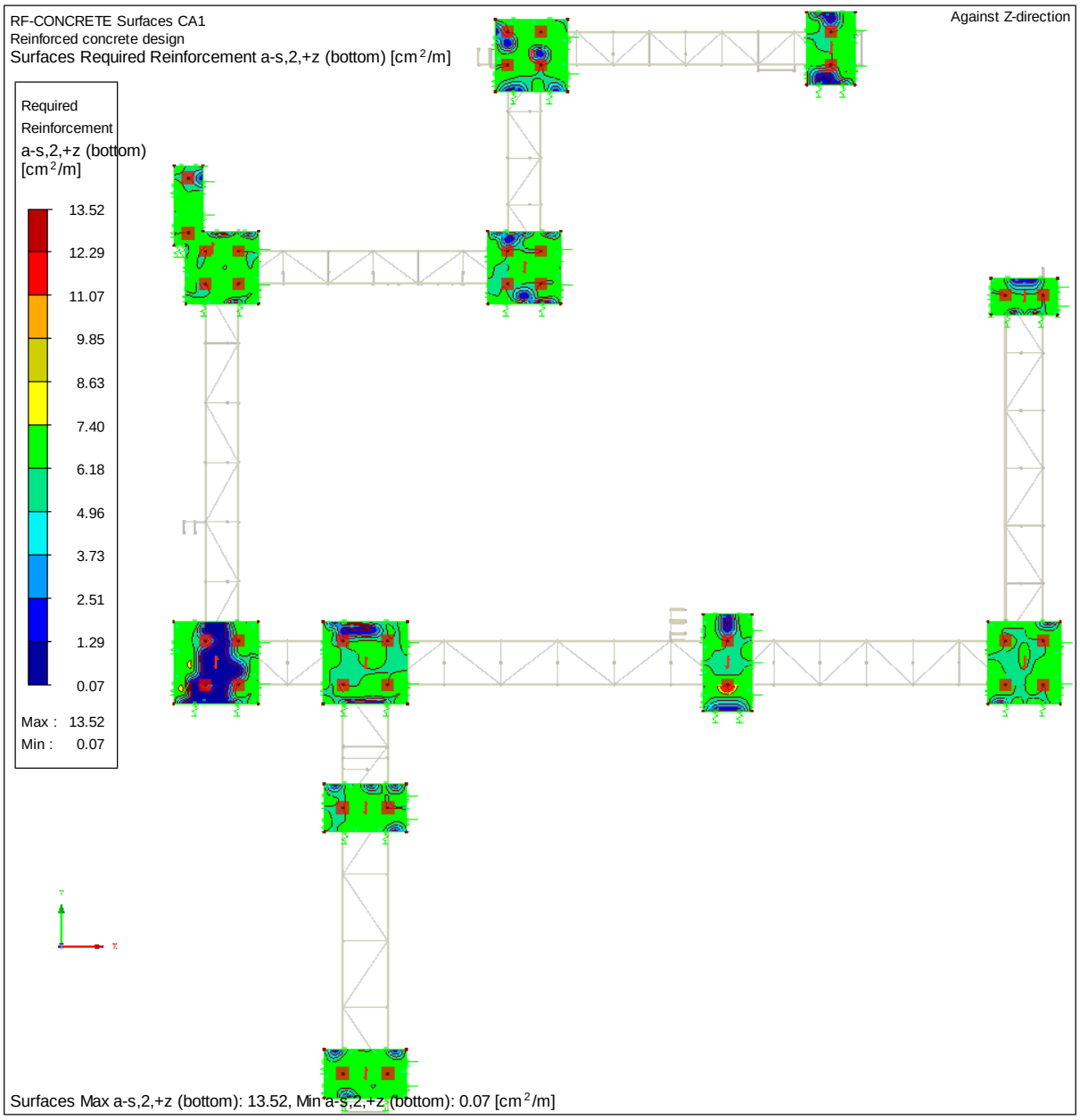


Figure 39 Required reinforcement As,2, bottom

Provide T16/150, As =13,41cm^2/m

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7. SKETCH OF FOUNDATION REINFORCEMENT

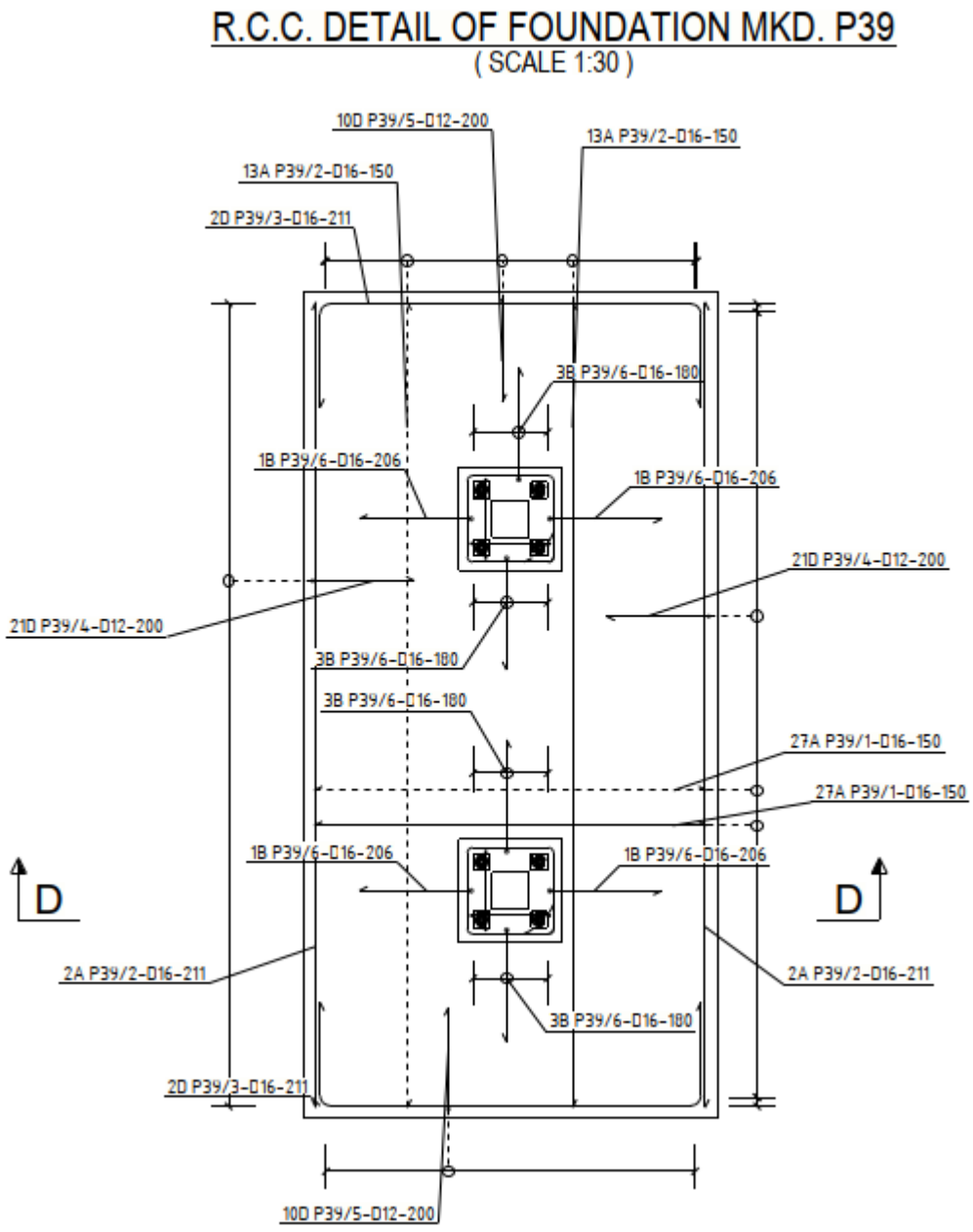


Figure 40 Top view

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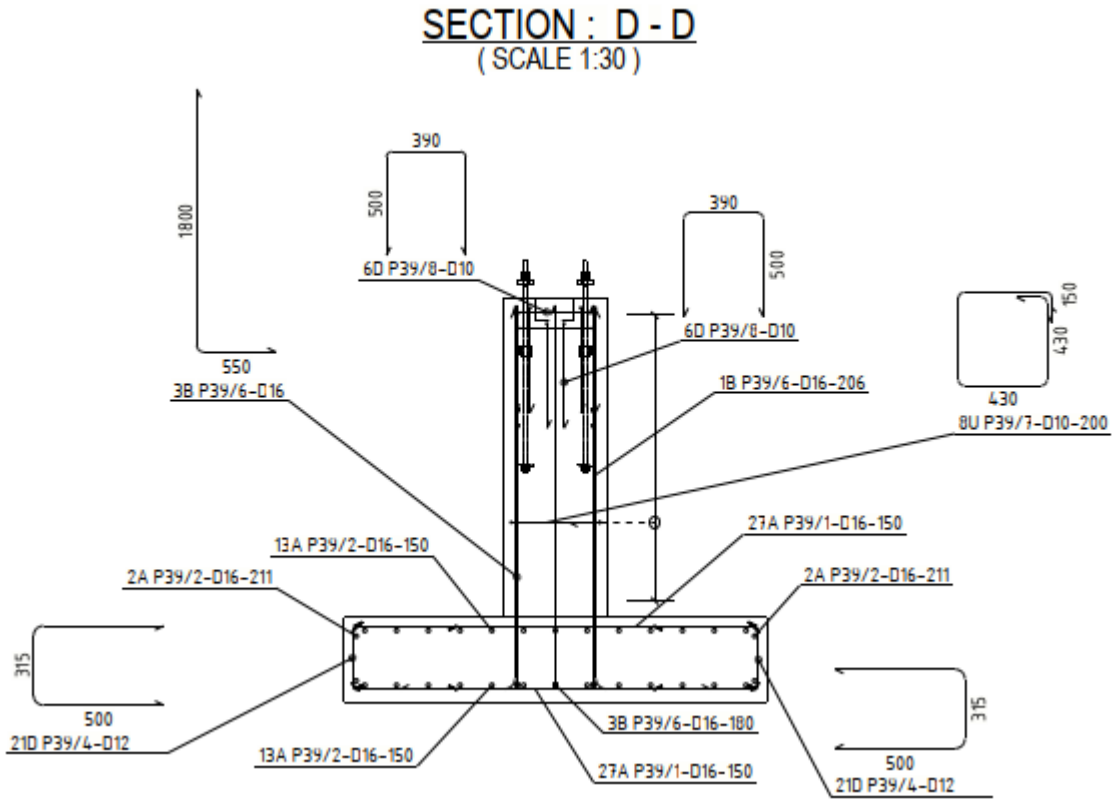


Figure 41 Section view

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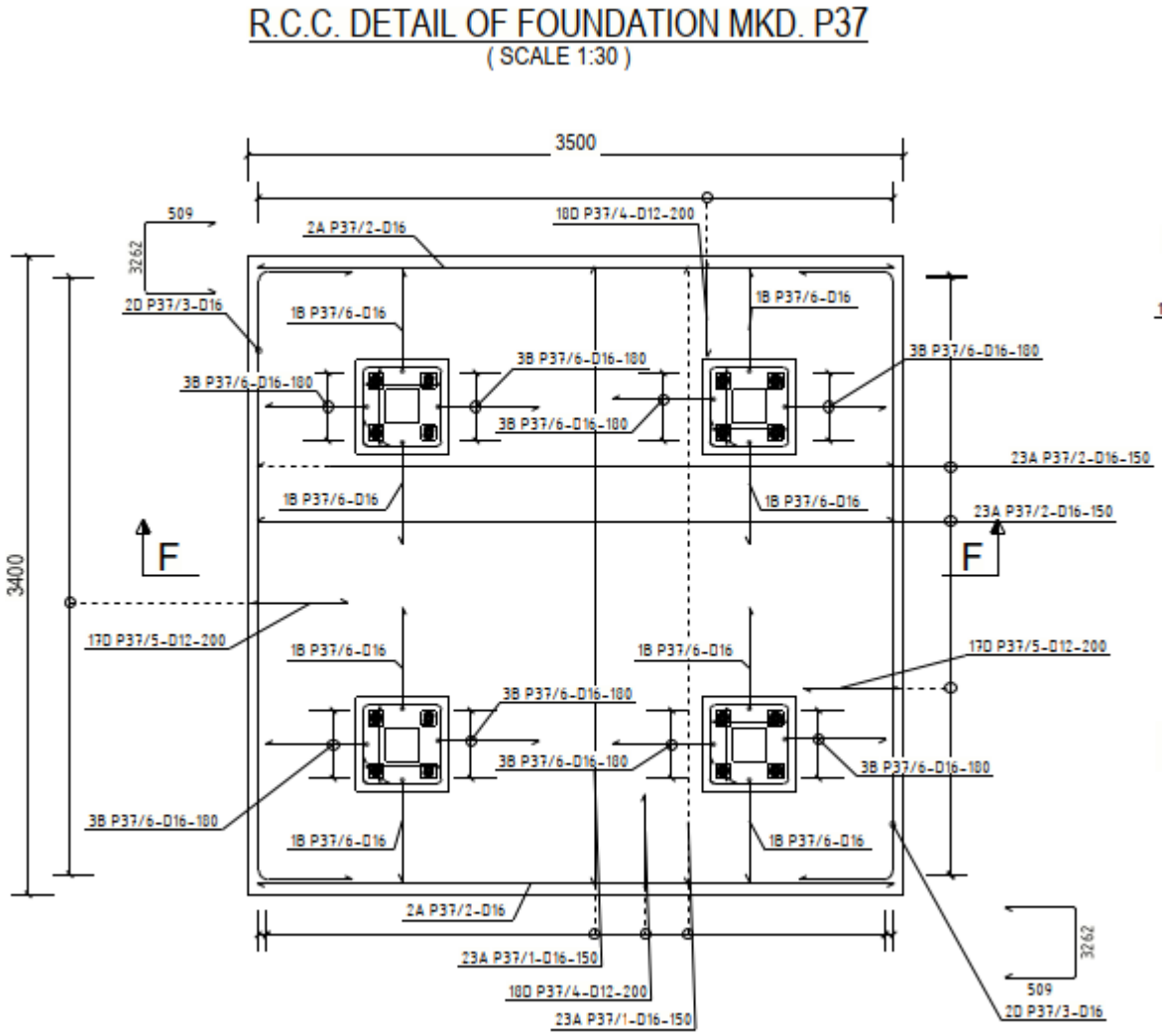


Figure 42 Top view

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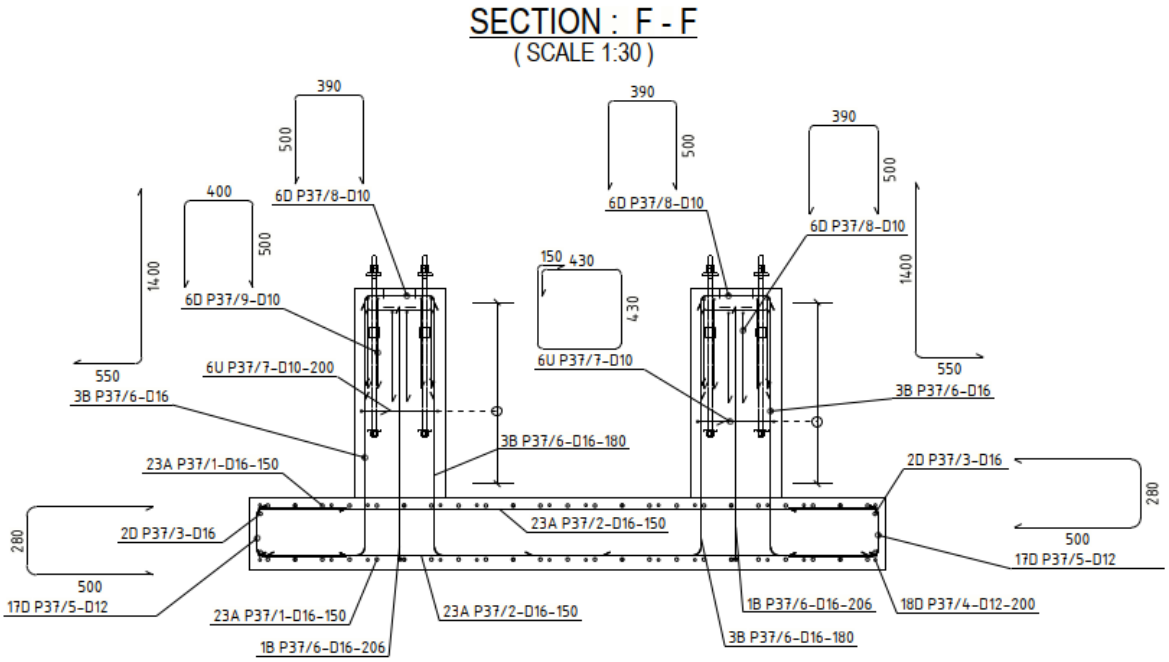
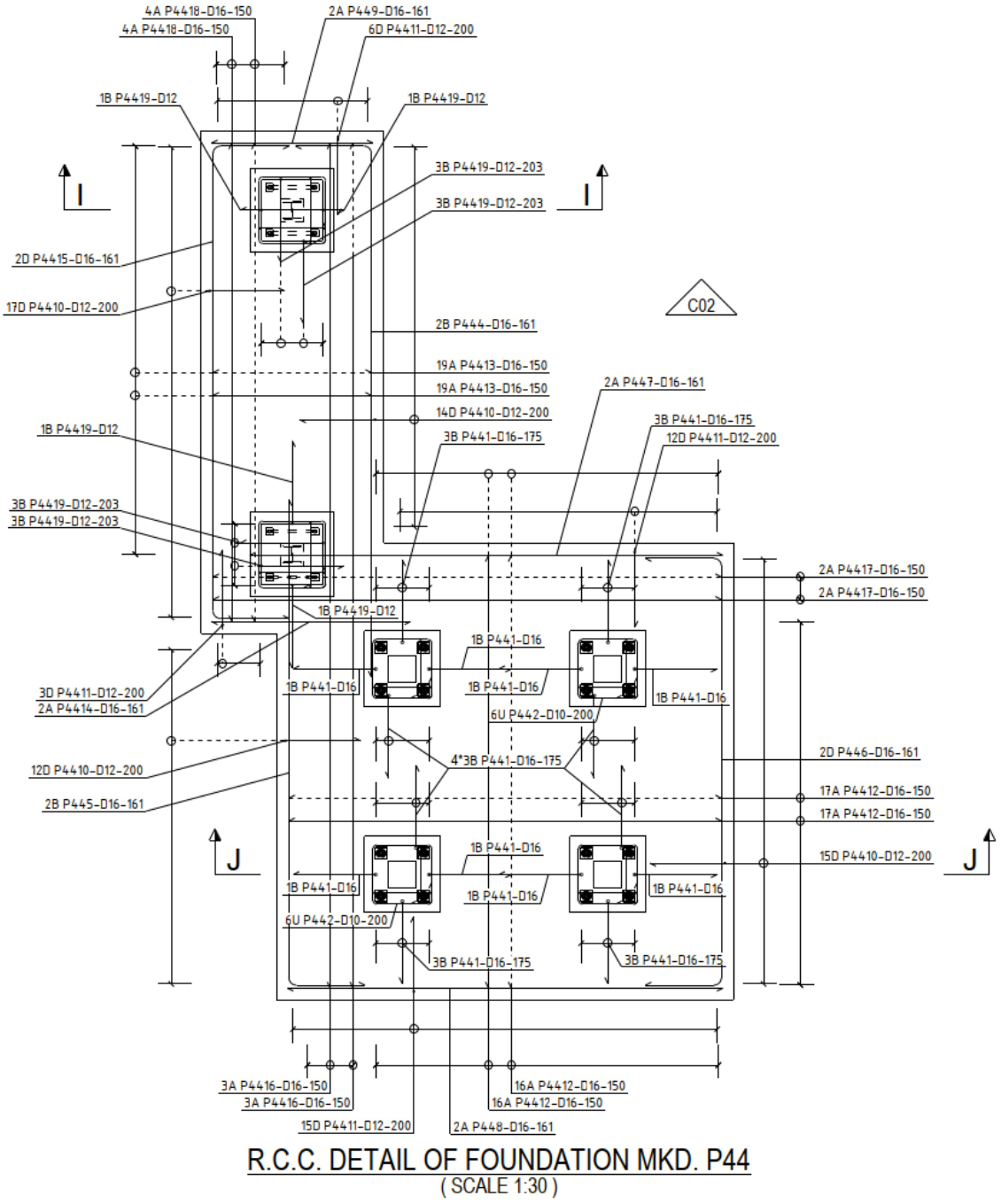


Figure 43 Section view

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R.C.C. DETAIL OF FOUNDATION MKD. P44
(SCALE 1:30)

Figure 44 Top view

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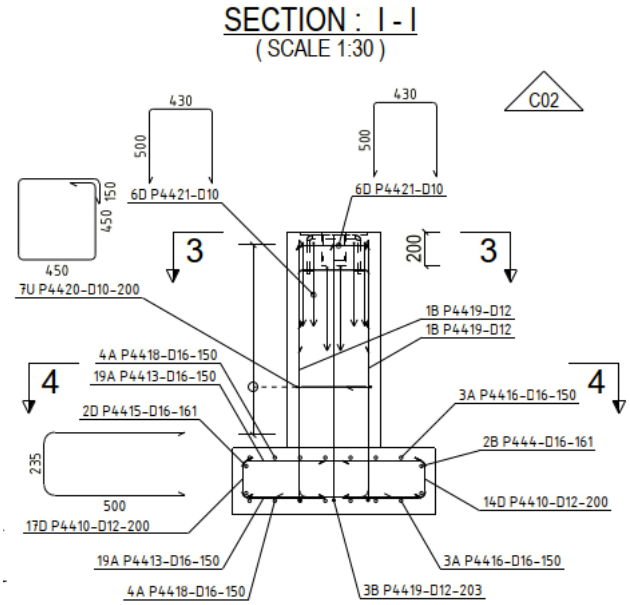


Figure 45 Section view I-I

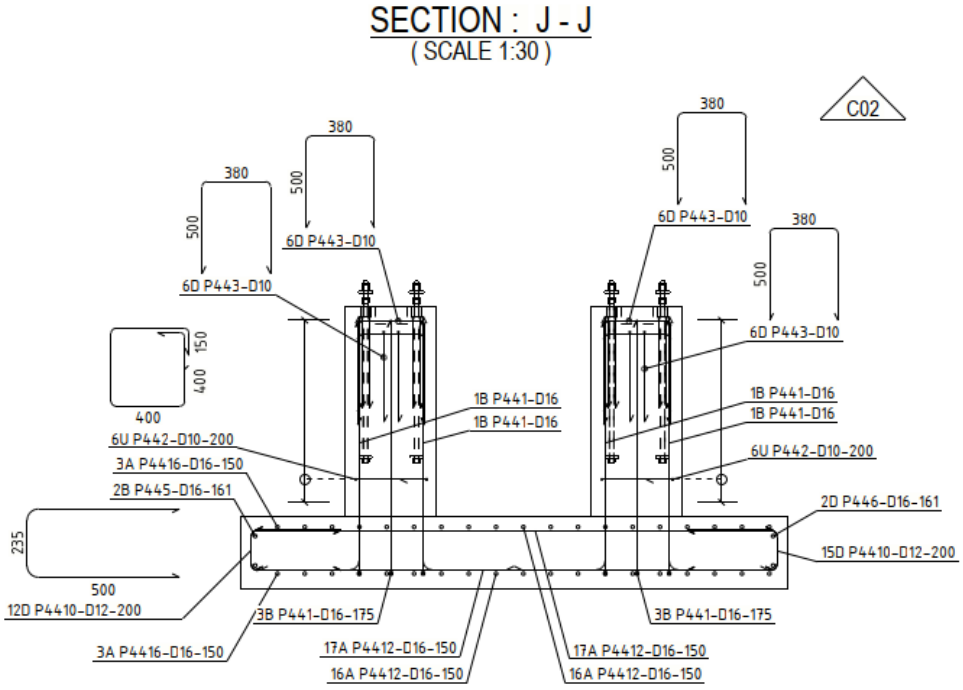


Figure 46 Section view J-J

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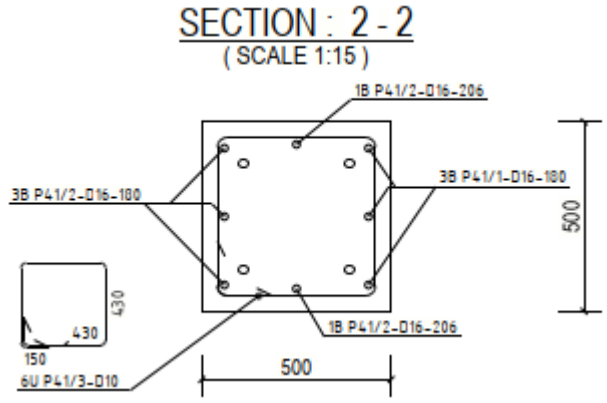


Figure 47 Reinforcement for pedestal cross section 500x500

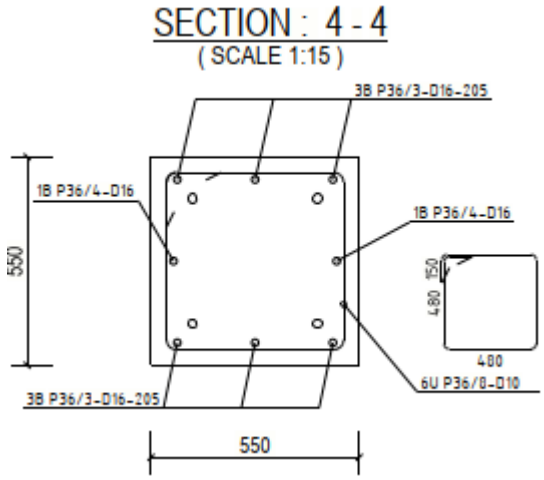


Figure 48 Reinforcement for pedestal cross section 550x550

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8. REFERENCE

[1] Saudi Building Code (SBC) 301, Structural - Loading and Forces.

[2] Saudi Building Code (SBC) 304, Structural – Concrete Structures.

[3] Project Design Criteria, MD-216-1000-EG-ST-CRT-1001-A03.

[4] Overall geotechnical report, MD-216-3000-EG-CV-RPT-1008-C01.

[5] General arrangement drawing, raw water treatment plan, 1-8/E-J, MD-216-49W1-EG-CV-GAD-1001

[6] General arrangement drawing, raw water treatment plan, 7-12/E-J, MD-216-49W1-EG-CV-GAD-1002

[7] General arrangement drawing, raw water treatment plan, 1-10/A-E, MD-216-49W1-EG-CV-GAD-1003

[8] Detail drawing, raw water treatment plan, Detail- and reinforcement drawing, Piperack foundation 1, MD-216-49W1-EG-CV-DED-1016

[9] Detail drawing, raw water treatment plan, Detail- and reinforcement drawing, Piperack foundation 2, MD-216-49W1-EG-CV-DED-1019

[10] General arrangement drawing, raw water treatment plant, structural steel, Pipebridge, plan view, MD-216-49W1-EG-ST-GAD-1006

[11] General arrangement drawing, raw water treatment plant, structural steel, Pipebridge, sections & details, MD-216-49W1-EG-ST-GAD-1006

[12] Ma'aden Engineering Structural Design Criteria MD-101-SMEM-EG-CRT-0001

[13] Super structure steel calculation MD-216-49W1-EG-ST-CAL-1001

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ATTACHMENT 1 INPUT LOAD DATA FROM PIPERACK

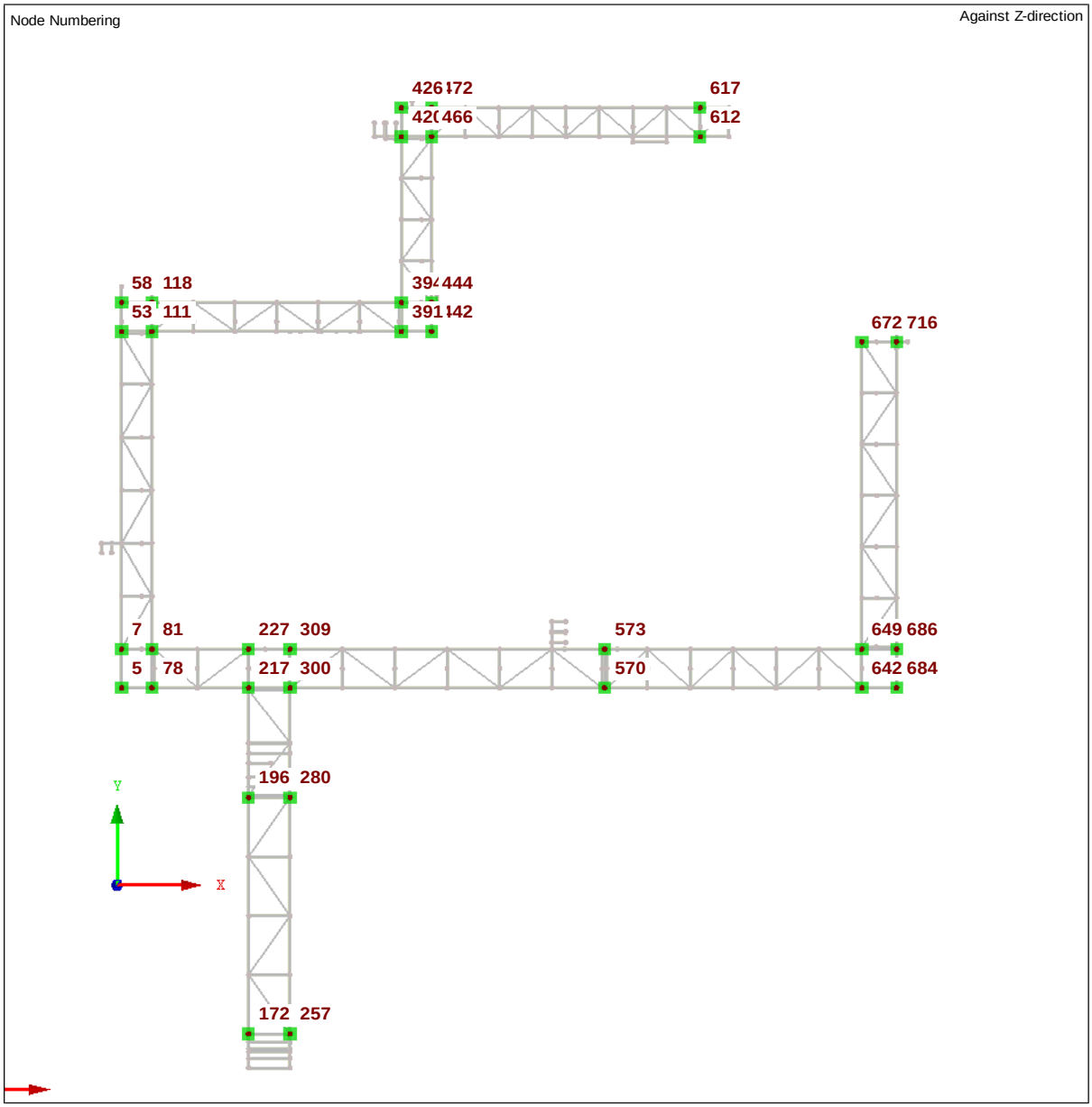


Figure 49 Node location in RFEM

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Node No.	LC/CO	Support Forces [kN]			Support Moments [kNm]		
		$P_{X'}$	$P_{Y'}$	$P_{Z'}$	$M_{X'}$	$M_{Y'}$	$M_{Z'}$
5	LC1	0.09	-2.70	-4.84	0.00	0.00	0.00
	LC2	0.38	-11.08	-14.74	0.00	0.00	0.02
	LC3	1.68	10.98	17.51	0.00	0.00	-0.18
	LC4	-1.68	-10.99	-17.51	0.00	0.00	0.18
	LC5	0.30	11.41	15.80	0.00	0.00	-0.02
	LC6	-0.30	-11.39	-15.77	0.00	0.00	0.02
	LC7	-1.34	0.22	-1.86	0.00	0.00	0.00
	LC8	1.34	-0.22	1.86	0.00	0.00	0.00
7	LC1	-0.03	0.16	-3.37	0.00	0.00	0.00
	LC2	-0.14	0.62	-11.25	0.00	0.00	0.03
	LC3	17.60	-0.61	31.00	0.00	0.00	-0.24
	LC4	-17.60	0.61	-31.00	0.00	0.00	0.24
	LC5	0.38	-0.08	-13.44	0.00	0.00	-0.01
	LC6	-0.38	0.08	13.41	0.00	0.00	0.01
	LC7	-0.33	-0.01	-1.75	0.00	0.00	0.43
	LC8	0.33	0.01	1.75	0.00	0.00	-0.43
53	LC1	-1.69	2.50	-4.93	0.00	0.00	-0.01
	LC2	-12.17	9.36	-28.96	0.00	0.00	-0.08
	LC3	13.09	-4.46	26.71	0.00	0.00	0.26
	LC4	-13.08	4.46	-26.71	0.00	0.00	-0.26
	LC5	7.52	9.11	30.12	0.00	0.00	0.13
	LC6	-7.52	-9.09	-30.09	0.00	0.00	-0.13
	LC7	-0.19	-0.51	-2.70	0.00	0.00	0.36
	LC8	0.19	0.51	2.70	0.00	0.00	-0.36
58	LC1	-1.39	0.04	-9.66	0.00	0.00	0.00
	LC2	-9.89	0.68	-41.98	0.00	0.00	0.01
	LC3	10.30	-0.02	28.52	0.00	0.00	0.13
	LC4	-10.30	0.02	-28.52	0.00	0.00	-0.13
	LC5	-9.00	-0.44	-33.62	0.00	0.00	0.09
	LC6	9.00	0.44	33.59	0.00	0.00	-0.09
	LC7	0.10	0.00	1.65	0.00	0.00	-0.01
	LC8	-0.10	0.00	-1.65	0.00	0.00	0.01
78	LC1	-0.07	-2.64	-7.13	0.00	0.00	-0.01
	LC2	-0.33	-8.65	-24.15	0.00	0.00	-0.03
	LC3	0.53	-11.30	-16.89	0.00	0.00	-0.18
	LC4	-0.53	11.30	16.89	0.00	0.00	0.18
	LC5	0.12	11.63	19.55	0.00	0.00	0.08
	LC6	-0.12	-11.59	-19.49	0.00	0.00	-0.08
	LC7	1.46	1.34	5.02	0.00	0.00	-0.34
	LC8	-1.46	-1.34	-5.02	0.00	0.00	0.34
81	LC1	0.00	0.12	-4.37	0.00	0.00	0.02
	LC2	0.01	0.37	-11.65	0.00	0.00	0.05
	LC3	-0.21	0.44	-29.60	0.00	0.00	-0.12
	LC4	0.21	-0.44	29.61	0.00	0.00	0.12
	LC5	-0.02	-0.10	-21.32	0.00	0.00	0.06
	LC6	0.02	0.10	21.26	0.00	0.00	-0.06
	LC7	0.26	0.17	0.46	0.00	0.00	-0.06
	LC8	-0.26	-0.17	-0.46	0.00	0.00	0.06
111	LC1	0.07	2.72	-2.64	0.00	0.00	0.00
	LC2	0.48	8.76	-18.08	0.00	0.00	0.01
	LC3	0.19	4.38	-27.49	0.00	0.00	0.25
	LC4	-0.19	-4.38	27.49	0.00	0.00	-0.25
	LC5	-0.30	12.40	22.11	0.00	0.00	0.10
	LC6	0.30	-12.38	-22.09	0.00	0.00	-0.10
	LC7	0.15	-1.02	1.32	0.00	0.00	0.01
	LC8	-0.15	1.02	-1.32	0.00	0.00	-0.01
118	LC1	0.08	-0.07	-8.27	0.00	0.00	0.00
	LC2	0.54	-0.23	-35.14	0.00	0.00	-0.04
	LC3	-0.21	-0.12	-26.86	0.00	0.00	0.09
	LC4	0.21	0.12	26.86	0.00	0.00	-0.09
	LC5	0.52	-0.05	-19.43	0.00	0.00	0.18
	LC6	-0.52	0.04	19.41	0.00	0.00	-0.18
	LC7	0.22	0.02	-0.07	0.00	0.00	-0.38
	LC8	-0.22	-0.02	0.07	0.00	0.00	0.38
172	LC1	-0.04	0.42	-9.63	0.00	0.00	0.00

	LC2	-0.50	4.16	-39.90	0.00	0.00	-0.26
	LC3	1.01	-0.60	49.58	0.00	0.00	1.87
	LC4	-1.01	0.60	-47.28	0.00	0.00	-1.89
	LC5	-0.01	1.76	0.42	0.00	0.00	-0.03
	LC6	0.02	-1.76	-0.39	0.00	0.00	-0.01
	LC7	4.39	0.01	-0.10	0.00	0.00	-1.89
	LC8	-4.39	-0.01	0.10	0.00	0.00	1.89
196	LC1	-0.03	0.34	-9.22	0.00	0.00	0.00
	LC2	-0.08	3.92	-25.86	0.00	0.00	0.04
	LC3	0.37	0.14	44.87	0.00	0.00	-0.14
	LC4	-0.37	-0.14	-42.62	0.00	0.00	0.15
	LC5	0.03	-8.19	2.65	0.00	0.00	-0.01
	LC6	-0.03	7.75	-2.81	0.00	0.00	0.01
	LC7	6.05	0.11	0.95	0.00	0.00	1.31
	LC8	-6.05	-0.11	-0.95	0.00	0.00	-1.31
217	LC1	-1.84	-0.95	-9.91	0.00	0.00	-0.03
	LC2	-11.19	-9.37	-54.09	0.00	0.00	-0.15
	LC3	14.55	0.92	22.37	0.00	0.00	0.06
	LC4	-14.53	-0.99	-22.45	0.00	0.00	-0.06
	LC5	4.95	31.56	54.90	0.00	0.00	0.06
	LC6	-5.10	-30.99	-54.37	0.00	0.00	-0.06
	LC7	-2.53	1.21	-1.85	0.00	0.00	-0.18
	LC8	2.53	-1.21	1.85	0.00	0.00	0.18
227	LC1	-2.08	0.03	-4.67	0.00	0.00	0.01
	LC2	-9.28	0.27	-7.95	0.00	0.00	0.03
	LC3	11.95	-0.03	14.12	0.00	0.00	0.03
	LC4	-11.94	0.03	-14.01	0.00	0.00	-0.03
	LC5	-6.57	-0.38	-57.23	0.00	0.00	0.01
	LC6	6.71	0.37	56.62	0.00	0.00	0.00
	LC7	-4.70	-0.03	-8.34	0.00	0.00	0.50
	LC8	4.70	0.03	8.34	0.00	0.00	-0.50
257	LC1	0.04	0.42	-9.63	0.00	0.00	0.00
	LC2	0.51	4.14	-39.93	0.00	0.00	0.22
	LC3	19.82	0.60	-49.58	0.00	0.00	1.13
	LC4	-19.12	-0.60	47.28	0.00	0.00	-1.14
	LC5	0.03	1.81	0.29	0.00	0.00	-0.04
	LC6	-0.03	-1.81	-0.26	0.00	0.00	0.08
	LC7	-4.43	-0.01	0.10	0.00	0.00	2.12
	LC8	4.43	0.01	-0.10	0.00	0.00	-2.12
280	LC1	0.03	0.35	-9.16	0.00	0.00	0.01
	LC2	-0.30	4.02	-23.73	0.00	0.00	0.04
	LC3	16.79	-0.14	-44.87	0.00	0.00	0.17
	LC4	-16.11	0.14	42.62	0.00	0.00	-0.16
	LC5	1.71	-7.97	-2.51	0.00	0.00	-0.12
	LC6	-1.80	7.52	2.59	0.00	0.00	0.12
	LC7	-5.37	-0.11	-0.95	0.00	0.00	-1.36
	LC8	5.37	0.11	0.95	0.00	0.00	1.36
300	LC1	0.25	-0.84	-7.16	0.00	0.00	0.04
	LC2	1.28	-7.03	-32.86	0.00	0.00	0.16
	LC3	-0.25	-0.27	-22.42	0.00	0.00	0.00
	LC4	0.25	0.34	22.50	0.00	0.00	0.00
	LC5	-0.35	28.22	41.65	0.00	0.00	0.10
	LC6	0.39	-28.11	-42.28	0.00	0.00	-0.09
	LC7	0.91	-1.31	-0.01	0.00	0.00	-0.32
	LC8	-0.91	1.31	0.01	0.00	0.00	0.32
309	LC1	0.29	0.01	-2.44	0.00	0.00	0.00
	LC2	1.50	-0.01	-0.30	0.00	0.00	0.00
	LC3	-0.75	0.02	-15.25	0.00	0.00	0.01
	LC4	0.75	-0.02	15.13	0.00	0.00	-0.01
	LC5	0.43	0.15	-39.89	0.00	0.00	0.08
	LC6	-0.49	-0.12	40.58	0.00	0.00	-0.09
	LC7	0.80	0.05	8.04	0.00	0.00	-0.13
	LC8	-0.80	-0.05	-8.04	0.00	0.00	0.13
391	LC1	1.37	-0.01	-2.60	0.00	0.00	0.00
	LC2	10.71	0.00	-14.17	0.00	0.00	0.00
	LC3	9.67	-0.03	19.36	0.00	0.00	-0.06
	LC4	-9.67	0.03	-19.36	0.00	0.00	0.06

	LC5	-3.59	-0.08	34.61	0.00	0.00	-0.37
	LC6	3.58	0.08	-34.64	0.00	0.00	0.37
	LC7	-0.51	-0.04	-5.31	0.00	0.00	-0.01
	LC8	0.51	0.04	5.31	0.00	0.00	0.01
394	LC1	1.60	0.14	-5.99	0.00	0.00	0.00
	LC2	10.67	1.89	-40.05	0.00	0.00	-0.01
	LC3	10.02	0.56	24.44	0.00	0.00	-0.04
	LC4	-10.02	-0.56	-24.45	0.00	0.00	0.04
	LC5	4.56	16.43	-32.39	0.00	0.00	-0.13
	LC6	-4.55	-16.43	32.40	0.00	0.00	0.13
	LC7	-1.20	-1.87	2.47	0.00	0.00	-0.35
	LC8	1.20	1.87	-2.47	0.00	0.00	0.35
420	LC1	0.00	0.03	-5.88	0.00	0.00	0.01
	LC2	0.51	-0.76	-24.03	0.00	0.00	0.07
	LC3	12.82	-0.54	31.95	0.00	0.00	0.08
	LC4	-12.77	0.55	-31.81	0.00	0.00	-0.09
	LC5	5.23	12.40	39.12	0.00	0.00	-0.43
	LC6	-5.23	-12.40	-39.12	0.00	0.00	0.43
	LC7	-0.94	-0.97	-5.56	0.00	0.00	0.08
	LC8	0.94	0.97	5.56	0.00	0.00	-0.08
426	LC1	0.01	0.01	-2.36	0.00	0.00	0.01
	LC2	-0.51	0.00	-10.02	0.00	0.00	0.03
	LC3	12.66	0.00	30.39	0.00	0.00	0.05
	LC4	-12.61	0.00	-30.30	0.00	0.00	-0.05
	LC5	-5.55	0.38	-40.05	0.00	0.00	0.10
	LC6	5.55	-0.38	40.05	0.00	0.00	-0.10
	LC7	0.51	-0.01	-0.10	0.00	0.00	-0.42
	LC8	-0.51	0.01	0.10	0.00	0.00	0.42
442	LC1	0.01	-0.01	-3.86	0.00	0.00	0.00
	LC2	0.04	-0.01	-20.76	0.00	0.00	-0.02
	LC3	0.30	0.00	-17.24	0.00	0.00	-0.04
	LC4	-0.30	0.00	17.24	0.00	0.00	0.04
	LC5	-0.02	0.30	16.22	0.00	0.00	-0.16
	LC6	0.02	-0.30	-16.30	0.00	0.00	0.16
	LC7	0.25	0.01	2.16	0.00	0.00	0.54
	LC8	-0.25	-0.01	-2.16	0.00	0.00	-0.54
444	LC1	-0.06	0.03	-7.66	0.00	0.00	0.00
	LC2	-0.26	0.12	-40.26	0.00	0.00	-0.01
	LC3	-0.73	0.51	-27.45	0.00	0.00	-0.05
	LC4	0.73	-0.51	27.45	0.00	0.00	0.05
	LC5	-0.15	4.33	-17.86	0.00	0.00	-0.17
	LC6	0.15	-4.40	17.94	0.00	0.00	0.17
	LC7	0.35	0.64	0.23	0.00	0.00	0.04
	LC8	-0.35	-0.64	-0.23	0.00	0.00	-0.04
466	LC1	0.02	-0.14	-8.61	0.00	0.00	0.00
	LC2	0.05	-1.02	-34.42	0.00	0.00	0.01
	LC3	0.33	-0.75	-35.15	0.00	0.00	0.09
	LC4	-0.33	0.74	35.00	0.00	0.00	-0.09
	LC5	-0.13	13.05	26.31	0.00	0.00	0.08
	LC6	0.13	-13.05	-26.31	0.00	0.00	-0.08
	LC7	0.33	2.07	9.82	0.00	0.00	0.28
	LC8	-0.33	-2.07	-9.82	0.00	0.00	-0.28
472	LC1	-0.02	0.01	-5.85	0.00	0.00	0.00
	LC2	-0.08	0.02	-28.23	0.00	0.00	0.02
	LC3	0.08	0.02	-26.47	0.00	0.00	0.04
	LC4	-0.08	-0.02	26.39	0.00	0.00	-0.04
	LC5	0.12	0.10	-26.04	0.00	0.00	0.11
	LC6	-0.12	-0.10	26.04	0.00	0.00	-0.11
	LC7	0.89	-0.06	-3.03	0.00	0.00	0.05
	LC8	-0.89	0.06	3.03	0.00	0.00	-0.05
570	LC1	0.01	0.06	-10.55	0.00	0.00	0.00
	LC2	-0.02	-0.83	-61.06	0.00	0.00	-0.01
	LC3	0.62	1.12	2.10	0.00	0.00	0.02
	LC4	-0.62	-1.12	-2.10	0.00	0.00	-0.02
	LC5	0.00	32.03	65.29	0.00	0.00	-0.03
	LC6	0.03	-33.37	-68.45	0.00	0.00	0.05
	LC7	-0.01	0.13	-0.15	0.00	0.00	-0.14

	LC8	0.01	-0.13	0.15	0.00	0.00	0.14
573	LC1	0.04	0.02	-11.48	0.00	0.00	0.00
	LC2	0.22	0.45	-57.06	0.00	0.00	-0.01
	LC3	0.37	-0.01	-2.20	0.00	0.00	0.02
	LC4	-0.37	0.01	2.20	0.00	0.00	-0.02
	LC5	0.00	0.00	-65.34	0.00	0.00	0.00
	LC6	-0.03	0.02	68.51	0.00	0.00	0.02
	LC7	-0.01	-0.32	0.25	0.00	0.00	0.14
	LC8	0.01	0.32	-0.25	0.00	0.00	-0.14
612	LC1	0.00	0.00	-6.30	0.00	0.00	0.00
	LC2	0.00	-0.04	-18.80	0.00	0.00	0.02
	LC3	0.63	0.48	0.93	0.00	0.00	-0.02
	LC4	-0.63	-0.48	-0.92	0.00	0.00	0.02
	LC5	0.00	14.15	40.56	0.00	0.00	-0.16
	LC6	0.00	-14.15	-40.56	0.00	0.00	0.16
	LC7	0.00	0.26	0.76	0.00	0.00	0.24
	LC8	0.00	-0.26	-0.76	0.00	0.00	-0.24
617	LC1	0.00	0.00	-6.16	0.00	0.00	0.00
	LC2	0.00	0.01	-18.43	0.00	0.00	0.02
	LC3	0.38	-0.01	-1.69	0.00	0.00	-0.01
	LC4	-0.38	0.01	1.68	0.00	0.00	0.01
	LC5	0.00	0.05	-40.51	0.00	0.00	-0.16
	LC6	0.00	-0.05	40.51	0.00	0.00	0.16
	LC7	0.00	0.01	-1.62	0.00	0.00	-0.28
	LC8	0.00	-0.01	1.62	0.00	0.00	0.28
642	LC1	1.82	-0.06	-2.34	0.00	0.00	0.00
	LC2	9.83	0.24	-9.40	0.00	0.00	0.00
	LC3	12.85	7.99	35.91	0.00	0.00	-0.06
	LC4	-12.85	-7.99	-35.90	0.00	0.00	0.06
	LC5	-2.68	13.12	21.29	0.00	0.00	-0.11
	LC6	2.79	-13.08	-21.04	0.00	0.00	0.11
	LC7	1.95	0.29	3.53	0.00	0.00	-0.39
	LC8	-1.95	-0.29	-3.53	0.00	0.00	0.39
649	LC1	1.49	0.01	-7.83	0.00	0.00	0.01
	LC2	8.13	-0.01	-28.27	0.00	0.00	0.05
	LC3	17.05	-0.28	27.02	0.00	0.00	-0.16
	LC4	-17.04	0.28	-27.01	0.00	0.00	0.16
	LC5	2.17	0.24	-22.28	0.00	0.00	-0.06
	LC6	-2.18	-0.24	22.24	0.00	0.00	0.06
	LC7	2.72	-0.01	7.80	0.00	0.00	0.03
	LC8	-2.72	0.01	-7.80	0.00	0.00	-0.03
672	LC1	0.01	0.00	-5.77	0.00	0.00	0.00
	LC2	0.01	0.00	-16.94	0.00	0.00	0.01
	LC3	14.44	0.00	34.32	0.00	0.00	0.25
	LC4	-14.44	0.00	-34.32	0.00	0.00	-0.25
	LC5	0.35	0.58	0.49	0.00	0.00	0.03
	LC6	-0.35	-0.58	-0.49	0.00	0.00	-0.03
	LC7	-0.29	0.00	-0.31	0.00	0.00	-0.34
	LC8	0.29	0.00	0.31	0.00	0.00	0.34
684	LC1	0.01	0.08	-4.82	0.00	0.00	0.00
	LC2	0.03	0.76	-17.93	0.00	0.00	-0.01
	LC3	0.59	-9.44	-38.87	0.00	0.00	-0.07
	LC4	-0.59	9.44	38.86	0.00	0.00	0.07
	LC5	-0.01	10.93	22.10	0.00	0.00	-0.06
	LC6	0.01	-10.90	-22.27	0.00	0.00	0.06
	LC7	0.22	-0.28	-4.65	0.00	0.00	0.02
	LC8	-0.22	0.28	4.65	0.00	0.00	-0.02
686	LC1	-0.02	0.01	-9.30	0.00	0.00	-0.01
	LC2	-0.15	-0.03	-35.47	0.00	0.00	-0.05
	LC3	-0.05	0.47	-24.72	0.00	0.00	-0.23
	LC4	0.04	-0.47	24.71	0.00	0.00	0.23
	LC5	-0.05	-0.16	-20.59	0.00	0.00	-0.07
	LC6	0.05	0.16	20.55	0.00	0.00	0.07
	LC7	0.07	0.01	-6.34	0.00	0.00	0.35
	LC8	-0.07	-0.01	6.34	0.00	0.00	-0.35
716	LC1	-0.01	-0.07	-6.74	0.00	0.00	0.00
	LC2	0.00	-0.71	-28.18	0.00	0.00	0.02

	LC3	0.06	0.00	-34.37	0.00	0.00	0.15
	LC4	-0.06	0.00	34.37	0.00	0.00	-0.15
	LC5	-0.01	0.60	-1.02	0.00	0.00	0.03
	LC6	0.01	-0.60	1.02	0.00	0.00	-0.03
	LC7	0.22	0.00	0.16	0.00	0.00	0.34
	LC8	-0.22	0.00	-0.16	0.00	0.00	-0.34

ATTACHMENT 2 PIPERACK DRAWING

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ATTACHMENT 3 COMMENT RESOLUTION SHEET

When comments are received from Employer, they need to be placed in a comment resolution table, and the filled table is always attached to the end of the document.

COMMENT TABLE	Document Name:	Document Number:	Revision No.	Comment Date:
General Comment:				

Sr. No	Chapter	Regarding	Comment	Resolution
1	6.5	settlement	Umax shall be 25mm per 6.3.1 of MD-101-SMEM-EG-ST-CRT-00001	corrected
2				
2				
3				
4				

1 Remember also to always mark the made changes with a triangle to the left side of the document.

You can also reply to the comments on an excel template, then it will be coded as a separate document and submitted to Employer together with the revised document itself. You can find the instruction and the excel template from “Document Codes and Numbering Procedure” [1].

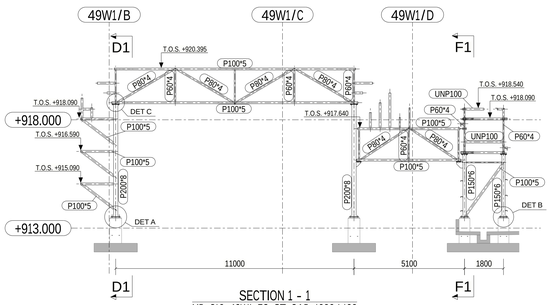
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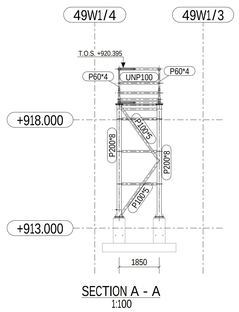
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DRAWING NUMBER: MD-216-49WL-EG-ST-GAD-1007 | REV: A01

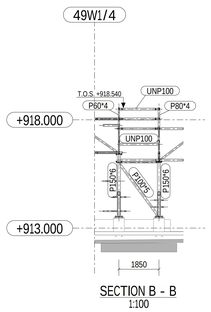
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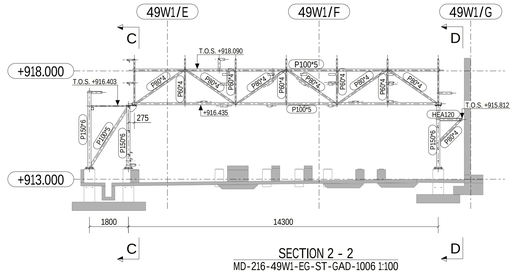
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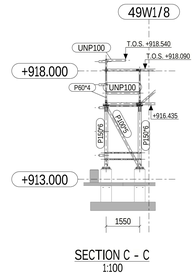
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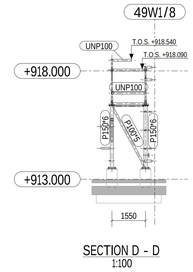
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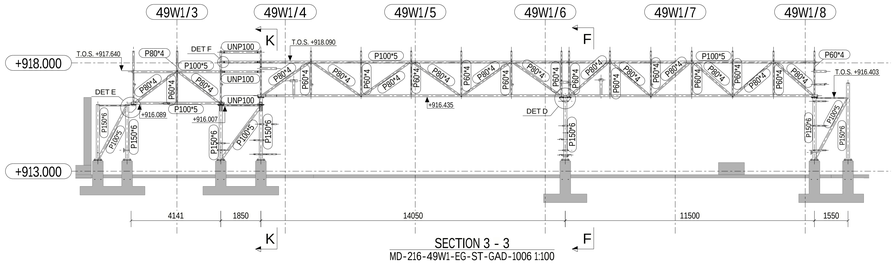
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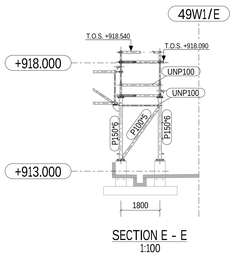
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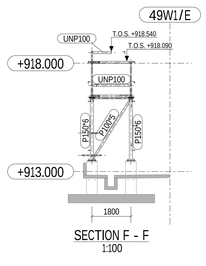
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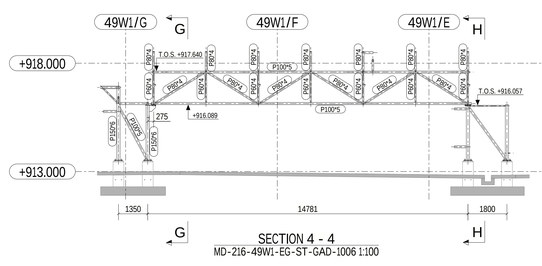
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MD-216-49W1-EG-ST-GAD-1006 1:100



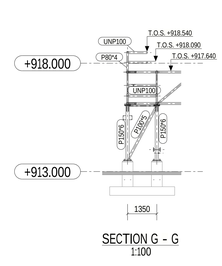
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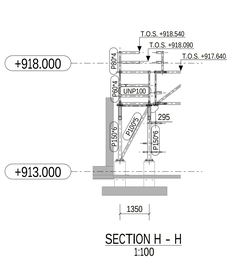
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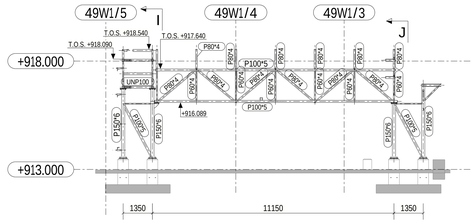
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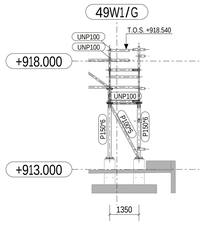
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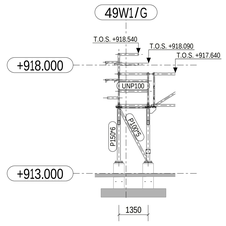
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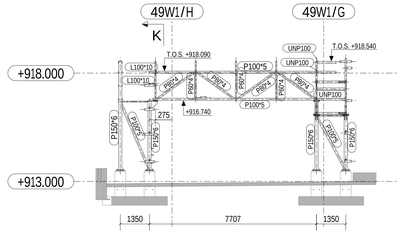
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MD-216-49W1-EG-ST-GAD-1006 1:100



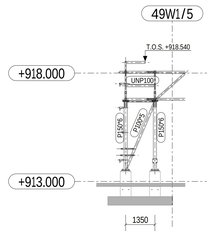
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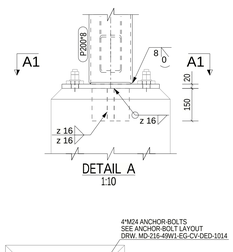
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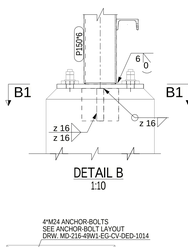


SECTION 6 - 6
MD-216-49W1-EG-ST-GAD-1006 1:100



SECTION K - K
1:100





WELDS:
All welds shall be in accordance with the project specification MD-101-SMEM-EG-ST-SPC-001 and AWS D1.1

THE SIZE OF A FILLET WELD REFERS TO THE LEG LENGTH

BOLTS:
HIGH STRENGTH BOLTS ASTM A325/A325M TYPE 1, UNLESS SPECIFIED OTHERWISE

MATERIALS:
- HOT ROLLED SECTIONS AND PLATES IN CONNECTIONS S275JR EN 10025
- RAILING AND LADDERS S235JR EN 10025

PAINT SYSTEMS:

PAINT FOR ACID SPLASH ENVIRONMENT:
ACID RESISTANCE PAINT REQUIRED
FOR STRUCTURE FOR ACID FUME

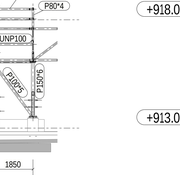
1) Emulsion clean to SSPC-SP 1
2) Abrasive blast clean to Sa 2 1/2
3) Primer: Hempadaur Multi Strength 45751, 1 coat 175 µm NDFT
4) Finish: Hempadaur Multi Strength 45751, 1 coat 175 µm NDFT or equivalent

COLOR CODING:

Steelwork: RAL 7040 Window Grey
Hoist beams: RAL 2004 Pure Orange
Railing and ladders: RAL 1023 Traffic Yellow

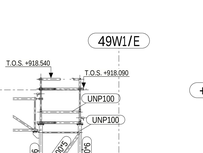
LEGEND:
TOS TOP OF STEEL
TYP TYPICAL
TOC TOP OF CONCRETE

NOTES:
1. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN METRES
2. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH CALCULATION REPORT
3. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS

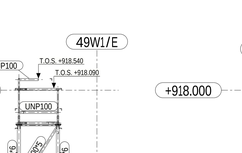


DETAIL C
1:10

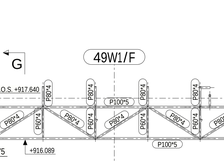
DETAIL D
1:10



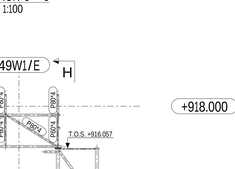
SECTION C1 - C1
1:10



SECTION D1 - D1
1:10



DETAIL E
1:10

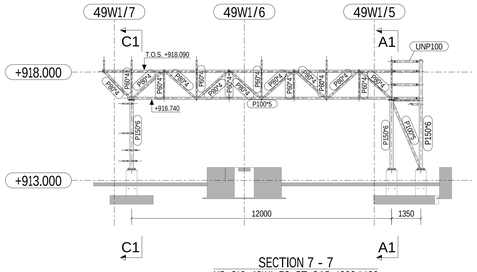


SECTION E1 - E1
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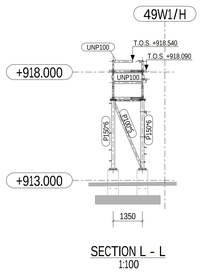
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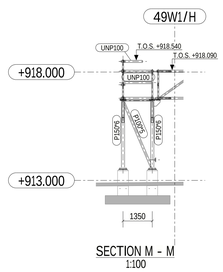
DETAIL F
1:10



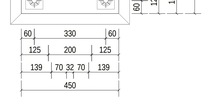
SECTION 7 - 7
MD-216-49W1-EG-ST-GAD-1006 1:100



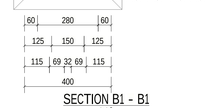
SECTION L - L
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SECTION M - M
1:100

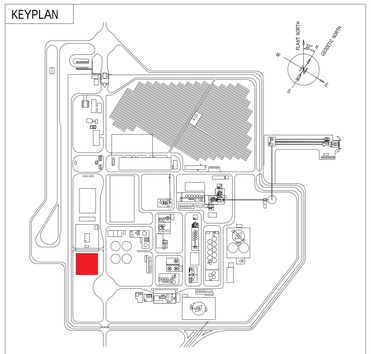


SECTION A1 - A1
1:10



SECTION B1 - B1
1:10

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-49W1-EG-CV-DED-1014	ANCHOR BOLT PLAN 1	MD-216-49W1-EG-CV-DED-1015	ANCHOR BOLT PLAN 2
MD-216-49W1-EG-CV-DED-1019	PIPERACK FOUNDATION 2	MD-216-49W1-EG-ST-GAD-1006	STEEL STRUCTURES OF PIPEBRIDGE - PLAN VIEW
MD-216-49W1-EG-ST-CAL-1001	CALCULATION REPORT OF BUFFER TANK'S STAIRCASE & PIPEBRIDGE	MD-216-49W1-EG-CV-DED-1016	PIPERACK FOUNDATION 1
MD-216-1000-EG-ST-CRT-1001	PROJECT DESIGN CRITERIA	MD-101-SMEM-EG-ST-SPC-0001	STEEL FABRICATION SPECIFICATION

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	30.11.2021	ISSUED FOR APPROVAL	JusRak	MauNot	LuiPiz	SouCha	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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L'ARSEN & TOUBRO DRAWING NUMBER	O19045_080_S_G_28110_007
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OUTOTEC DRAWING NUMBER	900707_GAU01_AG11_MTL01_49052
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REVISION	0

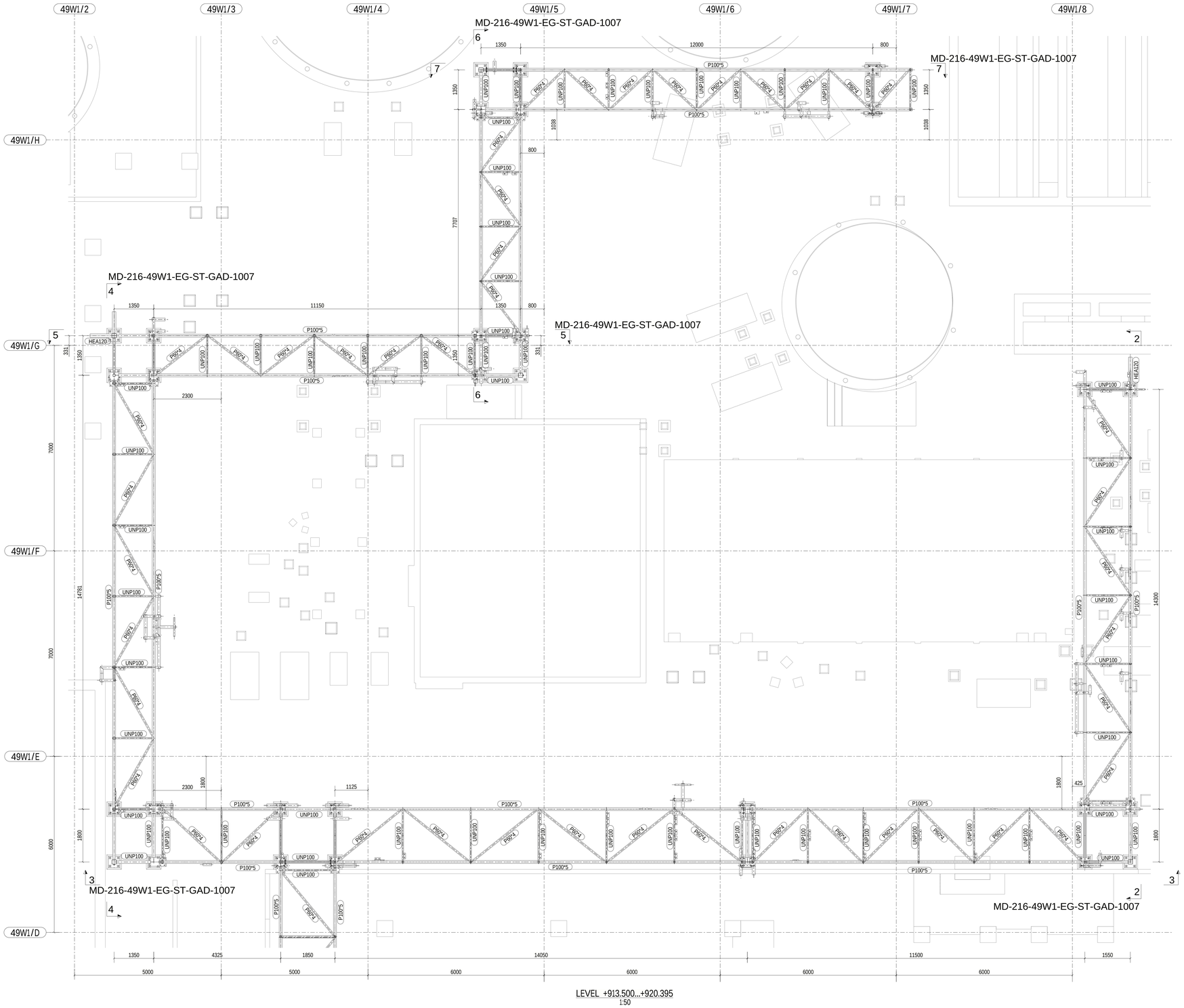
DRAWN	JusRak	30.11.2021
CHECKED	MauNot	30.11.2021
ENGINEER	LuiPiz	30.11.2021
APPROVED	SouCha	30.11.2021
EMPLOYER		
SCALE	1:100	A0

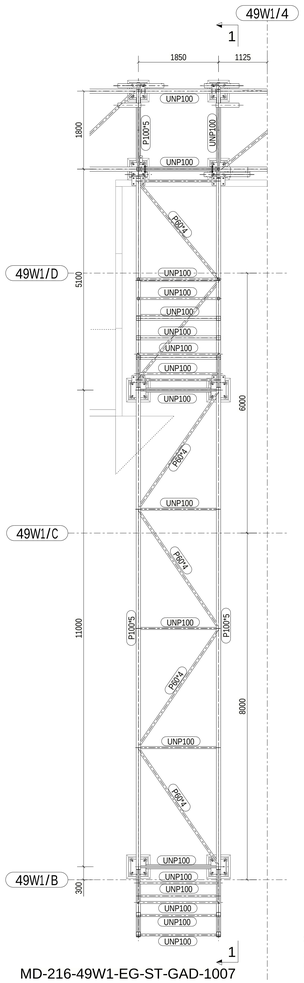
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
49W1 RAW WATER TREATMENT PLANT
STEEL STRUCTURES OF PIPEBRIDGE - SECTIONS & DETAILS

DRAWING NUMBER: MD-216-49W1-EG-ST-GAD-1007 | REV: A01

Tekla Structures

SCHEDULE OF MEMBERS		
Section	Profile	Material
s	P100*5	S275JR
s	P150*6	S275JR
s	P200*8	S275JR
u	UNP100	S275JR





WELDS:

All welds shall be in accordance with the project specification MD-101-SMEM-EG-ST-SPC-001 and AWS D1.1

THE SIZE OF A FILLET WELD REFERS TO THE LEG LENGTH

BOLTS:

HIGH STRENGTH BOLTS ASTM A325/A325M TYPE 1, UNLESS SPECIFIED OTHERWISE

MATERIALS:

- HOT ROLLED SECTIONS AND PLATES IN CONNECTIONS S275JR EN 10025
- RAILING AND LADDERS S235JR EN 10025

PAINT SYSTEMS:

PAINT FOR ACID SPLASH ENVIRONMENT:
ACID RESISTANCE PAINT REQUIRED
FOR STRUCTURE FOR ACID FUME

1) Emulsion clean to SSPC-SP 1
2) Abrasive blast clean to Sa 2 1/2
3) Primer: Hempadaur Multi Strength 45751 1 coat 175 µm NDFT
4) Finish: Hempadaur Multi Strength 45751, 1 coat 175 µm NDFT or equivalent

COLOR CODING:

Steelwork: RAL 7040 Window Grey
Hoist beams: RAL 2004 Pure Orange
Railing and ladders: RAL 1023 Traffic Yellow

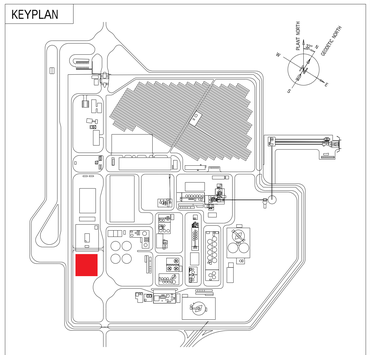
LEGEND:

TOS TOP OF STEEL
TYP TYPICAL
TOC TOP OF CONCRETE

NOTES:

1. ALL DIMENSIONS ARE IN MM AND LEVELS ARE IN METRES
2. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH CALCULATION REPORT
3. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
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MD-216-49W1-EG-CV-DED-1019	PIPERACK FOUNDATION 2	MD-216-49W1-EG-ST-GAD-1007	STEEL STRUCTURES OF PIPEBRIDGE - SECTIONS & DETAILS
MD-216-49W1-EG-ST-CAL-1001	CALCULATION REPORT OF BUFFER TANK'S STAIRCASE & PIPEBRIDGE	MD-216-49W1-EG-CV-DED-1016	PIPERACK FOUNDATION 1
MD-216-1000-EG-ST-CRT-1001	PROJECT DESIGN CRITERIA	MD-101-SMEM-EG-ST-SPC-0001	STEEL FABRICATION SPECIFICATION

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	30.11.2021	ISSUED FOR APPROVAL	JusRak	MauNot	LuiPiz	SouCha	

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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LARSEN & TOUBRO DRAWING NUMBER			REVISION
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OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707_GAU01_AG11_MTL01_49051		OU100732666	0
DRAWN	JusRak	30.11.2021	
CHECKED	MauNot	30.11.2021	
ENGINEER	LuiPiz	30.11.2021	
APPROVED	SouCha	30.11.2021	
EMPLOYER			
SCALE	1:50	A0	

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
49W1 RAW WATER TREATMENT PLANT
STEEL STRUCTURES OF PIPEBRIDGE - PLAN VIEW

DRAWING NUMBER MD-216-49W1-EG-ST-GAD-1006 REV A01

Tekla Structures